

Import Price Shocks, FX-Monetary Policies, and Real Income Stabilization

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Abstract

We characterize jointly optimal exchange rate and monetary policies in a heterogeneous-agent, import-dependent small open economy facing essential import-price spikes. When some households are borrowing constrained, exchange rate management becomes a real-income stabilization tool operating through the intertemporal margin, while monetary policy targets the labor wedge. In an open economy, these roles are not substitutable. A Ramsey planner internalizes how pecuniary general equilibrium effects from exchange rate movements redistribute real incomes: facing a spike, it leans against the depreciation, cutting exports and shifting resources to constrained households through real wages. Calibrated to Japan's 2022 energy-price path, optimal policy uses FX interventions, accepting costly interest parity deviations, to cut the exchange rate's co-movement with energy prices; monetary policy complements them with a disinflationary stance. Idiosyncratic risk adds a distributional time inconsistency; we characterize and solve optimal policies under timeless commitment and study how the exchange-rate-driven real-income stabilization motive affects equilibrium outcomes.

Keywords: Foreign exchange interventions; optimal monetary policy; heterogeneous agents; import price shocks; real income channel; energy shocks.

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1 Introduction

Recent years have repeatedly confronted import-dependent economies with large temporary spikes in the world prices of essential imports. The 2022 energy crisis, the renewed volatility in crude oil and LNG prices in 2026, and the rare-earth disruptions of 2025 all had the same economic structure: essential goods that were difficult to substitute away from in the short run became abruptly more expensive. For an import-dependent country, this is a terms-of-trade shock: the country must export more domestic tradeables to generate the purchasing power required to buy any given amount of imports. Because these imports are necessities, their short-run demand is highly price inelastic: demand falls little when the price rises, and the import bill goes up. This results in a real income loss for the country: more domestic output must be exported rather than consumed domestically to finance the higher import bill.

How much more of its output the country must hand over depends on the price at which foreigners buy it. When a country faces a downward-sloping demand for its goods, the world price of domestic tradeables must fall to induce foreigners to buy more and supply the requisite purchasing power. Consequently, an exogenous rise in the world price of an essential import can be accompanied by an endogenous decline in the world price of domestic output, which compounds the real-income losses: the country pays more for what it buys and receives less for what it sells. In a monetary economy, such real income losses inevitably take the form of domestic price inflation and nominal and real exchange rate depreciations: a real depreciation makes domestic output cheaper for foreigners, who buy more of it, and the export revenues thus generated enable the country to pay for its imports.

In this paper we address a policy-relevant question that directly relates to the mandate of monetary authorities in many countries: how should monetary and exchange rate policies be conducted by import-dependent economies facing temporary spikes in essential import prices¹? Since real exchange rate depreciations can amplify real income losses, should policymakers intervene in foreign exchange markets to resist depreciations, or should they let the real exchange rate move freely in response to such shocks? And what path of domestic inflation should be chosen while the shock lasts?

In the canonical representative agent small-open-economy New Keynesian framework, the conventional policy prescription is to use monetary policy to target nominal rigidities while allowing the real exchange rate to move freely to clear the foreign exchange market. In this literature the relevant frictions are nominal rigidities and the associated labor market distortions, which give monetary policy its role; the depreciation-induced real income losses that accompany an import price spike create no separate role for exchange rate policy to distort the path of the real exchange rate. That prescription rests on the assumption that agents have perfect access to international asset markets. Asset markets determine how much of the increased import bill is paid at each date and, therefore, how much the real exchange rate depreciates at each date and how closely its path tracks the path of import prices. The real exchange rate must eventually depreciate, but it need not do so contemporaneously with the import price spike. By running current account deficits

¹Exchange rate policy is a broad term. We define this policy as one where the monetary authority distorts cross-border financial flows to influence the path of the real exchange rate. This can be done through outright capital controls or foreign exchange interventions.

in high-price periods, the country can retain more of its output in these periods while exporting more in future when the shock has passed, thereby spreading the real income losses over time. A transitory import-price spike is then just a present-value real income loss. Policy cannot undo this loss, and has no separate reason to distort the timing of when the economy pays for it.

The recent heterogeneous-agent literature instead has departed from this assumption. When some households are liquidity constrained and cannot borrow against future incomes, the real income loss from an import price spike is no longer just a present value loss but also a current income loss: it falls on their real disposable income in the period the price rises. Restricted participation by households in asset markets then has general equilibrium consequences: constrained households would borrow against their future incomes during the spike if they could, and since they cannot, the economy borrows less in the aggregate than it would if every household were on its Euler path. More of the import bill must then be paid contemporaneously, more is exported in the high price periods, and the real exchange rate becomes overly correlated with import prices. The effective increase in the price of imports faced by the economy, in units of domestic output, is therefore amplified: not only does the exogenous world price go up, but the real exchange rate also depreciates simultaneously, and imports become dearer for both reasons. This feeds back and further erodes the real incomes of households, with the largest burden falling on those that are already borrowing constrained and consume out of current incomes.

The representative-agent prescription, monetary policy targeting nominal rigidities alongside an undistorted, freely floating exchange rate, need not remain optimal in this environment. When some households are borrowing constrained and cannot smooth income shocks intertemporally, how should exchange rate and monetary policies be conducted in response to temporary spikes in essential import prices? This paper studies this question in an import-dependent small open economy with heterogeneous households. Direct fiscal transfers are a natural way to redistribute real incomes; we study environments where that answer is unavailable: fiscal instruments capable of delivering lump-sum transfers to constrained households are assumed absent. Exchange rate and monetary policies cannot undo the real income loss in present-value terms; the question is whether they can influence how that loss is borne across households and across time, and to what extent they should.

Given this framework, this paper argues and illustrates that exchange rate policy takes on a prominent and distinctive role that it does not possess in a representative-agent world: stabilizing the real incomes of borrowing-constrained households. By altering the economy's net foreign asset position, directly through its own balance sheet or indirectly by changing the returns faced by participating households, the public agent can influence the path of the real exchange rate. Because the real exchange rate determines the purchasing power of domestic incomes over imports, this influence extends, through pecuniary general equilibrium effects, to the real incomes of all households, including those who are borrowing constrained. Facing a temporary import price spike, if additional borrowed foreign purchasing power is brought in while import prices are high and repaid after the shock has passed, less must be exported during the spike, the depreciation is smaller when the necessity is most expensive and larger after it has become cheap again, thereby reducing the excess co-movement of the real exchange rate with the shock. This lowers the effective

price of imports in domestic units and raises the real incomes of households across the economy when the adverse shock hits. The gain matters most for borrowing-constrained households, who have high marginal utility in these periods and would like to consume more but cannot borrow to do so. The inflows reverse once the shock has passed: the exchange rate depreciates, exports expand, and real incomes are lowered as the economy pays for its imports. By tilting the real exchange rate path, policy shifts real incomes toward the high-price periods and away from the low-price periods, mitigates the depreciation-induced amplification, and spreads the real incomes and consumption of borrowing-constrained agents more evenly across the price cycle. We argue that a public agent who cares about the welfare of constrained households finds this policy desirable relative to the conventional prescription that leaves financial markets and the resulting exchange rate path undistorted. We identify this as the open economy real-income stabilization channel of exchange rate policy delivered through the policy's influence on the intertemporal margin.

Monetary policy's role also extends beyond targeting nominal rigidities. However, monetary policy cannot perform the intertemporal task. The terms on which the economy borrows abroad are set in world markets, and domestic interest rate policy does not move them; the re-timing described above requires an instrument that acts on the economy's foreign financial flows and that role can be performed by exchange rate policy. What monetary policy does move, where nominal rigidities give it traction, is current employment and output, and through them the earnings of constrained households. In the spirit of the heterogeneous-agent New Keynesian (HANK) monetary policy literature, this is the familiar real-income earnings channel of monetary policy, though here it operates through the intratemporal consumption-leisure margin rather than the intertemporal margin. In an import-dependent economy, we illustrate that the strength and direction of this channel depend on how complementary imports are to domestic employment. When the two move closely together, expanding employment in the high-price periods can erode, rather than support, the real incomes of constrained households, and optimal monetary policy turns out tighter, not looser, than conventional representative-agent prescriptions imply.

We formalize these arguments in a small open economy that takes world interest rates and import prices as given but is not small in the market for its own exports, populated by two types of households: hand-to-mouth households, who consume their current income each period, and Ricardian households, who save and borrow freely in international asset markets. Firms produce the domestic tradeable good by combining labor with an essential imported input in fixed proportions, so that domestic activity and import demand move together, and the economy faces a downward-sloping, price-elastic export demand, so that export revenues in foreign units rise only with a real depreciation. Fiscal instruments capable of delivering lump-sum transfers to hand-to-mouth households are absent. In this environment we solve for jointly Ramsey-optimal monetary and exchange rate policies where monetary policy moves the labor wedge and exchange rate policy distorts the intertemporal margin. A two-period version of the model delivers the core results in closed form. We first study the model under inelastic labor supply, which shuts down the employment margin and isolates the role of exchange rate policy. With no hand-to-mouth households, optimal policy leaves the intertemporal margin undistorted for any path of import prices: this nests the conventional prescription as its representative-agent special case. With a positive mass of hand-to-mouth

households and a temporary import price spike, optimal policy instead distorts the intertemporal margin and induces stronger capital inflows in high-price periods. The resulting real exchange rate is more appreciated in high-price periods and depreciates more once the shock has passed, and is therefore more volatile across periods, while the domestic-units price of imported inputs, real incomes, and the consumption of hand-to-mouth households are all relatively stabilized.

The ability of policy to raise the real incomes of constrained households in the high-price periods, without transfers and without their participation in any asset market, rests on pecuniary general equilibrium effects. The intertemporal distortions, decentralized as a subsidy to Ricardian financial flows, induce them to borrow more and bring the borrowed foreign purchasing power to the foreign exchange market to bid for domestic output. Their bids crowd out foreigners and appreciate the real exchange rate, which lowers the domestic cost of the imported input and hence firms' marginal cost of production. Firms compete to expand imports and employment, and bid up real wages: the cost saving passes through into real wages. The wage gains accrue to all households, including hand-to-mouth households, whose consumption rises with their current labor income. Once the shock has passed and the borrowed resources are repaid, the same channel runs in reverse: the exchange rate depreciates, input costs rise, and real wages fall. Exchange rate policy thus uses the asset market access of Ricardian households to move the labor incomes of all households across periods, and it is this pass-through from the real exchange rate into real wages that delivers the real-income stabilization channel of exchange rate policy in our framework.

Allowing for an elastic labor supply introduces the employment margin and isolates the role of monetary policy. A familiar static terms-of-trade distortion gives monetary policy a role in this environment even without household heterogeneity: competitive firms do not internalize that their collective demand for imported inputs moves the real exchange rate against the economy, so employment and imports run higher than a price-taking economy would choose for itself, and optimal policy restrains them. This correction is already present in the representative-agent economy; what heterogeneity adds lies beyond this static correction. Because labor and the imported input are complementary, a marginal expansion of employment must be matched by additional imports; paying for the additional imports requires more export revenue, and hence a real depreciation. The depreciation creates a revaluation effect on Ricardian financial flows: in high import-price periods, when Ricardians are net borrowers, a depreciation revalues Ricardian financial flows so that the aggregate real income gains from higher employment are distributed disproportionately toward Ricardian households. The revaluation effect endogenously transfers purchasing power from the high-marginal-utility hand-to-mouth households toward the low-marginal-utility Ricardian households. A public agent who cares about the welfare of constrained households internalizes this endogenous redistribution and considers employment expansion in the high-price periods costlier than the representative-agent prescription does. Optimal monetary policy is therefore contractionary relative to the representative-agent-equivalent stance.

We embed these arguments in an infinite-horizon two-agent quantitative model with nominal wage rigidities and financial intermediation frictions, which provides the full decentralization of the intertemporal and labor wedges while preserving the economic intuition of the analytical framework. Monetary policy

sets the nominal interest rate; exchange rate policy is implemented by a public agent that shifts its asset positions between domestic and foreign markets, so intertemporal distortions take the form of costly interest parity gaps. We calibrate the model to Japan, which is among the most energy-import-dependent countries in the world and a clear case of exchange-rate-led amplification of domestic energy prices in the 2022 episode. We study perfect-foresight impulse responses together with a historical policy counterfactual on the realized 2022 path of world energy prices and compare the optimal policy response with conventional representative-agent prescriptions and with restricted optimal monetary policy, neither of which allows intertemporal distortions. We illustrate that representative-agent policy prescriptions make the real exchange rate co-move excessively with world energy prices: at the peak of the 2022 episode, 71 percent of the depreciation is attributable to heterogeneity, measured against a counterfactual economy in which no household is constrained. Optimal policy instead departs from conventional prescriptions and leans against the depreciation by worsening the public agent’s balance sheet, weakening the co-movement with world energy prices while accepting intertemporal distortions in the form of costly interest parity gaps. The gains appear in domestic energy prices and real wages: in our calibration, energy is 19.5 percent cheaper in domestic prices than under the conventional prescription at the peak of the shock and 29 percent of the peak real-wage loss is eliminated. Monetary policy complements the intervention with a disinflationary stance and prevents an excessive expansion of employment and imports by holding the domestic price level 2.5 percent lower at the peak than under the reference conventional policy. Monetary policy implemented optimally in isolation does little.

We extend our framework to a heterogeneous-agent economy with households facing idiosyncratic autonomous expenditure risk. As agents endogenously transition in and out of constrained borrowing states, the mass of constrained households responds to the shock and to policy itself. We characterize jointly optimal exchange rate and monetary policies in this environment and illustrate a distributional source of time inconsistency that must be accounted for. We formulate the Ramsey problem under timeless commitment and solve the nonlinear problem for the perfect-foresight response to an unanticipated import price spike. Our framework allows us to gauge the strength of the exchange-rate-driven real-income stabilization motive by comparing how the optimal policy differs from the next best policy regime that is blind to the real income motive. The qualitative arguments continue to hold: instead of depreciating strongly with world import prices, optimal policy leans against this depreciation and distorts the intertemporal margin by worsening the public agent’s external balance sheet. The exchange rate policy re-times and spreads the effect of the shock over time by cutting the contemporaneous co-movement of the exchange rate with import prices and depreciating more after the shock has passed. Monetary policy works in coordination and takes a stronger disinflationary stance. We decompose the policy-driven real-wage gains across the distribution and show that the gains accrue disproportionately to households at the borrowing limit.

The rest of the paper proceeds as follows. Section 2 relates the paper to the literature. Section 3 presents the analytical model, characterizes the constrained-efficient allocation, and shows how monetary policy and foreign exchange intervention decentralize the two policy margins. Section 4 develops the quantitative model, calibrates it to imported energy shocks, and compares optimal policy with restricted policy regimes.

Section 5 presents the heterogeneous-agent model with idiosyncratic expenditure risk and characterizes optimal policies under timeless commitment. Section 6 concludes.

2 Related Literature

How monetary and exchange rate policy should respond to import price shocks is a longstanding question in open economy macroeconomics. The conventional prescription grew out of the representative-agent small open economy New Keynesian framework of [Gali and Monacelli \(2005\)](#) and the classic case for flexible exchange rates ([Friedman 1953](#)): monetary policy addresses the distortions arising from nominal rigidities, international borrowing remains undistorted, and the exchange rate responds freely to shocks. With full participation in international asset markets, the depreciation that accompanies an import price spike is the expenditure switching required by a privately chosen current account path, and no separate role for exchange rate policy arises. We isolate the consequences of unequal household access to intertemporal markets. When some households cannot borrow against future income and targeted transfers are unavailable, we show that optimal intervention draws foreign purchasing power into high-price periods. The planner also chooses a tighter labor wedge than the representative-agent policy that already internalizes export-market power, because employment-driven import demand erodes the appreciation and shifts purchasing power toward unconstrained borrowers.

In heterogeneous-agent closed economies monetary policy redistributes through unequal earnings exposure, nominal balance-sheet revaluation, and unequal interest-rate exposure, and aggregate consumption responds because gains and losses accrue to households with different marginal propensities to consume ([Auclert 2019](#)). The earnings component of that transmission runs from the domestic real rate through aggregate demand to employment and reaches constrained households as higher labor income, and it outweighs intertemporal substitution ([Kaplan, Moll and Violante 2018](#)). In our economy, external financial flows pass through foreign exchange market clearing to the real exchange rate, imported-input costs, and real wages, and reach constrained households with employment held fixed. Monetary policy cannot reallocate foreign purchasing power across dates because it does not control the world return; intervention through external asset positions performs that task. Imported-input complementarity then gives employment expansion a distributional cost, because financing the associated imports depreciates the exchange rate and shifts purchasing power away from constrained households. Section 5.2 carries both margins into an economy in which idiosyncratic expenditure risk makes the mass of borrowing-constrained households an equilibrium outcome.

Optimal monetary policy in these economies has been characterized under a distributional objective. [Acharya, Challe and Dogra \(2023\)](#) derive how consumption inequality enters the tradeoff between price stability and productive efficiency. [McKay and Wolf \(2026\)](#) give the policymaker both a nominal interest rate and lump-sum transfers, and find that monetary policy carries limited distributional effects, so a distributional objective barely changes the optimal interest rate rule while progressive transfers do the redistributive work. Our economy removes that division of labor. Targeted transfers are unavailable, and

the two instruments reach constrained households together: intervention on the intertemporal margin, which monetary policy in an open economy does not control, moves foreign purchasing power across dates, and monetary policy sets employment and import demand, which govern how much of the appreciation arrives as a higher real wage.

In open economies with heterogeneous households, exchange rates carry first-order distributional consequences. Large devaluations raise the cost of living by more for low-income households (Cravino and Levchenko 2017). Household balance sheets and unequal access to international markets determine which exchange-rate regime constrained households prefer, and aggregate stabilization can come at the cost of wider consumption dispersion (de Ferra, Mitman and Romei 2020; Oskolkov 2023; Guo, Ottonello and Perez 2023). Auclert et al. (2021) quantify the real income channel of exchange rates: a depreciation raises import prices, cuts household real income, and lowers spending, which works against expenditure switching, and when short-run trade elasticities are low the real income channel dominates and a depreciation can be contractionary. Chen et al. (2023) derive optimal monetary policy in a two-agent open economy under alternative currency-pricing assumptions. Acharya and Challe (2025) find that with uninsured income risk and unequal access to bond markets, inefficient consumption inequality can warrant greater output and exchange-rate stabilization than in the representative-agent small open economy. In both of these policy problems monetary policy jointly moves domestic activity and the exchange rate. We instead study the real income channel as an object of policy: the public balance sheet changes the timing of the foreign purchasing power that finances an essential import and moves the real income loss into periods in which constrained households value income least, while monetary policy implements the labor wedge.

Building on the intermediary-based theory of exchange-rate determination in Gabaix and Maggiori (2015), research on foreign exchange intervention studies policy under imperfect international intermediation (Cavallino 2019; Basu et al. 2020). Itskhoki and Mukhin (2023) characterize optimal exchange rate policy in a representative-agent economy where nominal rigidities generate endogenous deviations from purchasing power parity and segmented financial markets generate deviations from interest parity: the efficient allocation is implemented with monetary policy closing the output gap and foreign exchange intervention eliminating the interest parity deviations. The representative-agent limit of our baseline model with transfers shares this assignment and restores interest parity. Unequal household access changes the welfare prescription, because the planner opens costly parity deviations to bring foreign purchasing power into periods in which constrained households have high marginal utility.

Within this literature Fanelli and Straub (2021) is the closest contribution. In their endowment economy a world interest rate shock moves Ricardian spending, which sets the relative price of tradable and nontradable goods. Hand-to-mouth households own fixed shares of both endowments and cannot trade assets, and that price therefore sets both the value of what they own and the cost of the subsistence tradables they must buy, leaving their consumption more volatile than it would be if they could borrow and save. A utilitarian planner distorts the Ricardian intertemporal margin to smooth it. Both planners distort the intertemporal choices of market participants on behalf of households shut out of asset markets, and the pecuniary mapping differs. Our relative price stands between two traded goods, the domestic good and the imported input, and it

clears the market for foreign purchasing power against the rest of the world. Ricardian borrowing supplies purchasing power that substitutes for export revenue along an upward-sloping schedule, the currency appreciates, the imported input becomes cheaper, and firms competing for labor bid up real wages. Policy diverts output from exports while imports are expensive and returns it once the shock has passed, re-timing a real income loss that constrained households cannot spread themselves. Their economy is real, with no nominal rigidity and no employment margin for monetary policy, and it therefore does not characterize the joint assignment of monetary and exchange rate policy. To our knowledge, this paper provides the first joint Ramsey characterization of monetary and foreign exchange policy in which unequal access to intertemporal markets itself generates the optimal intervention wedge.

A finite elasticity of export demand gives the economy market power over its own terms of trade (Johnson 1953; Corden 1974; Dixit and Norman 1980; De Paoli 2009). Costinot, Lorenzoni and Werning (2014) extend this motive to intertemporal policy and characterize optimal capital controls as dynamic terms-of-trade manipulation, driven by changes over time in a country’s incentive to influence its terms of trade. That motive operates in our economy whenever import demand responds to its own price, and our comparative statics recover it. When import demand is perfectly inelastic, the representative-agent intertemporal wedge is zero for every import-price path, even though the intertemporal instrument remains available. Our specifications admit this limit, and in it an intertemporal distortion introduced by unequal household access cannot be attributed to export-market power. Each exercise in the paper measures the real income stabilization motive against a counterfactual that leaves the market-power motive intact.

The macroeconomic effects of oil and energy price shocks and the monetary response to them have a long tradition (Bernanke, Gertler and Watson 1997; Blanchard and Galí 2010). Drechsel et al. (2026) compute welfare-optimal policy for commodity-exposed economies and find that where the commodity is a production input, optimal policy accommodates both the direct and the indirect effects on consumer prices and holds employment near its efficient level. Their household is representative, the input substitutes for labor with unit elasticity, and policy has no separate balance-sheet intervention instrument. Aktuğ and Rezghi (2026) bring oil shocks into the framework of Itskhoki and Mukhin (2023), where the motive for intervention is the financial friction and not the incidence of the loss across households. Chan, Diz and Kanngiesser (2024) find that when imported energy and labor are complementary, higher energy prices lower the labor share, and because constrained households cannot borrow against it aggregate demand falls and optimal monetary policy is less contractionary than in the representative-agent economy. This comparison reverses in our economy. The supply of foreign purchasing power slopes upward, imported-input complementarity ties employment expansion to depreciation, no transfer reaches constrained households, and a separate instrument manages the intertemporal margin; employment expansion in high-price periods therefore shifts purchasing power toward unconstrained borrowers. Auclert et al. (2024) find that in an energy-importing heterogeneous-agent economy monetary tightening has limited influence over imported inflation in isolation but becomes powerful when coordinated across importers, and energy-price subsidies insulate one economy while imposing adverse spillovers on others. Ari et al. (2022) document broad-based price-suppressing responses and recommend shifting toward income relief targeted to vulnerable households,

and support across OECD and European economies during the 2021-2022 energy crisis remained largely untargeted (Sgaravatti, Tagliapietra and Zachmann 2023; Hemmerlé et al. 2023). When fiscal targeting is incomplete, the exchange rate itself shapes the incidence of the loss, and our question is how external finance should be timed when fiscal instruments cannot reach constrained households.

Household heterogeneity also changes what commitment requires. Acharya, Challe and Dogra (2023) identify the temptation to exploit households' unhedged interest-rate exposure to reduce consumption inequality. Dávila and Schaab (2023) characterize this source of time inconsistency: the incentive to lower real returns in favor of indebted households with high marginal utility generates inflationary bias under discretion, while timeless implementation requires both inflation and distributional penalties. Their Ramsey problem carries the associated promises on forward-looking individual Bellman equations, alongside the familiar promise from the Phillips curve, which survives in their representative-agent limit. The same distributional incentive operates here through household Euler constraints. Because the public balance sheet sets the real return jointly with the real exchange rate, a re-optimizing public agent would move the announced intervention path, and commitment disciplines it. Domestic claims are indexed in our economy, which removes a distinct Fisherian temptation to revalue predetermined nominal positions through surprise inflation. Nuño and Thomas (2022) study the polar case in a small open economy where long-term claims are nominal and commitment front-loads inflation through the Fisher channel.

Dávila and Schaab (2023) compute nonlinear policy transitions and develop sequence-space perturbations for optimal policy. Other approaches expand in the scale of aggregate risk around a deterministic economy (Bhandari et al. 2021) or construct finite representations of idiosyncratic histories (Le Grand and Ragot 2023). Following Marcet and Marimon (2019), timeless commitment carries each household's inherited promise as a multiplier state. We show that this promise is affine in its inherited value and that the public agent's optimality conditions depend on promises through their conditional mean on assets and the expenditure requirement. This reduction retains the nonlinear aggregate transition and the full asset distribution while keeping the system in the two idiosyncratic states of the household problem. We solve the resulting perfect-foresight system in levels, preserving the movements in the constrained mass and in marginal utilities at the borrowing limit induced by the large import-price spike.

3 Analytical Framework

This section develops a two-period model of a small open economy that imports an essential intermediate input and is populated by households who are heterogeneous in their ability to participate in asset markets and smooth consumption intertemporally. A fraction of households are hand-to-mouth and consume their current income each period, while the remaining households can save and borrow freely in international asset markets. Given a temporary spike in import prices that erodes current household real incomes, we characterize optimal monetary and exchange rate policies for this economy.

The framework presented in this section is tractable enough to allow for a closed-form characterization of optimal policies. We describe a utilitarian Ramsey planner's optimization problem with only two policy

margins: the labor wedge and the intertemporal wedge. We interpret the labor wedge as the margin that monetary policy can move, and the intertemporal wedge as the margin that exchange rate policy can influence. This section takes a simplified, reduced-form approach to policy implementation and leaves the full decentralization with nominal rigidities and financial intermediation frictions to the quantitative model of Section 4.

We proceed as follows. Section 3.1 describes the environment and defines the equilibrium given policy instruments, and Section 3.2 sets up the Ramsey planner's problem. Section 3.3 characterizes optimal policy in the representative-agent special case. Section 3.4 turns to the heterogeneous-agent economy, first under inelastic labor supply to isolate the intertemporal dimension, then under elastic labor supply to characterize the jointly optimal intertemporal and labor wedges.

3.1 Environment

The economy operates over two periods, $t \in \{1, 2\}$. It produces a single final tradeable good and interacts with the rest of the world through three channels: it exports the domestic good, imports a productive input, and borrows or lends on international asset markets.

3.1.1 Goods and Relative Prices

Three tradeable goods circulate in the world economy. The first is the domestically produced good, produced only in the small open economy. It is consumed by domestic households and exported to the rest of the world. Let C_t , X_t , and Y_t denote aggregate domestic consumption, exports, and output. The resource constraint is given by:

$$C_t + X_t = Y_t \quad (1)$$

The second good is a foreign consumption good. It represents the foreign consumption basket. Domestic households do not consume it, and domestic firms do not use it in production. Bonds traded in international asset markets are claims denominated in units of this good. Moreover, this good also serves as the world numeraire: we normalize its foreign currency price to $P_t^* = 1$. One unit of foreign currency then equals one unit of this good, and all world prices are expressed in units of this good. The world gross real interest rate is assumed to be constant and denoted as R^* , i.e., one unit of the foreign good saved in period t returns R^* units in period $t + 1$. The economy is price-taking with respect to this interest rate.

The third good is an intermediate foreign good which the economy imports. The economy uses this good in the production of the domestic tradeable good. The world price of this input is denoted P_t^{T*} in units of the foreign good. The economy is also price-taking with respect to this relative price.

The domestically produced tradeable good is the only final good that the economy consumes. Let P_t denote the price of the domestic good in units of domestic currency and let e_t denote the nominal exchange rate, i.e., the price of the foreign currency in units of the domestic currency. With the world consumption good as the numeraire, e_t is also the price of the world consumption good in domestic currency units. Moreover,

since the domestic economy only consumes the domestic good, P_t also denotes the domestic price level. Then let ϵ_t denote the real exchange rate, simply given by $\epsilon_t = \frac{e_t}{P_t}$. By the law of one price for tradeables, the inverse of the real exchange rate, $\frac{1}{\epsilon_t}$, is the world price of the domestic good in units of the world numeraire. The economy's terms of trade is then given by $\frac{P_t^{T*}}{\epsilon_t} = \epsilon_t P_t^{T*}$, which is the price of the imported input in units of the domestic good. Finally, let W_t denote the nominal wage and $w_t = \frac{W_t}{P_t}$ denote the real wage in units of the domestic good.

3.1.2 Production Technology

A unit mass of identical firms produces the domestic good, Y_t , using labor, N_t , and the imported intermediate input, M_t , according to a Leontief technology:

$$Y_t = A_t \cdot \min \left\{ N_t, \frac{M_t}{\theta} \right\} \quad (2)$$

where $A_t > 0$ is total factor productivity and $\theta > 0$ is the input requirement per unit of labor employed. The Leontief specification captures an essential feature that motivates this paper: the imports are critical to the domestic economy, i.e., the imported and domestic inputs are strict complements in the short run, and a rise in the input price cannot be offset by substituting toward domestic factors. With no substitution effects, relative price changes create pure income effects, which is helpful in isolating how real income effects operate in the economy.

The firms' problem is to choose N_t and M_t to minimize the cost of producing Y_t subject to the production technology:

$$\min_{\{N_t, M_t\}} w_t N_t + \epsilon_t P_t^{T*} M_t \quad \text{s.t. } (2) \quad (3)$$

Given this Leontief technology, a cost-minimizing firm will always deploy the two inputs in fixed proportions to produce any given level of output:

$$M_t = \theta N_t \quad (4)$$

The output produced can be expressed as a linear function of labor input:

$$Y_t = A_t N_t \quad (5)$$

The firms' real marginal cost of production is therefore $\frac{w_t}{A_t} + \epsilon_t P_t^{T*} \frac{\theta}{A_t}$: each additional unit of output requires $\frac{1}{A_t}$ units of labor at wage w_t and $\frac{\theta}{A_t}$ units of the imported input at domestic cost $\epsilon_t P_t^{T*}$ per unit. Since firms are competitive, an equilibrium requires that firms make zero economic profits, i.e., the real marginal cost of production must equal one. This gives a linear expression for the real wage in terms of the price of the imported input and the real exchange rate:

$$w_t = A_t - \epsilon_t P_t^{T*} \theta \quad (6)$$

A_t denotes the total output produced per unit of labor employed, while $\epsilon_t P_t^T \theta$ denotes the output paid by the firm to buy θ units of the imported input that, together with a unit of labor, produces A_t units of output. The residual, $A_t - \epsilon_t P_t^T \theta$, is paid to labor. The real wage, therefore, has a linear one-for-one negative relationship with the price of the imported input and the real exchange rate. A rise in the price of the imported input or a real depreciation raises the marginal cost of production. At the prevailing wage, firms make losses and therefore decrease their demand for the input. Since inputs and labor are perfect complements, the demand for labor also falls. Competition for the released labor weakens until the wage has dropped one-for-one with the rise in input costs to restore zero profits. A fall in the price of the imported input or a real appreciation has the opposite effect: firms make positive profits and expand, bidding up wages until zero profits are restored.

Let Y_t^N denote the labor income that accrues to domestic workers in period t :

$$Y_t^N = w_t N_t = (A_t - \epsilon_t P_t^T \theta) N_t \quad (7)$$

Since Y_t^N is the share of gross final output that accrues to the primary domestic factors of production, it also represents the economy's GDP. The remainder, $\epsilon_t P_t^T \theta N_t$, is the import bill, i.e., the share of domestic output that firms sell in the foreign exchange market in return for buying θN_t units of the imported input. The real exchange rate determines the split: a depreciation raises the import bill and compresses GDP; an appreciation does the reverse.

3.1.3 Export Demand

As illustrated by the resource constraint (1), the domestic good is either consumed domestically or exported, where X_t denotes the share of output consumed externally in period t . Given the world price of the domestic good, $\frac{1}{\epsilon_t}$, we assume that the economy faces a downward-sloping, isoelastic export demand:

$$X_t = \zeta \epsilon_t^{1+\chi} \quad (8)$$

where $\zeta > 0$ is a scale parameter and $-(1+\chi)$ is the price elasticity of export demand. We assume that $\chi > 0$, i.e., the export demand is relatively price-elastic. Given this downward sloping demand structure, a real depreciation is required to induce the foreigners to buy more of the domestic good, and a real appreciation reduces the share of the domestic good sold abroad. Moreover, since the demand is relatively price elastic, the export revenue, $\frac{X_t}{\epsilon_t} = \zeta \epsilon_t^\chi$, in units of the foreign good is increasing in the real exchange rate. This is the Marshall-Lerner condition in this environment: a depreciation is required to increase the supply of foreign purchasing power in the domestic economy. χ therefore measures the elasticity of supply of foreign purchasing power (in units of the foreign good) with respect to the real exchange rate.

3.1.4 Households

The economy is populated by a unit mass of households of two types. A fraction $\lambda \in [0, 1)$ are hand-to-mouth (HtM) households and the remaining $1 - \lambda$ are Ricardian households. Both types consume only the

domestic good and supply labor. Let C_t^p and C_t^r denote the consumption of HtM and Ricardian households in period t .

All households supply N_t units of labor and take this quantity as given. In the analytical model presented in this section, we assume that N_t is a policy instrument. We discuss this further in Section 3.1.5.

The period utility of household $i \in \{r, p\}$ is:

$$u(C_t^i) - v(N_t)$$

where $u : \mathbb{R}_{++} \rightarrow \mathbb{R}$ and $v : \mathbb{R}_{++} \rightarrow \mathbb{R}$ are twice continuously differentiable, with $u' > 0$, $u'' < 0$, $v' > 0$, and $v'' > 0$. They satisfy the standard Inada conditions: $\lim_{C \downarrow 0} u'(C) = \infty$, $\lim_{C \rightarrow \infty} u'(C) = 0$, $\lim_{N \downarrow 0} v'(N) = 0$, and $\lim_{N \rightarrow \infty} v'(N) = \infty$. All households share the same preferences. The common discount factor is $\beta \in (0, 1)$, and we impose $\beta R^* = 1$ throughout.

Hand-to-mouth households. HtM households cannot access asset markets. Given labor incomes, $w_t N_t$, they choose consumption, C_t^p , to maximize lifetime utility subject to period-by-period budget constraints:

$$C_t^p = w_t N_t \tag{9}$$

Since they do not participate in asset markets, the budget constraints imply that they consume their current labor income each period, and income shocks therefore pass into their consumption one for one.

Ricardian households. Ricardian households, in addition to consuming the domestic good and supplying labor, also participate in international asset markets. Let f^* denote a Ricardian household's net foreign asset position in period 1, denominated in units of the foreign good, with $f^* > 0$ denoting saving and $f^* < 0$ borrowing. Given labor income, the real exchange rate, the world interest rate, and government policy (τ, T) , a Ricardian household's problem is to choose (C_1^r, f^*) to maximize lifetime utility subject to budget constraints:

$$\begin{aligned} \max_{\{C_1^r, C_2^r, f^*\}} \quad & u(C_1^r) + \beta u(C_2^r) \quad \text{s.t.} \\ C_1^r + \epsilon_1 f^* (1 + \tau) = & w_1 N_1 + T \\ C_2^r = & w_2 N_2 + \epsilon_2 R^* f^* \end{aligned} \tag{10}$$

where (τ, T) are policy instruments discussed in Section 3.1.5. The optimal asset choice satisfies the Euler equation:

$$u'(C_1^r) \epsilon_1 (1 + \tau) = \beta R^* u'(C_2^r) \epsilon_2 \tag{11}$$

3.1.5 Fiscal-Monetary Policy

We assume that public policy has access to two margins: the intertemporal wedge and the labor wedge. The policy that influences the intertemporal wedge is interpreted as the exchange rate policy and is implemented through a capital control tax τ on the gross foreign asset positions of Ricardian households. Tax revenue is

rebated lump-sum to Ricardian households as a transfer T . The public agent's budget constraint is:

$$(1 - \lambda) \tau \epsilon_1 f^* = (1 - \lambda) T \quad (12)$$

The public agent's ability to influence the labor wedge is modeled in a reduced-form manner: the public agent directly sets employment, N_t , and we interpret this policy as monetary policy. This keeps the analytical framework clean while preserving the trade-offs that monetary policy faces. We defer the full decentralization to Section 4 with nominal wage rigidities and financial intermediation frictions. There, monetary policy sets the inflation rate and exchange rate policy is implemented through FX interventions. The qualitative results from the analytical model carry over to that framework.

No direct fiscal transfers. The framework has no fiscal instrument capable of making lump-sum transfers to HtM households; the capital control revenue is rebated only to Ricardians. This modeling choice keeps the focus on redistribution through pecuniary general equilibrium effects created by exchange rate and monetary policies. Direct transfers to HtM households would trivially accomplish such redistribution.

3.1.6 Balance of Payments

The final piece in the model is the balance-of-payments (BoP) accounting condition, which is already implicit in the specified equations. Combining the household budget constraints, the firms' zero-profit condition (6), goods market clearing (1), and the fiscal budget constraint (12) yields the balance-of-payments condition, which is interpreted as the economy's external budget constraint. Denote aggregate consumption as $C_t = \lambda C_t^P + (1 - \lambda) C_t^R$ and let $F^* \equiv (1 - \lambda) f^*$ be the aggregate net foreign asset position of the economy. Finally, substitute in the export demand function, (8), to express the period-by-period balance of payments in units of the foreign good:

$$\begin{aligned} P_1^T \theta N_1 + F^* &= \zeta \epsilon_1^X \\ P_2^T \theta N_2 &= \zeta \epsilon_2^X + R^* F^* \end{aligned} \quad (13)$$

The BoP constraints have a natural interpretation of equating demand and supply of foreign purchasing power² in the domestic economy's foreign exchange market. Firms are always net buyers of foreign purchasing power: they produce the domestic good and must pay for the imported input, so they are always exchanging domestic output for foreign purchasing power. Foreigners are always net buyers of the domestic good and suppliers of foreign purchasing power. Ricardian households switch sides depending on the sign of financial flows. When $F^* < 0$, Ricardians are net buyers of the domestic good in the foreign exchange market: they borrow foreign purchasing power and use it to buy domestic goods, joining foreigners on the supply side. When $F^* > 0$, they are net sellers of the domestic good in the foreign exchange market: they surrender domestic output out of their current labor income in exchange for future claims on foreign

²We use the term foreign purchasing power because it does not matter whether agents exchange the foreign good or the intermediate input against the domestic good: the foreign good converts into the intermediate input in world markets at the price P_t^T . In other words, the foreign good is not a medium of exchange in the model. Equivalently, firms have access to a linear technology with productivity $\frac{1}{P_t^T}$ that converts the foreign good into the intermediate input.

purchasing power, effectively joining firms on the demand side. The real exchange rate clears this market each period: it moves until the demand for foreign purchasing power equals the supply.

Combining the two period-by-period BoP constraints yields the intertemporal budget constraint for the economy:

$$P_1^T \theta N_1 + \frac{1}{R^*} P_2^T \theta N_2 = \zeta \epsilon_1^X + \frac{1}{R^*} \zeta \epsilon_2^X \quad (14)$$

The left-hand side of (14) is the present value of import expenditure in units of the foreign good, while the right-hand side is the present value of export revenue in foreign units. The intertemporal balance of payments requires that the present value of import expenditure equals the present value of export revenue. The economy cannot be a net borrower or lender in the long run; it cannot run a trade deficit or surplus in present value terms.

Define the trade balance in units of the foreign good:

$$NX_t \equiv \zeta \epsilon_t^X - P_t^T \theta N_t \quad (15)$$

$NX_t > 0$ is a surplus and $NX_t < 0$ is a deficit. The trade balance is increasing in ϵ_t and decreasing in N_t : depreciation raises export revenue; higher employment raises the consumption of the intermediate input and hence increases the import bill. The intertemporal constraint requires $NX_1 + R^{*-1}NX_2 = 0$.

3.1.7 Equilibrium Definition

Given fiscal-monetary policy, (τ, T, N_t) , an equilibrium is a set of allocations and prices, $\{C_t^P, C_t^r, F^*, M_t, Y_t, X_t, w_t, \epsilon_t\}$ such that:

- i. HtM households satisfy their budget constraints, (9).
- ii. Ricardian households satisfy the Euler equation, (11) and the budget constraints in (10).
- iii. Firms satisfy the cost minimization conditions, (4), (5); and wage determination condition, (6).
- iv. Export demand satisfies (8).
- v. The fiscal budget constraint, (12), holds.
- vi. The domestic goods market clears, i.e., (1) holds.
- vii. The labor market clears, i.e., the labor demand by the firms equals the labor supply, N_t .
- viii. The balance of payments conditions, (13), hold.

3.2 Optimal Policy Problem

The equilibrium definition in Section 3.1.7 describes the set of all feasible decentralized equilibria, indexed by the public agent's choice of fiscal-monetary policy (τ, T, N_t) . We now describe the optimal policy problem of a utilitarian public agent that chooses a sequence of policies, $\{N_t, \tau, T\}$, to maximize social welfare subject to the equilibrium conditions listed in Section 3.1.7.

We take the primal approach and state the optimization problem of a Ramsey planner who directly chooses $\{N_t, \epsilon_t\}$ subject to the implementability constraints implied by household and firm optimization. We then illustrate that the Ramsey planner's solution can be implemented as an equilibrium with an appropriate choice of capital control policy, $\{\tau, T\}$.

Consider the following constrained optimization problem:

$$\begin{aligned} \max_{\{N_t, \epsilon_t\}} \sum_{t=1}^2 \beta^{t-1} [\lambda u(C_t^p(N_t, \epsilon_t)) + (1-\lambda)u(C_t^r(N_t, \epsilon_t)) - v(N_t)] \quad \text{s.t.} \\ NX_1(N_1, \epsilon_1) + \frac{1}{R^*} NX_2(N_2, \epsilon_2) = 0 \end{aligned} \quad (16)$$

where:

$$C_t^p(N_t, \epsilon_t) = Y_t^N(N_t, \epsilon_t) \quad (17)$$

$$C_t^r(N_t, \epsilon_t) = Y_t^N(N_t, \epsilon_t) - \frac{\epsilon_t NX_t(N_t, \epsilon_t)}{1-\lambda} \quad (18)$$

and $Y_t^N(N_t, \epsilon_t)$ and $NX_t(N_t, \epsilon_t)$ are given by labor income equation (7) and the trade balance equation (15), respectively. Equation (17) is obtained by combining the HtM households' budget constraint, (9), with wage determination equation (6). Equation (18) is obtained by combining the Ricardian households' budget constraints in (10) with the labor income equation, (7), and the balance of payments conditions, (13). This equation implies that Ricardian households are net borrowers in period t when $NX_t < 0$, consuming more than their period labor income, and net savers when $NX_t > 0$, consuming less. Moreover, since Ricardians save and borrow in units of the foreign good, the actual consumption in domestic units that such saving/borrowing can buy is a product of the real exchange rate, ϵ_t , and the trade balance in foreign units, NX_t . Finally, the constraint in (16) is the intertemporal budget constraint, (14): the present discounted value of net export revenue must equal zero.

We denote the solution to the optimization problem, (16), as the constrained efficient allocation. In Appendix A.1, we derive the first-order conditions for this problem and obtain the system of equations that characterize the solution. The solution in $\{N_t, \epsilon_t\}$ is characterized by an Euler-like intertemporal condition, (76), an intratemporal consumption-labor condition, (80) for $t = 1, 2$, and the intertemporal budget constraint in (16). In Appendix A.2 we discuss sufficient conditions that ensure that the solution is a unique maximum.

The constrained planning problem, (16), internalizes all equilibrium constraints except the Ricardian Euler equation, (11), and the goods-market-clearing condition, (1). The Ricardian Euler equation is not an implementability constraint for the planner because the planner can distort the intertemporal margin: the intertemporal wedge chosen by the Ramsey planner can be supported in the decentralized economy through an appropriate choice of τ . The goods-market-clearing condition is implied by Walras' identity. Lemma 1 formally shows that the solution to (16) can be supported as a decentralized equilibrium with an appropriate choice of policy, $\{\tau, T\}$. We define this decentralized equilibrium as the Ramsey equilibrium.

Lemma 1. *Let $\{N_t, \epsilon_t\}$ be a solution to the constrained optimization problem in (16). Then there exists a*

policy, $\{\tau, T\}$, such that the solution to (16) is implementable as an equilibrium defined in Section 3.1.7.

Proof. See Appendix A.3. □

Before we proceed to characterize the constrained efficient allocation under different assumptions on λ and import price sequences, $\{P_t^{T*}\}$, it is useful to state a few definitions. Define the average marginal utility, $\bar{u}'_t$, as the population-weighted average of the marginal utilities of the two types of households:

$$\bar{u}'_t \equiv \lambda u'(C_t^p) + (1 - \lambda) u'(C_t^r) \quad (19)$$

Define the labor wedge, τ_t^N , as the deviation of the average marginal rate of substitution between consumption and leisure from the real wage:

$$1 - \tau_t^N \equiv \frac{v'(N_t)}{w_t \bar{u}'_t} \quad (20)$$

Using the Ricardian Euler equation, (11), the intertemporal wedge corresponds to the capital control tax, τ :

$$1 + \tau \equiv \frac{\beta R^* u'(C_2^r) \epsilon_2}{u'(C_1^r) \epsilon_1} \quad (21)$$

In Appendix A.1, we use the first-order conditions of the Ramsey planner's problem to obtain closed-form expressions for the optimal labor and intertemporal wedges in terms of the optimal allocation, $\{N_t, \epsilon_t\}$. The optimal labor wedge is given by equation (82), and the optimal intertemporal wedge is given by equation (83).

Policy is said to be operative on a margin if it selects a non-zero wedge on that margin. On the other hand, an equilibrium is defined as laissez-faire if the policy chooses no distortion of the intertemporal and labor margins, i.e., sets the wedges to zero: $\tau = 0$ and $\tau_t^N = 0$. In what follows, we characterize the constrained efficient allocation under different assumptions on λ and import prices, P_t^{T*} , and compare this solution with laissez-faire equilibria and equilibria with sub-optimally chosen wedges.

3.3 Optimal Policy: Representative-Agent Benchmark

The representative-agent case is nested in the heterogeneous-agent problem by setting $\lambda = 0$: no household is borrowing constrained and the household's Euler equation always holds. Proposition 1 characterizes its constrained efficient allocation.

Proposition 1. *If $\lambda = 0$, then given any sequence of import prices and productivity, $\{P_t^{T*}, A_t\} > 0$, the constrained efficient allocation involves:*

- *No intertemporal distortions: $\tau = 0$.*
- *A strictly positive labor wedge that implements the optimal terms-of-trade:*

$$\tau_t^N = \frac{1}{\chi} \frac{\epsilon_t P_t^{T*} \theta}{w_t} > 0$$

Proof. See Appendix A.4. □

First, with a single household type that participates freely in international asset markets, there is no role for policy to distort the intertemporal margin. The proposition establishes that the undistorted Ricardian Euler equation coincides with the planner's optimality condition: the agent's borrowing and saving already smooth consumption optimally for any path of import prices and productivities, and the planner has no incentive to distort the intertemporal price. Therefore, in the decentralized economy, the optimal capital control tax is zero.

Second, given that the economy faces an upward-sloping supply of foreign purchasing power in the form of export revenues, the model features a classic terms-of-trade distortion: competitive firms do not internalize the effect of their import demand on the real exchange rate. Each additional unit of import demand must depreciate the real exchange rate to induce foreigners to supply additional foreign purchasing power. This depreciation increases the firms' costs in the form of domestic output paid to foreigners: not only for the marginal unit of foreign purchasing power supplied by them but also for all inframarginal units. Noting that exports, X_t , can be written as a product of the real exchange rate, ϵ_t , and the foreign good supplied by foreigners, $\zeta \epsilon_t^X$, we have:

$$dX_t = \epsilon_t \chi \zeta \epsilon_t^{X-1} d\epsilon_t + \zeta \epsilon_t^X d\epsilon_t$$

where the first term is the direct quantity effect: a real depreciation increases the quantity of the foreign good supplied at the margin; and the second term is the inframarginal price effect: a real depreciation increases the price of the foreign good for all inframarginal units supplied by foreigners. Holding NX_t constant, taking the total differential of (15) yields:

$$P_t^{T*} \theta dN_t = \chi \zeta \epsilon_t^{X-1} d\epsilon_t$$

Together this gives us:

$$dX_t = \epsilon_t P_t^{T*} \theta dN_t + \frac{1}{\chi} \epsilon_t P_t^{T*} \theta dN_t$$

where the first and second terms have the same interpretation as above. The second term is the terms-of-trade effect, which constitutes a resource loss for the economy that price-taking firms do not internalize. Such terms-of-trade effects have been discussed in the classic optimal trade policy literature at least since Johnson (1953), are closely related to the general-equilibrium argument for optimal tariffs in Dixit and Norman (1980), and arise in a small open economy whenever export demand has finite elasticity (Corden 1974; De Paoli 2009). A Ramsey planner internalizes such terms-of-trade effects and considers the marginal cost of expanding employment and imports to be higher than private agents do. Define $\hat{w}_t$ as the effective marginal product of labor, i.e., the net resources retained by the domestic economy when employment is increased by a marginal unit after accounting for the terms-of-trade effect:

$$\hat{w}_t \equiv A_t - \frac{dX_t}{dN_t} = w_t - \frac{1}{\chi} \epsilon_t P_t^{T*} \theta$$

The planner values the marginal unit of labor at $\hat{w}_t$, and the constrained efficient allocation therefore

restricts employment and imports below the laissez-faire level, which reduces output and export supply and results in a more appreciated real exchange rate in equilibrium relative to the competitive outcome³. The optimal labor wedge is proportional to the inframarginal price effect term, $\frac{1}{\chi} \epsilon_t P_t^{T*} \theta$, derived above.

This classic terms-of-trade distortion is neither a focus nor a requirement for the main results of the paper. It naturally arises in our framework due to the upward-sloping export supply assumption. Moreover, to isolate the impact of heterogeneity on optimal policy, it is useful to identify and separate out the effect of this terms-of-trade distortion in the overall policy response, and the representative-agent framework serves as the benchmark for that purpose.

Before proceeding to the heterogeneous-agent model, it is useful to conduct a comparative statics exercise with respect to the import price, P^{T*} . Lemma 2 characterizes the constrained efficient allocation for the case where import prices are constant over time, $P_1^{T*} = P_2^{T*} = P^{T*}$, and then discusses how an increase in the period 1 price induces households to borrow in international asset markets to smooth this income shock.

Lemma 2. *Let $\lambda = 0$, $A_t = A > 0$ for $t \in \{1, 2\}$, and $\beta R^* = 1$. Let all other parameters satisfy the conditions of Lemma A.1.*

*i. **Symmetric equilibrium.** If $P_1^{T*} = P_2^{T*} = P^{T*}$, then the constrained efficient allocation satisfies*

$$N_1 = N_2, \quad \epsilon_1 = \epsilon_2, \quad NX_1 = NX_2 = 0$$

*ii. **Comparative statics.** In addition, assume that utility over consumption satisfies $u'(C) = C^{-\sigma}$ with $\sigma \geq \frac{1}{1+\chi}$. Fix $P_2^{T*} = P^{T*}$ and let $P_1^{T*} > P^{T*}$. Then the constrained efficient allocation satisfies*

$$NX_1 < 0 < NX_2$$

Proof. See Appendix A.5. □

When import prices are equal across periods, part (i) establishes that the constrained efficient allocation is symmetric: employment, the real exchange rate, and the trade balance are identical in the two periods, and trade is intratemporally balanced. With $\beta R^* = 1$ and no asymmetry between the two periods, the representative agent has no incentive to shift resources intertemporally, and equal consumption across periods is optimal. The labor wedge is active in each period, statically correcting the terms-of-trade distortion, but the intertemporal wedge is undistorted.

³In our framework import-export tariffs are unavailable. However, under Leontief production imports move one for one with employment, and the labor wedge therefore internalizes the terms-of-trade distortion and implements the same allocation an optimal tariff would implement. With isoelastic export demand the distortion is purely static: the labor wedge corrects it period by period independently, and the intertemporal wedge provides no additional leverage for this purpose. It follows that if labor supply is perfectly inelastic, the planner has no instrument with which to internalize the distortion, yet the optimal intertemporal wedge is still zero and the constrained efficient allocation coincides with the competitive laissez-faire allocation. Section 3.5 shows that this is special to Leontief production: under a general CES technology facing unequal import prices, neither wedge alone corrects the distortion and the planner distorts both margins. The Leontief production framework is therefore a special and analytically useful case because any intertemporal distortions chosen when $\lambda > 0$ reflect the role of household heterogeneity alone.

Part (ii) considers a temporary increase in the import price in period 1, with $P_1^{T*} > P^{T*}$. The high input price reduces real wages and compresses real income in period 1. The representative agent responds by borrowing freely in international asset markets at the world interest rate to smooth consumption across periods, running a trade deficit in period 1 and a surplus in period 2. The planner sets $\tau = 0$: the privately optimal current account path is also socially optimal, and there is no role for intertemporal policy. The terms-of-trade wedge continues to operate in each period independently.

The representative-agent benchmark therefore delineates the role of each policy instrument. Given the absence of import-export tariffs, the labor wedge through monetary policy internalizes the terms-of-trade distortion period by period, and the optimal capital control tax is zero for any sequence of import prices. This outcome is a close analog of conventional policy prescriptions in representative-agent small open economy models: since there are no frictions in asset markets, real income shocks do not call for interventions in foreign exchange markets. In response to import price spikes, the exchange rate moves freely as households optimally smooth consumption. However, when a fraction of households cannot borrow against future income to smooth consumption, asset markets need not be left undistorted. The following subsections discuss optimal policies in a heterogeneous-agent economy with a non-zero mass of hand-to-mouth households.

3.4 Heterogeneous Agents ($\lambda \in (0, 1)$)

We now turn to the case $\lambda \in (0, 1)$. A fraction λ of households cannot borrow against future income. When import prices are temporarily high, Ricardian households respond by borrowing in international asset markets, insulating their consumption from the income shock. HtM households have no such recourse: they consume their current labor income each period and bear the full income loss contemporaneously. This asymmetry in asset market access creates a possible role for policy.

To isolate the intertemporal dimension of optimal policy, we begin with the case of inelastic labor supply, $N_t = \bar{N}$. With employment fixed exogenously, the only instrument available to the public agent is the capital control tax, τ , and the Ramsey planner's problem reduces to choosing a point on the feasible frontier defined by the intertemporal budget constraint. The labor wedge and its interaction with the intertemporal wedge are taken up in Section 3.4.2.

3.4.1 Inelastic Labor Supply ($N_t = \bar{N}$)

Equal import prices. We begin with the case $P_1^{T*} = P_2^{T*} = P^{T*}$. With $\beta R^* = 1$ and no asymmetry between the two periods, neither household type has an incentive to shift resources intertemporally. The privately optimal asset position for Ricardian households is $f^* = 0$, implying balanced trade in each period. With $NX_1 = NX_2 = 0$, both household types consume their current labor income in each period. Since the income path is identical across periods, consumption is equal over time for both types, and since trade is balanced in each period, Ricardian and HtM households also consume equally within each period: $C_t^p = C_t^r$ for $t \in \{1, 2\}$.

This allocation is optimal for both household types. HtM households would choose it even if they had access to asset markets, and in this special case their inability to smooth consumption is inconsequential. The planner therefore has no incentive to distort the intertemporal margin, and the constrained efficient allocation coincides with laissez-faire: $\tau = 0$ and the exchange rate path is undistorted. This result is the direct counterpart of part (i) of Lemma 2: with equal prices and $\beta R^* = 1$, the intertemporal wedge is zero both in the representative-agent economy and in the heterogeneous-agent economy. Lemma 3 formally summarizes this result.

Lemma 3. *Assume inelastic labor supply: $N_t = \bar{N} > 0$. Let $\lambda \in (0, 1)$, $A_t = A > 0$ for $t \in \{1, 2\}$, and $\beta R^* = 1$. Let $P_1^{T*} = P_2^{T*} = P^{T*} > 0$. Then the constrained efficient allocation involves no intertemporal distortions: $\tau = 0$, i.e., it coincides with the laissez-faire equilibrium, and satisfies:*

$$\epsilon_1 = \epsilon_2, \quad NX_1 = NX_2 = 0, \quad C_1^r = C_1^p = C_2^r = C_2^p$$

Proof. Follows as a Corollary to Proposition 5. □

The laissez-faire equilibrium under $P_1^{T*} > P_2^{T*}$. Now consider the case where $P_1^{T*} > P_2^{T*} = P^{T*}$. Before turning to optimal policy we characterize the laissez-faire equilibrium, which provides the reference point against which optimal policy is measured. The high input price in period 1 compresses real labor incomes, and Ricardian households borrow in international asset markets to smooth consumption over their lifetime. As in the representative-agent case of Lemma 2, the economy runs a trade deficit in period 1 and a surplus in period 2: $NX_1 < 0 < NX_2$.

The heterogeneity in asset market access, however, introduces a dimension that is absent in the representative-agent economy. Ricardian households supplement their compressed period-1 labor income with the resources borrowed from abroad, consuming strictly more than their current incomes. HtM households, by contrast, consume their current labor income each period. In period 1, Ricardians consume strictly more than HtM households, $C_1^r > C_1^p$; in period 2 the pattern reverses, $C_2^r < C_2^p$, as Ricardians service their debt while HtM households enjoy the full recovery of labor incomes. The laissez-faire equilibrium therefore features a consumption differential between the two household types that varies over the income cycle. Lemma 4 formally establishes these properties.

Lemma 4. *Assume inelastic labor supply: $N_t = \bar{N} > 0$. Let $\lambda \in (0, 1)$, $A_t = A > 0$, for $t \in \{1, 2\}$, and $\beta R^* = 1$. Let $P_2^{T*} = P^{T*} > 0$ and $P_1^{T*} > P^{T*}$. Consider the laissez-faire equilibrium, which imposes $\tau = 0$. The equilibrium satisfies:*

$$NX_1 < 0 < NX_2, \quad C_1^r > C_1^p, \quad C_2^r < C_2^p$$

Proof. See Appendix A.6. □

A perturbation along the feasible frontier. Before characterizing optimal policy, it is useful to examine how a capital control tax affects equilibrium outcomes. The intertemporal budget constraint, $NX_1 +$

$R^{*-1}NX_2 = 0$, defines the set of feasible decentralized equilibria that can be reached for some choice of τ . We take the laissez-faire equilibrium characterized in Lemma 4 as the reference point and consider the impact of an increase in the capital control tax⁴. By distorting Ricardian households' saving decisions, this policy affects the intertemporal path of the real exchange rate, and through it both the sharing of resources between the domestic economy and the rest of the world and the distribution of domestically retained resources between the two household types.

A higher capital control tax, $d\tau > 0$, raises the private cost of Ricardian foreign saving and equivalently lowers the cost of foreign borrowing, inducing additional period-1 borrowing. Let $B \equiv -NX_1 > 0$ denote period-1 borrowing; the tax generates an additional $dB > 0$. By the intertemporal budget constraint, this additional borrowing must be repaid: $dNX_2 = R^*dB > 0$. In what follows, we trace the equilibrium consequences of a perturbation, $dB > 0$.

The period-1 balance of payments condition can be written as:

$$P_1^{T*}\theta N = \zeta\epsilon_1^X + B$$

Written this way, the BoP condition is the market-clearing condition for foreign purchasing power in the domestic economy's foreign exchange market. The left-hand side is firms' demand for foreign purchasing power to pay for the imported input, which is perfectly inelastic in equilibrium because labor supply is fixed. The right-hand side is the supply of foreign purchasing power: foreigners supply $\zeta\epsilon_1^X$ by purchasing the domestic good, and Ricardians supply B through their borrowing in international asset markets. The purchasing power supplied by foreigners is increasing in ϵ_1 : a real depreciation induces them to buy more of the domestic good and consequently supply more in return, and a real appreciation does the reverse. Since the net demand by firms is perfectly inelastic, a rise in B crowds out export revenues one-for-one. In other words, Ricardians bring additional foreign purchasing power and seek to exchange it for the domestic good, creating an excess supply that they clear by bidding down the real exchange rate. Taking the total differential of the BoP condition shows that the equilibrium requires a real appreciation:

$$d\epsilon_1 = -\frac{dB}{\chi\zeta\epsilon_1^{X-1}} < 0 \quad (22)$$

By inducing Ricardians to borrow more, the public agent substitutes borrowed funds for export revenues as the source of foreign purchasing power for firms, reducing the need to sell domestic output to foreigners in the current period. To trace how this changes the flow of domestic output in the foreign exchange market, we multiply the period-1 BoP condition by ϵ_1 to express it in domestic units:

$$\epsilon_1 P_1^{T*}\theta N = \zeta\epsilon_1^{1+X} + \epsilon_1 B$$

The left-hand side is the share of domestic output sold by firms in the foreign exchange market in return for buying the foreign input, which represents a share of firms' production costs. The first term on the

⁴While we use the laissez-faire allocation as the starting point, the economic intuition discussed in this subsection holds for any point on the intertemporal budget constraint satisfying $NX_1 < 0 < NX_2$, given any import price sequence $\{P_1^{T*}, P_2^{T*}\}$.

right-hand side represents exports, X_1 , and the second term, $\epsilon_1 B$, is the domestic output purchased by Ricardians with borrowed foreign resources. Taking the total differential gives:

$$P_1^T \theta N d\epsilon_1 = (1 + \chi) \zeta \epsilon_1^X d\epsilon_1 + (\epsilon_1 dB + B d\epsilon_1) \quad (23)$$

Since $d\epsilon_1 < 0$, the left-hand side of (23) measures the total cost savings for firms. As Ricardians bid down the real exchange rate, the domestic-good cost of the imported input, $\epsilon_1 P_1^{T*}$, and therefore firms' marginal cost of production, falls. At the prevailing real wage, hiring a unit of labor and buying θ units of input to produce A units of output now creates a positive profit opportunity. Firms attempt to expand production and demand more inputs, and since inputs and labor are perfect complements, the demand for labor also expands. With the labor endowment fixed at N in equilibrium, competition for workers instead bids up the real wage until positive profit opportunities have been eliminated. The firms' zero-profit condition, (6), implies that the real wage must rise one-for-one with a real appreciation. In other words, cost savings by firms are passed through to workers in the form of higher real labor incomes. The increase in labor income is therefore:

$$dY_1^N = -P_1^T \theta N d\epsilon_1 > 0 \quad (24)$$

Workers gain these resources, and the right-hand side of (23) shows who gives them up. The net cost reduction for firms can be decomposed into two components. The first expression is $dX_1 = (1 + \chi) \zeta \epsilon_1^X d\epsilon_1$, the decrease in exports. Using (22) we have:

$$dX_1 = - \left(\epsilon_1 dB + \frac{1}{\chi} \epsilon_1 dB \right) < 0 \quad (25)$$

The decrease in exports itself decomposes into a quantity effect, $\epsilon_1 dB$, and an inframarginal price effect, $\frac{1}{\chi} \epsilon_1 dB$. The quantity effect is the domestic output no longer sold to foreigners because Ricardian borrowing replaces foreign-supplied purchasing power at the margin. The price effect arises from the upward-sloping export supply curve: a real appreciation decreases the domestic output paid to foreigners at the margin and against all inframarginal units of foreign purchasing power supplied by them.

The second expression on the right-hand side of (23) represents the net change in the quantity of the domestic good bought by Ricardians with the borrowed foreign good. The quantity effect, $\epsilon_1 dB$, is positive: an increase in borrowing allows Ricardians to buy more of the domestic good at any given price. The price effect, $B d\epsilon_1$, is negative: a real appreciation lowers the domestic-good value of all foreign resources exchanged by Ricardians, not only the marginal unit. The sign of the net change depends on whether the quantity effect dominates the price effect or vice versa.

Substituting (24) and (25) into (23), the net increase in labor income can be expressed as the sum of two inframarginal price effects associated with a real exchange rate appreciation:

$$dY_1^N = \left(\frac{\epsilon_1}{\chi} - B \frac{d\epsilon_1}{dB} \right) dB > 0 \quad (26)$$

The first term is the inframarginal price effect on foreigners: an appreciation lowers the domestic output

paid against all inframarginal units of foreign purchasing power supplied through exports. The second term is the inframarginal price effect on Ricardians: an appreciation lowers the domestic-good value of all inframarginal foreign resources borrowed by Ricardians⁵. Both household types therefore gain through higher labor income, financed by foreigners and Ricardians giving up their claims on domestic resources.

Thus, a subsidy to Ricardian borrowing induces a real exchange rate appreciation, which, through cost reduction for firms, results in an endogenous transfer of real resources toward labor incomes and away from foreign exchange market transactions. Since labor incomes are gained by both Ricardians and HtM households, this policy redistributes through pecuniary general equilibrium effects. Exports fall, while HtM consumption rises one-for-one with labor income. The change in Ricardian consumption is ambiguous: the labor-income gain and the positive quantity effect of additional borrowing are offset partly or fully by the inframarginal price effect:

$$\begin{aligned} dC_1^p &= dY_1^N > 0 \\ dC_1^r &= dY_1^N + \frac{1}{1-\lambda} [\epsilon_1 dB + B d\epsilon_1] \end{aligned}$$

However, in period 2 Ricardians face a larger debt burden. The intertemporal resource constraint requires that an appreciation in period 1 must be followed by a depreciation in period 2. The economy must export more in period 2 to pay back the larger period-1 borrowing. A real depreciation in period 2 does an analogous endogenous resource transfer but in the opposite direction: it reduces labor incomes and increases the share of output sold to foreigners. Using the period-2 BoP condition with $dNX_2 = R^* dB > 0$:

$$d\epsilon_2 = \frac{R^* dB}{\chi \zeta \epsilon_2^{\chi-1}} > 0 \quad (27)$$

A larger repayment obligation depreciates the real exchange rate in period 2. This raises the domestic-good price of the imported input and hence firms' marginal cost, inducing them to reduce production. At the prevailing wage, firms make losses. Labor demand falls, and the real wage declines until zero profits are restored. Hence labor income falls:

$$dY_2^N = -P_2^T \theta_N d\epsilon_2 < 0 \quad (28)$$

Since HtM households consume current labor income, their consumption falls in period 2. Ricardians are worse off as well: labor income falls and they face a higher repayment burden in domestic units, and both the quantity and price effects are unambiguously negative:

$$\begin{aligned} dC_2^p &= dY_2^N < 0 \\ dC_2^r &= dY_2^N - \frac{1}{1-\lambda} [\epsilon_2 R^* dB + R^* B d\epsilon_2] < 0 \end{aligned}$$

The net fall in labor income can be decomposed in a way that parallels the period-1 expression:

$$dY_2^N = - \left[\frac{\epsilon_2}{\chi} - B \frac{d\epsilon_2}{dB} \right] R^* dB \quad (29)$$

⁵The positive quantity effect on Ricardians cancels against the negative quantity effect on foreigners. The net gain in labor income is the sum of the two inframarginal price effects.

The first term is the inframarginal export-price effect: the depreciation increases domestic output sold to foreigners against all inframarginal units of foreign purchasing power supplied by them. The second is the inframarginal price effect on Ricardian repayment obligations: a depreciation increases the domestic-good value of their existing debt burden. Since $B > 0$, this effect places part of the burden directly on Ricardians and dampens the fall in labor income.

Thus, a marginal increase in period-1 borrowing redistributes resources toward labor income in period 1 and away from labor income in period 2. Exports fall in period 1 and rise in period 2, so HtM households, who live off current labor income, gain in the first period and lose in the second, while Ricardians face an ambiguous period-1 effect and an unambiguous period-2 loss. This intertemporal endogenous redistribution of resources is driven by the path of the real exchange rate, which policy can choose by distorting the intertemporal margin.

The planner's first-order condition. To characterize optimal policy, we return to the primal problem, (16), where a utilitarian Ramsey planner directly chooses the optimal exchange rate path by internalizing the pecuniary general equilibrium effects of exchange rate movements. In Appendix A we derive the Ramsey planner's first-order condition with respect to ϵ_t , which we restate below to make explicit the net marginal benefits of an appreciation at time t :

$$(\beta R^*)^{t-1} \left[\bar{u}'_t \left(\frac{\epsilon_t}{\chi} - NX_t \frac{d\epsilon_t}{dNX_t} \right) + u'(C_t^r) \epsilon_t + u'(C_t^r) NX_t \frac{d\epsilon_t}{dNX_t} \right] = \mu \quad (30)$$

where $\bar{u}'_t$ is average marginal utility, defined in (19), and μ is the Lagrange multiplier on the intertemporal budget constraint. The left-hand side captures the net benefit of a marginal appreciation in period t . The first term is the labor-income gain valued at average marginal utility. The term in parenthesis is identical to the net change in labor income decomposed into two inframarginal price effects, as illustrated in equations (26) and (29) above. The second term is the direct quantity effect on the Ricardian foreign asset position. The third is the price effect: an appreciation lowers the domestic-good value of all inframarginal foreign resources exchanged by Ricardians. The first and second terms have positive signs. The third term's sign depends on the sign of NX_t : if the Ricardians are net borrowers ($NX_t < 0$), the price effect is negative, and if they are net savers ($NX_t > 0$), the price effect is positive.

Equation (30) can be rearranged to group appreciation-induced resource transfers among agents into two net transfers. First, a net transfer between foreigners and the domestic economy, and, second, a net transfer between the two household types within the domestic economy:

$$(\beta R^*)^{t-1} \left[u'(C_t^r) \epsilon_t \left(1 + \frac{1}{\chi} \right) + \lambda (u'(C_t^p) - u'(C_t^r)) \left(\frac{\epsilon_t}{\chi} - NX_t \frac{d\epsilon_t}{dNX_t} \right) \right] = \mu \quad (31)$$

The first term is interpreted as a net resource transfer from foreigners to domestic agents valued at Ricardian marginal utility: an appreciation reduces exports at the margin and lowers the domestic output paid against inframarginal foreign purchasing power. The second term captures the net resource transfer between domestic households. It represents the pecuniary general equilibrium effect of a real appreciation on HtM

households through their labor incomes. Since a share λ of the overall increase in labor income accrues to HtM households, it is effectively equivalent to a net resource transfer from Ricardians to HtMs. The planner values this transfer at the marginal utility difference between the two types: how much more HtM households value these resources relative to Ricardians.

The Lagrange multiplier μ captures the marginal cost of a real exchange rate appreciation in period t through a tightening of the intertemporal budget constraint. Feasibility dictates that an appreciation in period t must be followed by a depreciation in period $t + 1$. A marginal depreciation at time $t + 1$ relaxes the intertemporal budget constraint, the marginal utility benefit of which is also μ . An optimizing planner equates the benefits of inducing a marginal appreciation at time t to the costs of inducing a marginal depreciation at time $t + 1$. This yields the planner's intertemporal optimality condition:

$$\epsilon_1 u'(C_1^r) + \Omega_1 = \beta R^* [\epsilon_2 u'(C_2^r) + \Omega_2] \quad (32)$$

where

$$\Omega_t \equiv \frac{\lambda [u'(C_t^p) - u'(C_t^r)] \left(\frac{\epsilon_t}{X} - NX_t \frac{d\epsilon_t}{dNX_t} \right)}{\frac{1+X}{X}} \quad (33)$$

Ω_t is the redistributive wedge that captures the social value of a net resource transfer between domestic households per unit of the resource transfer between foreigners and the domestic economy induced by a marginal real exchange rate change at time t .

Optimal policy under $P_1^{T*} > P_2^{T*}$. The intertemporal optimality condition (32) and the intertemporal budget constraint characterize the constrained efficient allocation for any import price sequence. From Lemma 4, the laissez-faire allocation satisfies $NX_1^{LF} < 0 < NX_2^{LF}$, which implies $C_1^{r,LF} > C_1^{p,LF}$ and $C_2^{r,LF} < C_2^{p,LF}$. HtM households therefore have strictly higher marginal utility than Ricardians in period 1 and strictly lower marginal utility in period 2, giving $\Omega_1^{LF} > 0$ and $\Omega_2^{LF} < 0$. Moreover, the laissez-faire allocation satisfies the Euler equation: $\epsilon_1^{LF} u'(C_1^{r,LF}) = \beta R^* \epsilon_2^{LF} u'(C_2^{r,LF})$. Evaluating the left- and right-hand sides of (32) at the laissez-faire allocation, we get:

$$\epsilon_1^{LF} u'(C_1^{r,LF}) + \Omega_1^{LF} > \epsilon_2^{LF} u'(C_2^{r,LF}) + \Omega_2^{LF}$$

Thus the marginal benefit of a real exchange rate appreciation in period 1 exceeds the marginal cost of the required depreciation in period 2 at the laissez-faire allocation whenever $P_1^{T*} > P_2^{T*}$. The inability of HtM households to intertemporally smooth consumption in response to a transitory fall in real incomes makes their income and consumption excessively volatile from the perspective of a utilitarian planner. By inducing a real exchange rate appreciation in period 1, and a depreciation in period 2, the planner can tilt the economy's access to resources toward the high-input price period and away from the low-input price period. This motive is absent when all households participate in asset markets; with a mass of constrained households, the same tilt induces an endogenous resource transfer toward HtM households in the high-input price period and away from them in the low-input price period.

The period-1 appreciation reduces the domestic-good cost of the imported input, raises firms' residual profits at the prevailing wage, and through the zero-profit condition (6) bids wages up until zero profits are restored. Labor income rises in period 1, and since HtM households consume current labor income one-for-one, their consumption rises. In period 2, the depreciation raises input costs, compresses labor incomes, and reduces HtM consumption. By tilting the exchange rate path, the planner makes HtM consumption intertemporally smoother than under laissez-faire: the compression in period 1 is partially offset, and the recovery in period 2 is partially dampened. For Ricardian households, the change in consumption in period 1 is ambiguous: the labor income gain and the positive quantity effect are partially or fully offset by the appreciation-induced reduction in the domestic-good value of their foreign borrowing. In period 2 their consumption unambiguously falls, as they face both lower labor income and a higher domestic-good cost of repayment. The net effect on social welfare is strictly positive, however: at utilitarian Pareto weights, a marginal appreciation at $t = 1$ with the required depreciation at $t = 2$ always raises welfare starting from the laissez-faire allocation.

Therefore, optimal policy involves a more appreciated real exchange rate in period 1 and a more depreciated real exchange rate in period 2 relative to laissez-faire: $\epsilon_1^O < \epsilon_1^{LF}$ and $\epsilon_2^O > \epsilon_2^{LF}$. Moreover, the trade deficit must be larger in period 1 and the trade surplus must be larger in period 2 relative to laissez-faire. At the same time, this appreciation partially offsets the exogenous spike in the world input price by compressing its domestic-good equivalent in the high-price period at the cost of a higher domestic equivalent in the low-price period, thereby stabilizing the country's terms of trade. Since real labor income, Y_t^N , i.e., the country's GDP, moves one-for-one against the terms of trade, it is also relatively stabilized under the conditions of Lemma A.2: higher in the high-input price period and lower in the low-input price period under the constrained efficient allocation. Proposition 2 summarizes these results.

Finally, the constrained efficient allocation involves a strictly positive intertemporal wedge, i.e., in a decentralized equilibrium, this allocation is implementable through a strictly positive capital control tax on Ricardian asset holdings, or equivalently a subsidy on Ricardian borrowing. The tax on Ricardian saving raises their private cost of accumulating foreign assets, inducing them to borrow more in period 1 than they would choose under laissez-faire. The public agent in effect uses the Ricardian households' asset market participation to redistribute resources, operating through the pecuniary general equilibrium effects of real exchange rate changes. In Proposition 2 we establish that whenever $P_1^{T*} > P_2^{T*}$, the constrained efficient allocation is decentralized with a strictly positive capital control tax at $t = 1$, i.e., $\tau > 0$:

$$\tau = \frac{\Omega_1^O - \beta R^* \Omega_2^O}{\epsilon_1^O u'(C_1^{r,O})} > 0 \quad (34)$$

Proposition 2. *Assume inelastic labor supply: $N_t = \bar{N} > 0$. Let $\lambda \in (0, 1)$, $A_t = A > 0$ for $t \in \{1, 2\}$, $\beta R^* = 1$, $0 < \chi \leq 1$, $P_2^{T*} = P^{T*} > 0$, and $P_1^{T*} > P^{T*}$. Let $(\epsilon_t^O, C_t^{r,O}, C_t^{p,O}, NX_t^O, Y_t^{N,O})$ denote the constrained efficient allocation and $(\epsilon_t^{LF}, C_t^{r,LF}, C_t^{p,LF}, NX_t^{LF}, Y_t^{N,LF})$ denote the laissez-faire allocation. The constrained efficient allocation involves:*

- i. *A strictly positive intertemporal wedge: $\tau > 0$.*

ii. A relatively volatile real exchange rate: Exchange rate is strictly more appreciated in period 1 and strictly more depreciated in period 2:

$$\epsilon_1^O < \epsilon_1^{LF}, \quad \epsilon_2^O > \epsilon_2^{LF}$$

iii. A relatively volatile trade balance: The trade deficit is strictly larger in period 1 and the trade surplus is strictly larger in period 2:

$$NX_1^O < NX_1^{LF} < 0 < NX_2^{LF} < NX_2^O$$

iv. Input price in domestic tradeable units is strictly lower in period 1 and strictly higher in period 2:

$$\epsilon_1^O p_1^{T*} < \epsilon_1^{LF} p_1^{T*}, \quad \epsilon_2^O p_2^{T*} > \epsilon_2^{LF} p_2^{T*}$$

v. Real labor income is strictly higher in period 1 and strictly lower in period 2:

$$Y_1^{N,O} > Y_1^{N,LF}, \quad Y_2^{N,O} < Y_2^{N,LF}$$

vi. Under sufficiently weak conditions established in Lemma A.2, input prices and labor incomes are relatively stabilized:

$$|\epsilon_1^O p_1^{T*} - \epsilon_2^O p_2^{T*}| < |\epsilon_1^{LF} p_1^{T*} - \epsilon_2^{LF} p_2^{T*}|$$

$$\left| Y_2^{N,O} - Y_1^{N,O} \right| < \left| Y_2^{N,LF} - Y_1^{N,LF} \right|$$

Proof. See Appendix A.7. □

In the high-input price period, exchange rate policy suppresses exports below their laissez-faire level and retains a larger share of domestic output within the economy. Due to inframarginal price effects on foreigners and Ricardians, the resources thus retained flow disproportionately toward labor incomes, and since HtM households consume their current labor income without any recourse to asset markets, this channel raises their consumption in states where import prices are high. In future states when the shock has subsided, the exchange rate depreciates to service the accumulated external obligation, reversing the resource transfer: the export share rises, the domestic share contracts, and real wages fall. The net effect is a partial smoothing of the income and consumption path of HtM households relative to laissez-faire: exchange rate policy provides a degree of implicit insurance against transitory import price shocks even in the absence of direct fiscal transfers or explicit insurance markets. Proposition 3 summarizes the corresponding changes in output shares.

Proposition 3. *Assume that conditions of Proposition 2 hold. Relative to the laissez-faire equilibrium, the Ramsey equilibrium involves:*

- a lower export share of output, a higher domestic share, a higher HtM share of output, and an ambiguous change in the Ricardian share of output in the high input price period, i.e., at $t = 1$.

- *a higher export share of output and a lower domestic share, a lower HtM and Ricardian share of output in the low input price period, i.e., at $t = 2$.*

Proof. See Appendix A.8. □

We interpret the intertemporal policy characterized in this subsection as the real-income stabilization channel of exchange rate policy, and we refer to the public agent's reason for distorting the intertemporal margin as the real-income stabilization motive. By distorting the intertemporal margin, the public agent changes the net supply of foreign purchasing power in the domestic economy's foreign exchange market, which in turn changes the real exchange rate and imported-input costs. Firms pass these cost changes into wages, thereby affecting the real incomes of constrained households. The resources arrive indirectly: these pecuniary general equilibrium effects originate in the actions of Ricardians in the foreign exchange market and reach households who do not themselves respond to changes in intertemporal prices.

The economics of our framework naturally shares its flavor with the real income channel of monetary policy in closed-economy HANK models, delivered there through monetary policy's influence on the intertemporal margin. In both cases, policy has no direct way of reaching borrowing-constrained households and must influence their real incomes and consumption through indirect pecuniary general equilibrium effects. There, monetary policy moves the domestic real interest rate, unconstrained households move spending across time, and the resulting change in aggregate demand influences employment and with it the real incomes of borrowing-constrained households. Our open economy framework differs in several ways. First, general equilibrium effects reach constrained households through changes in relative prices rather than through the level of employment or production. By altering the intertemporal price faced by Ricardians, policy influences their participation in the foreign exchange market, and with it the real exchange rate, the imported-input costs faced by firms, and therefore the real wages received by all workers through the zero-profit condition. These effects are operative, and in fact strongest, when employment is perfectly inelastic, as we illustrate in Section 3.5. Second, the source of the indirect general equilibrium effect on constrained households in our framework has no counterpart in closed-economy HANK models: an upward-sloping export-revenue supply curve. When Ricardians borrow more, they crowd out foreigners and appreciate the real exchange rate, which lowers firms' marginal cost and, through the zero-profit condition, feeds into higher real wages. Policy can therefore divert resources away from exports and toward domestic households by inducing stronger Ricardian participation in asset markets. The elasticity of export revenue supply determines how far foreigners are crowded out and how much the real exchange rate responds: the more elastic the supply, the smaller the exchange rate response to Ricardian participation and the weaker the general equilibrium effect on real wages. Third, in an economy small with respect to global asset markets, the intertemporal price is disciplined by those markets and intertemporal distortions lie beyond the limits of monetary policy. Delivering the real income gains therefore necessitates a separate policy instrument that directly distorts the intertemporal margin, which in our analytical model is a capital control tax and, as we illustrate in Section 4, is equivalently implementable through foreign exchange interventions. Our framework illustrates that the real income influence of policies distorting the intertemporal margin extends beyond the monetary

policy setting in which HANK models study it.

Monetary policy in a small open economy still plays a role in distorting the labor wedge and therefore influences real outcomes. In the next subsection, we relax the inelastic labor supply assumption to isolate this role, and show that household heterogeneity modifies the optimal labor wedge beyond the static terms-of-trade correction and that import-employment complementarity requires monetary and exchange rate policies to be set in coordination.

3.4.2 Elastic Labor Supply

The preceding subsection fixes employment at $\bar{N}$ and characterizes the optimal intertemporal wedge in isolation. In this subsection we endogenize the level of employment and input utilization and characterize the two margins jointly in our heterogeneous-agent economy. We interpret the labor wedge as implementable through monetary policy in a decentralized economy with nominal rigidities, while our analytical framework maintains a reduced-form specification where the public agent directly chooses employment. We illustrate that the interaction between household heterogeneity and employment-import complementarity creates a novel role for policy to distort the labor wedge.

The representative-agent framework of Section 3.3 establishes that monetary policy already has a role: the labor wedge corrects for the terms-of-trade distortion that arises because competitive firms do not internalize the effect of their employment and import decisions on the real exchange rate. Facing an upward-sloping supply of foreign purchasing power, a change in their collective import demand depreciates the real exchange rate and the economy must give up more domestic output to foreigners against all inframarginal units of foreign purchasing power supplied by them. The public agent internalizes this and restricts the economy's employment and import levels below laissez-faire with a strictly positive labor wedge, $\tau_t^N > 0$. There we showed that a Ramsey planner values a marginal unit of employment at the effective marginal product, i.e., the net output retained by the economy for a marginal increase in employment, defined as $\hat{w}$:

$$\hat{w}_t = w_t - \frac{\epsilon_t P_t^{T*} \theta}{\chi}$$

where the second term is the terms-of-trade correction. The planner's first-order condition with respect to employment in the representative-agent economy is given by $v'(N_t) = \hat{w}_t u'(C_t)$. To isolate any distortion to the labor wedge beyond the static terms-of-trade correction, we define the terms-of-trade-adjusted labor wedge as:

$$1 - \hat{\tau}_t^N \equiv \frac{v'(N_t)}{\hat{w}_t u'(C_t)} \quad (35)$$

Proposition 1 establishes that the representative-agent constrained efficient allocation satisfies $\hat{\tau}_t^N = 0$ and $\tau = 0$ for any sequence of import prices.

Define the RA-equivalent policy as the one that imposes these same restrictions in the heterogeneous-agent economy: $\hat{\tau}_t^N = 0$ for $t \in \{1, 2\}$ and $\tau = 0$. In other words, the RA-equivalent policy satisfies the undistorted Ricardian Euler equation and implements a labor wedge that just corrects for the static terms-of-trade

distortion. Together with the intertemporal budget constraint, it satisfies:

$$\epsilon_1 u'(C_1^r) = \beta R^* \epsilon_2 u'(C_2^r) \quad (36)$$

$$v'(N_t) = \hat{w}_t \bar{u}'_t \quad t \in \{1, 2\} \quad (37)$$

In what follows, we derive the jointly optimal intertemporal and labor wedges in the heterogeneous-agent economy and compare them to the RA-equivalent policy wedges. The optimal intertemporal wedge carries the same sign as in the previous subsection, and the optimal terms-of-trade-adjusted labor wedge is strictly positive, $\hat{\tau}_t^N > 0$, whenever $P_1^{T*} \neq P_2^{T*}$.

A perturbation to N_t . Before characterizing the planner's optimality condition with respect to employment, it is useful to consider the partial effect of a perturbation to N_t , $dN_t > 0$, on equilibrium outcomes, holding the trade balance, NX_t , fixed. Since the trade balance is fixed, the perturbation is purely intratemporal.

An increase in the net labor supply, through labor-market clearing, requires that firms hire all additional labor and produce more. Since labor and imported inputs are perfect complements, the labor market and the foreign exchange market must clear simultaneously. An increased labor supply creates downward pressure on the real wage, lowering marginal costs and opening up positive profit opportunities that draw firms into demanding more labor and more imported inputs.

Holding NX_t fixed, the foreign purchasing power needed to pay for the additional imports must come from higher export revenues. Because the supply of foreign purchasing power is increasing in the real exchange rate, financing the larger import bill requires a real depreciation. Taking the total differential of the trade balance, $NX_t = \zeta \epsilon_t^X - P_t^{T*} \theta N_t$ and imposing $dNX_t = 0$ gives:

$$d\epsilon_t = \frac{P_t^{T*} \theta}{\chi \zeta \epsilon_t^{X-1}} dN_t > 0 \quad (38)$$

Firms' import bill in domestic units goes up, not only because imports have increased at the margin, but also because the real exchange rate depreciates and increases the cost of all inframarginal imports. Exports expand with the depreciation, and Ricardian households face a pecuniary general equilibrium effect: their net financial flows, NX_t , are revalued:

$$\epsilon_t P_t^{T*} \theta dN_t + d\epsilon_t P_t^{T*} \theta N_t + d\epsilon_t NX_t = dX_t \quad (39)$$

In equilibrium, zero profits must be restored: as the real exchange rate depreciates, the real wage must fall one-for-one until positive profit opportunities are eliminated:

$$dw_t = -P_t^{T*} \theta d\epsilon_t < 0 \quad (40)$$

In other words, the inframarginal increase in firms' costs is paid for entirely by workers, who give up real wages. The net effect on labor income is therefore ambiguous. The direct effect of higher employment raises

labor income, and the indirect general-equilibrium effect through the real exchange rate and real wage lowers it:

$$dY_t^N = w_t dN_t + N_t dw_t = (A_t - \epsilon_t P_t^{T*} \theta) dN_t - d\epsilon_t P_t^{T*} \theta N_t \geq 0 \quad (41)$$

where the first term is the direct effect at the prevailing wage and the second the general-equilibrium effect through the real exchange rate. Combining with (39) we get:

$$A_t dN_t - dX_t = dY_t^N - d\epsilon_t NX_t \quad (42)$$

where the left-hand side, $A_t dN_t - dX_t = \hat{w}_t dN_t$, is the net gain to the domestic economy: marginal output retained domestically after accounting for increased exports, and the right-hand side shows how this additional output is distributed between labor income and Ricardian revaluation effects.

Since HtM households consume their current labor income, $dC_t^P = dY_t^N$. Ricardians face the same change in labor income and, in addition, face the revaluation effect:

$$dC_t^r = dY_t^N - \frac{1}{1-\lambda} NX_t d\epsilon_t$$

where the sign of the revaluation effect depends on NX_t : a depreciation raises the domestic-good value of the foreign resources Ricardians bring into the economy when they are net borrowers, and raises the domestic-good cost of the resources they send abroad when they are net savers. Combining household budgets with Equation (42), the net changes in consumption of the two household types can be expressed as:

$$dC_t^P = \hat{w}_t dN_t + NX_t d\epsilon_t \quad (43)$$

$$dC_t^r = \hat{w}_t dN_t - \frac{\lambda}{1-\lambda} NX_t d\epsilon_t \quad (44)$$

The above equations imply that the revaluation effect on Ricardians is equivalent to a net transfer of resources between the two household types. For a marginal increase in employment, the domestic economy gains $\hat{w}_t$ units of output, but the distribution of these resources between the two household types is determined endogenously through the real exchange rate and the sign of the country's trade balance. If the country is a net importer/net borrower ($NX_t < 0$), a real depreciation implies an endogenous transfer from HtM households to Ricardians, and if the country is a net exporter/net saver ($NX_t > 0$), the direction of the transfer is reversed.

To summarize, a marginal increase in employment, holding the trade balance fixed, induces a real depreciation and a fall in real wages. This depreciation raises exports and allows firms to import and produce more, and the domestic economy potentially gains resources from this increased production. However, the real depreciation creates a general-equilibrium revaluation effect on Ricardian financial flows, due to which such gains are unequally distributed between the two household types. In other words, by increasing employment a planner can potentially generate more resources for the domestic economy, but the distribution of these resources is asymmetric and determined endogenously when households differ in their asset market participation.

The planner's first-order condition. A utilitarian Ramsey planner internalizes both the resources a marginal unit of employment creates for the economy and how they are distributed. In Appendix A we derive the planner's first-order condition with respect to employment, which we restate below:

$$v'(N_t) = \hat{w}_t \bar{u}'_t + \lambda [u'(C_t^p) - u'(C_t^r)] NX_t \frac{d\epsilon_t}{dN_t} \Big|_{NX_t} \quad (45)$$

The left-hand side is the social marginal disutility of labor. The first term on the right-hand side is the social value of the resources a marginal unit of employment creates: after accounting for the terms-of-trade distortion, the economy retains $\hat{w}_t = A_t - \frac{1+\chi}{\chi} \epsilon_t P_t^T \theta$ units of output, valued at the average marginal utility, i.e., as they would be valued if every household received an equal share of them. The second term corrects for the fact that they do not, since the depreciation that pays for the additional imports affects Ricardians through a revaluation effect on their financial flows. As discussed previously, this transfer runs from HtM households to Ricardians when they are net borrowers ($NX_t < 0$) and in the opposite direction when they are net savers ($NX_t > 0$). The planner values it at the marginal utility difference between the two types. This revaluation effect disappears in a representative-agent economy, where $\lambda = 0$; when trade is intratemporally balanced, $NX_t = 0$; or when the two household types consume the same amount of the good, $C_t^p = C_t^r$.

A strictly positive terms-of-trade-adjusted labor wedge under $NX_t \neq 0$. Given any sequence of import prices and productivity, $\{P_t^T, A_t\}$, the constrained efficient allocation satisfies the intertemporal budget constraint, $NX_1 + R^{*-1}NX_2 = 0$, together with the optimal intertemporal condition, (32), derived in the previous subsection, and the intratemporal optimality conditions, (45), for each t . The optimal terms-of-trade-adjusted labor wedge is the incidence correction measured against the resource gain it offsets:

$$\hat{\tau}_t^N = - \frac{\lambda [u'(C_t^p) - u'(C_t^r)] NX_t \frac{d\epsilon_t}{dN_t} \Big|_{NX_t}}{\hat{w}_t \bar{u}'_t} \quad (46)$$

If the solution satisfies $NX_t = 0$, employment choice just accounts for the terms-of-trade distortion and the wedge is zero. If it involves $NX_t \neq 0$, then the wedge is strictly positive. This is because, with strictly concave utility, $u(\cdot)$, the sign of $[u'(C_t^p) - u'(C_t^r)] NX_t$ is always negative whenever $NX_t \neq 0$. This follows from $C_t^r - C_t^p = -\epsilon_t NX_t / (1 - \lambda)$. If Ricardians are net borrowers in a period ($NX_t < 0$), then they consume more than HtM households, and consequently have a lower marginal utility. Similarly, if Ricardians are net savers in a period ($NX_t > 0$), then they consume less than HtM households, and have a higher marginal utility. The Ramsey planner internalizes the exchange-rate-induced revaluation effects on Ricardian financial flows that arise due to employment-imports complementarity and endogenously redistribute resources away from the higher marginal utility household type. For a utilitarian planner, this endogenous redistribution imposes an additional welfare cost of increasing employment at the margin beyond the labor disutility cost. Consequently, whenever $NX_t \neq 0$, optimal policy involves a strictly positive labor wedge correction beyond the terms-of-trade correction. Proposition 4 formally summarizes this result.

Proposition 4. *Given any $\{P_t^{T*}, A_t\}$, and $\lambda \in (0, 1)$, consider the constrained efficient allocation that solves (16). Assume that it satisfies $\hat{w}_t > 0$ for $t \in \{1, 2\}$. Then this allocation involves a strictly positive terms-of-trade-adjusted labor wedge, $\hat{\tau}_t^N = 1 - \frac{v'(N_t)}{\hat{w}_t \bar{u}_t} > 0$, whenever $NX_t \neq 0$.*

Proof. See Appendix A.9. □

Optimal policy under $P_1^{T*} = P_2^{T*}$. The preceding discussion allows us to characterize optimal policy in this case almost immediately because it is analogous to the representative-agent case discussed in Proposition 1 and Lemma 2 of Section 3.3. When import prices are equal across periods and $\beta R^* = 1$, the primitives of the planner's problem are symmetric across the two dates. Given this symmetry, a date-asymmetric allocation would only create dispersion in consumption and labor across otherwise identical periods. Averaging such an allocation across the two dates preserves feasibility and, by concavity of consumption utility and convexity of labor disutility, weakly raises welfare. Hence the constrained efficient allocation is symmetric: $N_1 = N_2$ and $\epsilon_1 = \epsilon_2$. Therefore, the two periods must then have the same trade balance; the intertemporal budget constraint implies that $NX_1 = NX_2 = 0$. Ricardians neither borrow nor save, so both household types consume their current labor income in each period and $C_t^P = C_t^r$ for $t \in \{1, 2\}$.

With balanced trade and equal consumption across household types, the revaluation effect term in the planner's first-order condition with respect to employment, (45), vanishes. The employment optimality condition reduces to $v'(N_t) = \hat{w}_t \bar{u}_t'$, which is identical to the RA-equivalent policy condition (37), i.e., the solution involves a strictly positive labor wedge that accounts for the static terms-of-trade distortion: $\tau_t^N = \frac{1}{\chi} \frac{\epsilon_t P_t^{T*} \theta}{w_t} > 0$, and the terms-of-trade-adjusted labor wedge is zero: $\hat{\tau}_t^N = 0$. Since the planner chooses a symmetric allocation, it satisfies the undistorted Ricardian Euler equation, so $\tau = 0$. The constrained efficient allocation therefore coincides with the RA-equivalent policy in this case. Proposition 5 formally summarizes optimal policy with symmetric primitives.

Proposition 5. *Let $\lambda \in (0, 1)$, $A_t = A > 0$, $P_t^{T*} = P^{T*} > 0$ for $t \in \{1, 2\}$, and $\beta R^* = 1$. Assume the conditions of Lemma A.1. Consider the constrained efficient allocation that solves (16). This solution involves:*

- *No intertemporal distortion: $\tau = 0$.*
- *A strictly positive labor wedge implementing the optimal terms-of-trade: $\tau_t^N = \frac{1}{\chi} \frac{\epsilon_t P_t^{T*} \theta}{w_t} > 0$, i.e., the terms-of-trade-adjusted labor wedge is zero: $\hat{\tau}_t^N = 0$.*
- *Symmetric equilibrium across periods: $N_1 = N_2$, $\epsilon_1 = \epsilon_2$, $NX_1 = NX_2 = 0$, and $C_1^P = C_1^r = C_2^P = C_2^r$.*

Proof. See Appendix A.10. □

Optimal policy under $P_1^{T*} > P_2^{T*}$. When the import price is temporarily higher in period 1, the RA-equivalent policy no longer implements the constrained efficient allocation. With borrowing-constrained

households, the utilitarian planner instead chooses both a positive intertemporal wedge and a positive terms-of-trade-adjusted labor wedge. The intertemporal wedge moves the real exchange rate path, redistributing resources toward labor incomes and away from exports in the high-price period and reversing this pattern in the low-price period. The labor wedge moves employment and imported-input demand within each period, accounting for terms-of-trade effects and for exchange-rate-induced revaluation effects on Ricardian financial flows that redistribute resources between the two household types. Proposition 6 formally summarizes that implementing the constrained efficient allocation requires distorting both margins. The optimal policy prescription therefore differs from the representative-agent benchmark: it requires a combination of exchange rate policy that distorts the intertemporal margin and monetary policy that distorts the consumption-leisure margin, both performing distinct roles in optimally redistributing resources across time and between household types.

Proposition 6. *Let $\lambda \in (0, 1)$, $A_t = A > 0$, $P_1^{T*} > P_2^{T*} = P^{T*} > 0$, and $\beta R^* = 1$. Assume the conditions of Lemma A.1. Consider the constrained efficient allocation that solves (16). Assume that this solution satisfies $\hat{w}_t > 0$ for $t \in \{1, 2\}$. Assume further that $0 < \chi \leq 1$ and preferences satisfy $u'(C) = C^{-\sigma}$ with $\sigma > 1/(1+\chi)$. Then the constrained efficient allocation involves:*

- A strictly positive intertemporal wedge, $\tau > 0$.
- A strictly positive terms-of-trade-adjusted labor wedge, $\hat{\tau}_t^N > 0$ for $t \in \{1, 2\}$.

Proof. See Appendix A.11. □

Proposition 6 only allows comparison between optimal policy and the RA-equivalent policy in proportionate terms: signs and magnitudes of the two wedges. These wedge comparisons do not imply an unambiguous ranking of employment levels across policy regimes. The revaluation term in (45) implies that monetary policy tends to restrain employment in period 1. At the same time, a positive intertemporal wedge implies that exchange rate policy tends to appreciate the period-1 real exchange rate relative to the RA-equivalent allocation, which raises $\hat{w}_1$ and tends to push employment in the opposite direction. The ranking of N_1 and similarly N_2 across regimes is therefore ambiguous. While not globally provable, using numerical simulations in the next section, we show that under reasonable parameterizations, the statements of Propositions 2 and 3 hold true even under elastic labor supply: optimal policy appreciates the real exchange rate and worsens the trade balance in the high-price period and depreciates the exchange rate and improves the trade balance in the low-price period, making them intertemporally more volatile when compared to the RA-equivalent allocation. At the same time, optimal policy stabilizes the domestic-units price of the imported input, i.e., the terms of trade, $\epsilon_t P_t^{T*}$, and consequently stabilizes domestic labor income, Y_t^N , making them less volatile across periods relative to the RA-equivalent allocation. Finally, optimal policy lowers the export share of output and raises the HtM share of output in the high-price period, and reverses this pattern in the low-price period, relative to the RA-equivalent allocation.

Employment-import complementarity together with household heterogeneity requires the two policies to be set in coordination. This is because both policies are linked through the same equilibrium object,

the real exchange rate. In the high-price period the intertemporal distortion induces Ricardians to borrow more and to supply additional foreign purchasing power to the domestic foreign exchange market. This borrowed purchasing power attempts to substitute for export revenues and appreciate the real exchange rate. However, as Ricardians attempt to bid down the real exchange rate, firms see positive profit opportunities and attempt to expand their labor and import demand. In equilibrium, how much the real exchange rate appreciates and consequently how much the real wage rises depends on how much monetary policy allows labor supply to expand in response to higher demand. The more monetary policy allows employment and imports to expand, the weaker is the appreciation that exchange rate policy is able to deliver and the smaller is the gain in real wages of workers. Workers gain real incomes but by working longer hours, while a relatively depreciated real exchange rate disproportionately benefits Ricardian borrowers, whose foreign borrowing becomes more valuable in domestic units due to a monetary-policy-induced relative depreciation. By contrast, if monetary policy restricts labor supply, expanded firm demand instead bids up real wages and the real exchange rate appreciates more. Workers gain real incomes in the form of higher earnings per hour while Ricardian borrowing loses value in domestic units. Conditional on exchange rate policy, optimal monetary policy chooses employment to balance the marginal costs of labor disutility and revaluation-induced redistribution against the marginal benefits of higher net output. Conditional on monetary policy, the optimal intertemporal policy determines the net external financial flows of the economy. Both policies together determine the path of employment and the real exchange rate.

We have established a real income redistribution role for monetary policy that parallels in many ways the earnings channel of monetary policy emphasized in closed-economy HANK models, albeit with important differences. By influencing the level of inflation, monetary policy distorts the labor wedge and directly affects employment levels and therefore the real incomes of households. In these models, monetary expansions after adverse productivity shocks can raise employment and current earnings of constrained households. The implication is that monetary policy is relatively more expansionary than in a representative-agent world. In our import-dependent open economy studied here, employment is complementary to imported inputs. In states with high import prices, higher employment raises the import bill; with an upward-sloping supply of foreign purchasing power, the required increase in imports depreciates the real exchange rate. The depreciation lowers real wages and revalues Ricardian financial flows, which distributes the aggregate real income gains from higher employment disproportionately toward borrowers, i.e., Ricardian households, and away from HtM households who are borrowing constrained. A utilitarian planner internalizes this endogenous redistribution and considers employment expansion in high-price states to be costly.

So far we have assumed a specific production and import structure in our framework, which has helped with analytical tractability. In the next subsection we consider alternative assumptions and present a comparative statics exercise.

3.5 Comparative Statics

The previous subsections characterized how heterogeneity in asset market access shapes the optimal intertemporal and labor wedges under utilitarian welfare when the economy faces a transitory import price

spike, in a framework tractable enough to deliver the constrained efficient allocation in closed form. This subsection relaxes the two assumptions that carry that tractability. First, we replace the Leontief technology with a CES technology in labor and imported inputs that nests it, and vary the elasticity of substitution and the inverse Frisch elasticity of labor supply. Second, we consider an economy in which imports are final goods instead of production inputs. Both exercises trace how the optimal wedges vary with the degree of essentiality of imports, and both show that the wedges attributable to heterogeneity are largest when imports are least substitutable.

3.5.1 CES Production Technology

As an alternative to the Leontief technology described in Section 3.1.2, we consider a CES production technology that combines labor and imported inputs to produce the domestic output:

$$Y_t = A_t \left[(1 - \alpha) N_t^{\frac{\xi-1}{\xi}} + \alpha \left(\frac{M_t}{\theta} \right)^{\frac{\xi-1}{\xi}} \right]^{\frac{\xi}{\xi-1}} \quad (47)$$

where $\xi > 0$ is the elasticity of substitution between labor and imported inputs; the Leontief technology of the previous subsections is nested as the limit $\xi \rightarrow 0$. For the numerical simulations, households' utility is $u(C) = C^{1-\sigma}/(1-\sigma)$ and labor disutility is $v(N) = \eta N^{1+\phi}/(1+\phi)$, where ϕ is the inverse Frisch elasticity of labor supply and $\phi \rightarrow \infty$ is the perfectly inelastic case considered previously.

Appendix A.12 states the utilitarian Ramsey planner's primal problem under this technology and shows that ξ and ϕ jointly govern the degree of essentiality of imports: as $\xi \rightarrow 0$ and $\phi \rightarrow \infty$, the price elasticity of import demand approaches zero. We vary these two parameters and solve for the optimal wedges under an illustrative calibration described in the appendix.

Figures A.1, A.2, and A.3 in Appendix A.12 report the optimal intertemporal and labor wedges over (ξ, ϕ) for representative-agent ($\lambda = 0$) and heterogeneous ($\lambda = 0.35$) economies under equal ($P_1^{T*} = P_2^{T*}$) and unequal ($P_1^{T*} > P_2^{T*}$) import price paths. First, when import prices are equal across periods, the optimal intertemporal wedge is zero for every pair (ξ, ϕ) , for $\lambda = 0$ and $\lambda > 0$ alike: the no-intertemporal-distortion result survives across the entire parameter space. With import and export tariffs absent, the planner corrects the static terms-of-trade distortion through the labor wedge alone; the distortion arises because additional export revenue is earned only at a lower export price, and this correction is positive throughout and increases as the elasticity of substitution approaches zero. Second, when import prices are unequal, the no-intertemporal-distortion result does not extend to general CES production: neither wedge alone can fully correct the terms-of-trade distortion, and the planner distorts both margins simultaneously. The intertemporal motive is the capital-control motive of Costinot, Lorenzoni and Werning (2014): the economy takes the world interest rate as given yet holds market power in its own exports, and intertemporal distortions are generated by the time variation in the incentive to exercise that market power, which the import price spike creates and equal prices remove. Leontief production is the special case in which this motive disappears, as the analytical framework established and the exercise confirms. Imports then move one for one with employment, so the period-by-period labor wedge implements the entire terms-of-trade

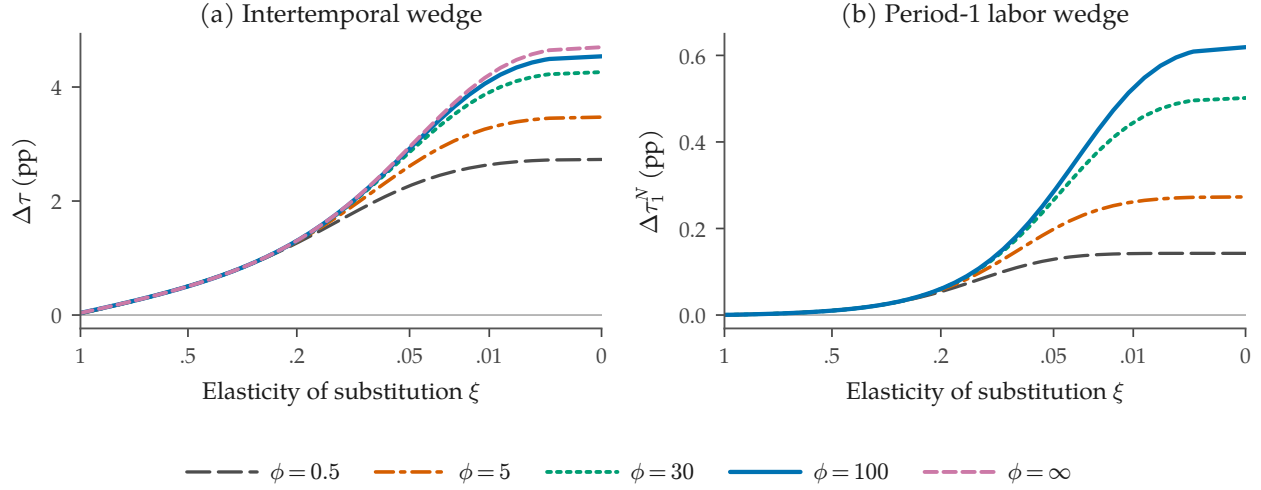


Figure 1. Optimal wedges: isolating the role of heterogeneity

Note: The horizontal axis uses a cube-root scale. The $\phi = \infty$ series appears only in the left panel because the labor margin is absent.

correction, leaving no motive on the intertemporal margin; with perfectly inelastic labor supply the labor wedge is unavailable, yet the planner leaves the intertemporal margin undistorted despite the time-varying incentive, and the constrained efficient allocation coincides with laissez-faire. Proposition 1 establishes this result analytically; any intertemporal distortion that arises with $\lambda > 0$ under Leontief production therefore reflects the role of heterogeneity alone.

Away from the Leontief limit, both wedges are nonzero even with $\lambda = 0$: the level of a wedge in the heterogeneous-agent economy mixes the corrections a representative-agent planner would already choose with the part contributed by heterogeneity. Differencing the wedges in the two economies facing the same import price path therefore helps isolate the role of heterogeneity. Let $\Delta\tau$ and $\Delta\tau_1^N$ denote the differences between the optimal intertemporal and period-1 labor wedges of the heterogeneous economy and those of the representative-agent economy. Figure 1 plots both differences against the elasticity of substitution for a range of labor supply elasticities. Both differences are positive throughout and largest as $\xi \rightarrow 0$ and $\phi \rightarrow \infty$ jointly, the limits in which imports become essential to domestic activity. Intertemporal distortions, in the form of either capital controls or foreign exchange interventions, are therefore not equally desirable across imported goods. For imports with close substitutes or a price-elastic demand the additional wedges are small, and under Cobb-Douglas production negligible: the case for intertemporal distortions rests strongly on the necessity character of imports. Similarly, the labor wedge difference is increasing in the degree of essentiality, and the complementarity between the policies that distort these margins strengthens with it: the intertemporal policy appreciates the real exchange rate, and the more tightly imports move with employment, the larger the wedge monetary policy must set to keep an employment and import expansion from undoing the appreciation, and the more of the cost saving reaches constrained households through real wages. Heterogeneity thus widens the scope of both instruments selectively: the more essential the import, the larger the distortion each of them is called on to deliver.

3.5.2 Imports as Final Goods

The discussion so far has treated imports as intermediate inputs. In this subsection we depart from that assumption and instead consider imports as final goods. We illustrate that the real-income stabilization motive of exchange rate policy, operating through the intertemporal margin, survives this departure, while the role of monetary policy becomes much more limited. The model structure is unchanged except that production of domestic output is linear in labor and households consume both domestic and imported goods, with Stone-Geary preferences over the two:

$$u(C, M) = \frac{[C^{1-\omega}(M - \underline{M})^\omega]^{1-\sigma}}{1 - \sigma}, \quad M \geq \underline{M}$$

In Appendix A.13 we describe the environment with these assumptions, state the constrained problem, and present the comparative statics results. We conduct a comparative statics exercise to illustrate how the optimal intertemporal and labor wedges vary with the degree of essentiality of imports measured by the Marshallian price elasticity of demand for imported goods. We show that this elasticity is governed by the subsistence level $\underline{M}$ and the share parameter ω : with $\underline{M} = 0$ preferences are Cobb-Douglas and the elasticity is one; with $\underline{M} > 0$, as $\omega \rightarrow 0$ the elasticity falls toward zero and demand becomes perfectly inelastic at the subsistence level. We illustrate that the optimal intertemporal wedge attributable to heterogeneity rises with the degree of essentiality and is largest in the limit where demand is perfectly price inelastic. The real-income stabilization motive of exchange rate policy delivered through the intertemporal margin continues to operate: policy supports the real incomes of constrained households by cutting exports and retaining more output for domestic consumption while import prices are high, and reverses the pattern once they fall. The role for stabilization is increasing in the degree of essentiality of imports, supporting a stronger case for intertemporal distortions when imports are necessities. The optimal labor wedge attributable to heterogeneity, however, is small in this environment and smallest when import demand is perfectly inelastic. The role for monetary policy beyond correcting nominal rigidities and the static terms-of-trade distortion is limited in this case. The complementarity between the two policies that arises when imports are necessary inputs in production does not extend to the case where imports are final goods.

For the remainder of the paper we return to the original specification, in which imports are production inputs under the Leontief technology. The next section embeds the analytical framework in a two-agent, infinite-horizon quantitative model with nominal wage rigidities and financial intermediation frictions. In this extended framework, FX interventions decentralize the intertemporal policy, while monetary policy sets the inflation rate and controls the labor wedge. The quantitative model carries over the economic intuition and compares optimal policy to alternative policies under MIT shocks to import prices.

4 Quantitative Model

The analytical framework characterizes the two policy margins in closed form under a reduced-form policy structure, in which the public agent sets employment directly and distorts the intertemporal margin through

a capital control tax. In this section we develop an infinite-horizon two-agent economy in which those margins are carried by the instruments that monetary and fiscal authorities actually operate. Monetary policy sets a nominal interest rate and moves the labor wedge through nominal wage rigidity, and exchange rate policy takes the form of foreign exchange intervention through the public balance sheet, where the intertemporal wedge appears as a deviation from interest parity. Holding that deviation open costs the economy a carry loss, and the size of the distortion that policy should choose therefore becomes a quantitative question.

We solve the Ramsey problem of this economy, establish the conditions under which its solution is implemented by the two instruments, and calibrate the model to Japan as an energy importer. We then compare Ramsey optimal policy with restricted optimal counterfactual policies, to isolate what each instrument accomplishes and whether either can reproduce the other's role in its absence. Two imported-energy price paths carry the exercise: a transitory spike, which separates the two roles, and the realized 2021-2023 path of world energy prices, along which we ask what the two policies would have prescribed.

4.1 Environment

Time is discrete and indexed by $t \geq 1$. The economy is deterministic and faces an exogenously specified path of imported input prices and total factor productivity, $\{P_t^T, A_t\}_{t \geq 1}$ where $P_t^T > 0$ and $A_t > 0$ for all t and bounded. The preferences, household types, final domestic good production technology, export demand, and relative-price definitions are as in Section 3. The intertemporal policy is implemented through foreign exchange interventions in the presence of financial intermediation frictions, while the labor wedge is implemented through monetary policy in the presence of nominal wage rigidities.

4.1.1 Households

As before, a fraction $(1 - \lambda)$ of households are Ricardian and a fraction λ are HtM. Ricardian households now trade one-period domestic nominal bonds. Let $B_{t+1}^{M,r}$ denote nominal bond holdings chosen in period t , and let R_{t+1}^M denote the gross nominal return paid on bonds carried into period $t + 1$. Ricardian households solve

$$\begin{aligned} \max_{\{C_t^r, B_{t+1}^{M,r}\}_{t \geq 1}} \sum_{t=1}^{\infty} \beta^{t-1} [u(C_t^r) - v(N_t)] \quad \text{s.t.} \\ P_t C_t^r + B_{t+1}^{M,r} = W_t N_t + R_t^M B_t^{M,r} + T_t \end{aligned} \quad (48)$$

Let $\Pi_{t+1} \equiv P_{t+1}/P_t$ and $R_{t+1} \equiv R_{t+1}^M/\Pi_{t+1}$ denote gross inflation and the gross real interest rate in domestic units. The solution to the Ricardian household problem is characterized by an Euler equation:

$$\frac{u'(C_t^r)}{\beta u'(C_{t+1}^r)} = R_{t+1} \quad (49)$$

and a no-Ponzi-transversality condition:

$$\lim_{T \rightarrow \infty} \frac{B_{T+1}^r}{\prod_{s=2}^T R_s} = 0 \quad (50)$$

where $B_{t+1}^r \equiv B_{t+1}^{M,r}/P_t$ denotes the Ricardian household's bond holdings in real domestic units. Ricardian households participate in domestic asset markets but not directly in international asset markets. HtM households remain excluded from all asset markets and direct fiscal transfers. They consume their current labor incomes each period:

$$C_t^P = w_t N_t \quad (51)$$

As before, both household types work the same number of hours in equilibrium, N_t , and do not choose those hours themselves. The labor aggregator and the unions described next determine them.

4.1.2 Labor Aggregator

The final good production technology is inherited from the analytical model, with firms combining a final labor service and imported inputs to produce the final good. The final labor service, N_t , is produced by a competitive labor packer who aggregates over $k \in [0, 1]$ tasks performed by task-specific labor, $N_{k,t}$, using a CES technology:

$$N_t = \left(\int_0^1 N_{k,t}^{\frac{\varepsilon_w - 1}{\varepsilon_w}} dk \right)^{\frac{\varepsilon_w}{\varepsilon_w - 1}}, \quad \varepsilon_w > 1$$

Let $W_{k,t}$ denote the nominal wage for task k . Cost minimization gives task demand and the aggregate wage index:

$$N_{k,t} = \left(\frac{W_{k,t}}{W_t} \right)^{-\varepsilon_w} N_t, \quad W_t = \left(\int_0^1 W_{k,t}^{1-\varepsilon_w} dk \right)^{\frac{1}{1-\varepsilon_w}} \quad (52)$$

4.1.3 Unions

We adapt the nominal wage setting and rigidity structure from [Erceg, Henderson and Levin \(2000\)](#) and [Auclert, Rognlie and Straub \(2024\)](#). For each task, $k \in [0, 1]$, workers are represented by a union. Taking the aggregator's labor demand function for the task k as given, the union sets the nominal wage, $W_{k,t}$, and calls on all members to work the same number of hours, $N_{k,t}$. For each task, k , workers are representative of the population: a mass λ are HtM and $(1 - \lambda)$ are Ricardian. The union seeks to maximize the population-weighted utility of its members. The union also faces quadratic utility costs for changing nominal wages. The union's optimization problem is to maximize:

$$\sum_{s \geq 0} \beta^s \left(\lambda u(C_{t+s}^P) + (1 - \lambda) u(C_{t+s}^r) - v(N_{t+s}) - \frac{\psi_{nr}}{2} \left(\frac{W_{k,t+s}}{W_{k,t+s-1}} - 1 \right)^2 \right) \quad (53)$$

where $\psi_{nr} > 0$ parametrizes the degree of nominal wage rigidity. Each union is infinitesimal and only takes into account its marginal effect on every household's consumption and labor supply. Since each union's optimization problem is symmetric, in equilibrium all unions will have the same optimal wage and labor

supply decisions: $W_{k,t} = W_t$ and $N_{k,t} = N_t$ for all k . In Appendix B.1 we derive the unions' optimal wage and labor supply decisions that yield the aggregate wage Phillips curve of the form:

$$v'(N_t) = (1 - \tau_t^N) w_t \bar{u}'_t \quad (54)$$

where $(1 - \tau_t^N)$ represents the labor wedge and is given by:

$$1 - \tau_t^N = \frac{1}{\mu_w} + \frac{\left(\frac{w_t}{w_{t-1}} \Pi_t - 1 \right) \frac{w_t}{w_{t-1}} \Pi_t - \beta \left(\frac{w_{t+1}}{w_t} \Pi_{t+1} - 1 \right) \frac{w_{t+1}}{w_t} \Pi_{t+1}}{\frac{\varepsilon_w}{\psi_{nr}} w_t N_t \bar{u}'_t} \quad (55)$$

where $\mu_w \equiv \varepsilon_w / (\varepsilon_w - 1)$ is the wage markup and $\bar{u}'_t$ is the average marginal utility as defined in equation (19).

4.1.4 Financial Intermediaries

The financial intermediation structure in our model is adapted from Gabaix and Maggiori (2015). The world is populated by a unit mass of identical financial intermediaries who intermediate borrowing and lending transactions between domestic and world financial markets. These intermediaries are assumed to be owned outside the small open economy and take offsetting asset positions in domestic and world markets:

$$\frac{F_{t+1}}{e_t} + F_{t+1}^* = 0 \quad (56)$$

where F_{t+1} is the domestic asset position in units of domestic currency and F_{t+1}^* is the asset position in the international asset markets, in units of the world numeraire, i.e., the foreign good. A positive position denotes a saving and a negative position denotes a borrowing. The intermediaries' gross returns in foreign good units, v_{t+1} , are given by:

$$v_{t+1} = R_{t+1}^M \frac{F_{t+1}}{e_{t+1}} + R^* F_{t+1}^* \quad (57)$$

There is a limited repayment commitment on the part of intermediaries: each period, immediately after taking their asset positions, an intermediary can divert a portion $\Gamma |F_{t+1}^*|$, with $\Gamma > 0$, of the position it intermediates: $|F_{t+1}^*|$. If the intermediary diverts the funds, the position is unwound and the lenders recover a portion $(1 - \Gamma |F_{t+1}^*|)$ of their claims $|F_{t+1}^*|$. Since the lenders anticipate the incentives of the intermediary to divert funds, the intermediary is subject to a credit constraint such that discounted returns from intermediation are weakly higher than the returns from diverting the funds:

$$\frac{1}{R^*} v_{t+1} \geq \Gamma |F_{t+1}^*| \times |F_{t+1}^*| \quad (58)$$

The intermediary's constrained optimization problem is then given by:

$$\max_{\{F_{t+1}^*, F_{t+1}\}} \frac{1}{R^*} v_{t+1} \quad \text{s.t. to (56), (58)}$$

The solution to this optimization problem yields the intermediary's asset demand rule in the world asset markets:

$$F_{t+1}^* = \frac{1}{\Gamma} \left[1 - \frac{R_{t+1}^M}{R^*} \frac{e_t}{e_{t+1}} \right] = \frac{1}{\Gamma} \left[1 - \frac{R_{t+1}}{R^*} \frac{e_t}{e_{t+1}} \right] \quad (59)$$

Equation (59) shows that the intermediary's asset demand is linearly related to the interest parity gap $R_{t+1} \frac{e_t}{e_{t+1}} - R^*$. Define the domestic real interest rate in foreign units as $\tilde{R}_{t+1}$:

$$\tilde{R}_{t+1} \equiv R_{t+1} \frac{e_t}{e_{t+1}} \quad (60)$$

If $\tilde{R}_{t+1} = R^*$, intermediary asset demand is zero. When $\tilde{R}_{t+1} < R^*$, the intermediary saves in foreign markets and borrows in domestic markets; when $\tilde{R}_{t+1} > R^*$, the intermediary borrows in foreign markets and saves in domestic markets.

4.1.5 Monetary Policy

Monetary policy sets the path of domestic nominal interest rates, $\{R_{t+1}^M\}_{t \geq 1}$ where the nominal rate satisfies the Fisher equation:

$$R_{t+1}^M = R_{t+1} \Pi_{t+1} \quad (61)$$

4.1.6 Fiscal Policy

The public agent can directly participate in domestic as well as foreign asset markets, bypassing any private intermediaries. Let $B_{t+1}^{M,G}$ denote government domestic bond holdings in domestic currency units and let B_{t+1}^{G*} denote government's foreign asset position in units of the foreign good. The period-by-period government budget constraint in domestic currency units is given by:

$$B_{t+1}^{M,G} + e_t B_{t+1}^{G*} + (1 - \lambda) T_t = R_t^M B_t^{M,G} + e_t R^* B_t^{G*}$$

where T_t denotes lump-sum taxes/transfers to Ricardian household. As in the analytical model, HtM households remain excluded from direct fiscal transfers by assumption. Define $B_{t+1}^G \equiv \frac{B_{t+1}^{M,G}}{P_t}$ as the government's domestic bond holdings in real domestic units. The government budget constraint in real domestic units is then given by:

$$B_{t+1}^G + e_t B_{t+1}^{G*} + (1 - \lambda) \frac{T_t}{P_t} = R_t B_t^G + e_t R^* B_t^{G*} \quad (62)$$

The government is subject to no-Ponzi conditions in both markets:

$$\lim_{T \rightarrow \infty} \frac{B_{T+1}^G}{\prod_{s=2}^T R_s} = 0, \quad \lim_{T \rightarrow \infty} \frac{B_{T+1}^{G*}}{R^{*T-1}} = 0 \quad (63)$$

together with a consistency condition on interest rates which ensures that government asset positions satisfy no-Ponzi conditions in domestic as well as foreign units:

$$0 < \inf_{T \geq 2} \prod_{s=2}^T \frac{\tilde{R}_s}{R^*} \leq \sup_{T \geq 2} \prod_{s=2}^T \frac{\tilde{R}_s}{R^*} < \infty \quad (64)$$

4.1.7 Domestic Bonds Market Clearing

Domestic bonds market clearing requires that the asset positions by Ricardians, government, and financial intermediaries must sum to zero: $(1 - \lambda)B_{t+1}^{M,r} + B_{t+1}^{M,G} + F_{t+1} = 0$. Define the aggregate Ricardian position in real domestic units as $B_{t+1} \equiv (1 - \lambda)B_{t+1}^r$. Using the intermediary balance sheet condition, (56), and expressing the bonds market clearing condition in real domestic units, we have:

$$B_{t+1} + B_{t+1}^G = \epsilon_t F_{t+1}^* \quad (65)$$

4.1.8 Balance of Payments and Intertemporal Budget Constraint

The balance of payments condition is implicitly satisfied, given household, firm, government budget constraints, and goods and domestic bonds market clearing conditions are satisfied. In Appendix B.2, we derive the period-by-period balance of payments condition which is given by:

$$P_t^{T*} \theta N_t + F_{t+1}^* + B_{t+1}^{G*} = \zeta \epsilon_t^X + R_t \frac{\epsilon_{t-1}}{\epsilon_t} F_t^* + R^* B_t^{G*} \quad (66)$$

The left-hand side represents the demand for foreign purchasing power by firms, private intermediaries, and the government at time t . The right-hand side represents the supply of foreign purchasing power from export revenues and inherited foreign positions. The BoP conditions represent the effective external budget constraint of the economy. The period-by-period constraints can be combined to obtain the intertemporal budget constraint:

$$\sum_{t=1}^{\infty} \frac{1}{R^*} \left(1 - \frac{\tilde{R}_{t+1}}{R^*}\right)^2 + \sum_{t=1}^{\infty} \frac{P_t^{T*} \theta N_t}{R^{*t-1}} = \sum_{t=1}^{\infty} \frac{\zeta \epsilon_t^X}{R^{*t-1}} + R^* B_1^{G*} + R_1 \frac{\epsilon_0 F_1^*}{\epsilon_1} \quad (67)$$

See Appendix B.2 for the derivation of this condition. The first term on the left-hand side represents the present discounted value of the carry cost associated with allowing interest parity deviations, i.e., allowing intermediaries to take non-zero asset positions in the domestic bonds market. This cost arises due to the assumption that intermediaries are owned outside the small open economy and must be compensated over and above the world interest rate for participating in the domestic bonds market. The second term represents the present discounted value of the import bill and first term on the right-hand side represents the present discounted value of the export revenues. The second and third terms represent inherited government and intermediary claims at $t = 1$.

4.1.9 Decentralized Equilibrium

Given the initial states: $\{R_1^M, B_1, B_1^{G*}, \epsilon_0, F_1^*, w_0\}$, exogenous sequence of import prices and productivity, $\{P_t^T, A_t\}_{t=1}^\infty$, and fiscal-monetary policy: $\{B_{t+1}^G, B_{t+1}^{G*}, R_{t+1}^M, \frac{T_t}{P_t}\}_{t=1}^\infty, R_1$, a decentralized equilibrium consists of an allocation: $\{C_t^T, C_t^P, B_{t+1}, N_t, M_t, Y_t, F_{t+1}^*, X_t\}_{t=1}^\infty$ and prices $\{\epsilon_t, w_t, \Pi_t, R_{t+1}\}_{t=1}^\infty$ that satisfy all optimality and market clearing conditions. We define the decentralized equilibrium and list all equations that characterize the equilibrium in Appendix B.3.

4.2 Optimal Policy Problem

Analogous to the approach in the analytical section, we consider the primal problem of a utilitarian Ramsey planner who directly chooses $\{N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^T, C_t^P\}_{t=1}^\infty$ to maximize welfare subject to implementability constraints implied by the decentralized equilibrium. The general Ramsey problem features a time-inconsistency problem which arises from the financial intermediation structure in the model: at $t = 1$, the claim due to the intermediaries in foreign units is given by $\frac{R_1^M}{\Pi_1} \frac{\epsilon_0 F_1^*}{\epsilon_1}$. Since the intermediaries lent in domestic currency units, Π_1 and ϵ_1 can be chosen to influence the ex-post foreign-units value of the inherited claim. For the quantitative exercise, we initialize the economy at the Ramsey steady state, in which A_t and P_t^T are constant over time. In that steady state, $R = R^*$ and hence $F_1^* = 0$. This initialization removes the planner's incentive at $t = 1$ to revalue an inherited intermediary position. However, it does not eliminate time inconsistency more generally: following a shock, the commitment allocation can generate nonzero intermediary positions that a future planner would inherit. We therefore interpret the solution to (114), stated in Appendix B.4, as a full-commitment Ramsey allocation initialized at the Ramsey steady state⁶.

Lemmas B.1, B.2, B.3 stated in Appendix B.4 together establish conditions under which the constrained efficient allocation, i.e., the solution to (114), is implementable as a decentralized equilibrium. Under these restrictions, the planner can choose N_t directly, and the wage Phillips curve specified by equations (54), (55) enters only in the decentralization. As in the analytical framework, choosing N_t directly chooses the optimal labor wedge. Inflation and the nominal interest rate path, $\{\Pi_t, R_{t+1}^M\}$ can be constructed ex-post to implement the constrained efficient allocation in a decentralized equilibrium. The planner chooses the intertemporal wedge through $\tilde{R}_{t+1}$. The quantitative model implements this wedge through public foreign exchange interventions. By choosing domestic and foreign asset positions, $\{B_{t+1}^G, B_{t+1}^{G*}\}$, the public agent implements the constrained efficient allocation with optimal intertemporal distortions. In the analytical framework the same wedge is represented by a direct capital control tax. Intertemporal distortions are costly here: the carry cost term in the intertemporal budget constraint, (67), that arises due to the financial intermediation friction represents a cost paid by the economy for departing from interest parity, a cost the capital control tax of the analytical framework does not carry.

⁶Given $F_1^* = 0$ and $\beta R^* = 1$, if $A_t = A > 0$ and $P_t^T = P^T > 0$ for all t , the solution to (114) is a steady state: $C_t^T = C^T$, $C_t^P = C^P$, $N_t = N$, $\epsilon_t = \epsilon$, and $\tilde{R}_{t+1} = R^*$. We verify this property numerically. The resulting allocation has $F_{t+1}^* = 0$ and $B_{t+1}^{G*} = B_1^{G*}$ in every period, so reoptimization reproduces the same allocation when the economy remains in this no-shock steady state. This stationarity property does not imply time consistency along a transition after a shock.

4.3 Quantitative Application: Energy Price Shocks and Optimal Policy

We have studied optimal policy in response to essential import price shocks in general terms; its leading application is fossil fuel price spikes in an energy-importing economy. Fossil fuels fit the theory's definition of an essential import. They are necessary to household and business energy needs, substitution away from them is limited in the short run, and their prices are set in US dollars in global markets that no single importer can influence, so a price spike raises an import-dependent country's energy expenditure almost in proportion to its energy intensity. The 2022 energy crisis brought Brent crude and European natural gas to multiples of their pre-pandemic levels ([International Energy Agency 2022](#); [U.S. Energy Information Administration 2026](#); [European Securities and Markets Authority 2023](#)), and the 2026 disruption to crude and LNG flows through the Strait of Hormuz reproduced the pattern within months ([International Energy Agency 2026](#)).

We calibrate the model to Japan. Japan is among the world's largest energy importers: it imports nearly 90 percent of its primary energy, was the largest importer of liquefied natural gas in 2022, and ranks fifth in global crude oil consumption with imports accounting for 97 percent of demand ([Agency for Natural Resources and Energy 2024](#)). In 2022, fossil fuel import bills rose to roughly 4 percent of GDP and contributed to the largest annual trade deficit on record at 20 trillion yen ([Nippon.com 2023](#)). Japan also fits the particular sense in which the model economy is small and open. It is small with respect to world asset markets and world energy markets: it takes world interest rates and dollar energy prices as given. It is not small in the market for its own exports: Japanese output is differentiated, and foreign demand for it is downward sloping, so earning additional foreign purchasing power within a period requires selling more cheaply.

Japan's experience during the 2022 spike illustrates the amplification the theory studies. The dashed line in the left panel of Figure 2 plots an index of world energy prices in US dollars, which peaked 172 percent above its January 2021 level. Over the same period the yen depreciated by 42 percent against the dollar, and Japan experienced unusually high inflation, with the CPI peaking 5.4 percent above its January 2021 level. The right panel combines the two: the CPI-deflated yen-dollar exchange rate, a measure of the real exchange rate, peaked 36 percent above its initial level. The solid line in the left panel plots the price of imported energy in units of the Japanese consumption basket: at the peak, imported energy cost 1.27 times what it would have cost had only the world price risen with no real depreciation⁷.

We use the calibrated economy to examine the assignment of the two instruments: what exchange rate policy should do and through which margin, what monetary policy should do conditional on the exchange rate policy, and whether either instrument can reproduce the other's role in its absence. Two imported-energy price paths serve these questions. The first is an MIT shock, a perfect-foresight impulse response to a doubling of the energy price that decays at its estimated AR(1) persistence, which isolates the two roles. The second is a historical simulation along the realized path of world energy prices through the 2022 crisis, a deterministic hump-shaped sequence over 2021-2023 that peaks 157 percent above the pre-shock level,

⁷The figure is descriptive, and we do not attribute the real depreciation, in whole or in part, to the energy price spike. Under a downward-sloping export demand, however, a real depreciation accompanying an essential import price spike is what the theory predicts.

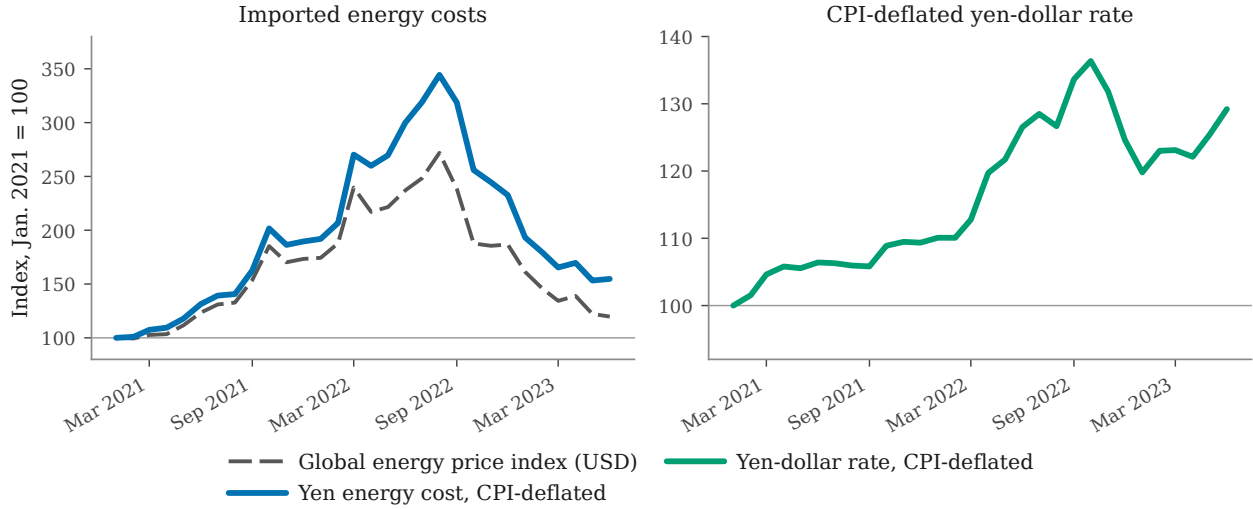


Figure 2. Energy prices and the yen, 2021-2023

Note: The left panel plots the IMF global energy price index in U.S. dollars and the same index converted into CPI-deflated yen. The right panel plots the CPI-deflated yen-dollar exchange rate. Monthly series are normalized to 100 in January 2021 and run through June 2023. Sources: IMF global energy price index and yen-dollar exchange rate from FRED; Japan CPI from the Statistics Bureau of Japan.

which gauges their quantitative importance. The exercise asks whether restricted asset market participation materially amplifies the domestic consequences of an energy price shock and how each instrument should respond. The model calibration is described in Appendix B.5.

We compare four policy regimes, each defined by the question it answers. The RA-equivalent benchmark sets both the intertemporal wedge and the terms-of-trade-adjusted labor wedge to zero: it applies the policy corrections that are optimal without household heterogeneity. The interest-parity Ramsey policy chooses the labor wedge freely while suppressing the intertemporal wedge; it asks what monetary policy can accomplish when it cannot change the timing of foreign purchasing power. The FX-only policy chooses the intertemporal wedge freely while suppressing the terms-of-trade-adjusted labor wedge; it asks what exchange rate policy can accomplish when employment is not independently managed. The optimal policy chooses both.⁸

The representative-agent economy of Appendix B.6, in which no household is borrowing constrained and every household is on its Euler equation, fixes the reference point. Figures B.1 and B.2 report its allocation and implementation paths under the MIT shock. The two instruments separate along the lines of *Itskhoki and Mukhin (2023)*: exchange rate policy is assigned to the financial friction, and monetary policy to the frictions in domestic production. The public agent's foreign asset position takes on the financial flows that intermediaries would otherwise price, so interest parity holds, no carry losses are incurred, and the

⁸We do not present laissez-faire policies in the comparison. A laissez-faire intertemporal policy is dominated even without household heterogeneity: private financial flows intermediated under the friction open interest parity gaps and carry losses that the public balance sheet can undo at no cost, which is why optimal policy in the representative-agent economy discussed below restores interest parity; *Itskhoki and Mukhin (2023)* characterize optimal exchange rate and monetary policy in a representative-agent economy with this class of financial frictions. The comparison here therefore focuses on how heterogeneity changes the role of policy beyond what representative-agent models already establish.

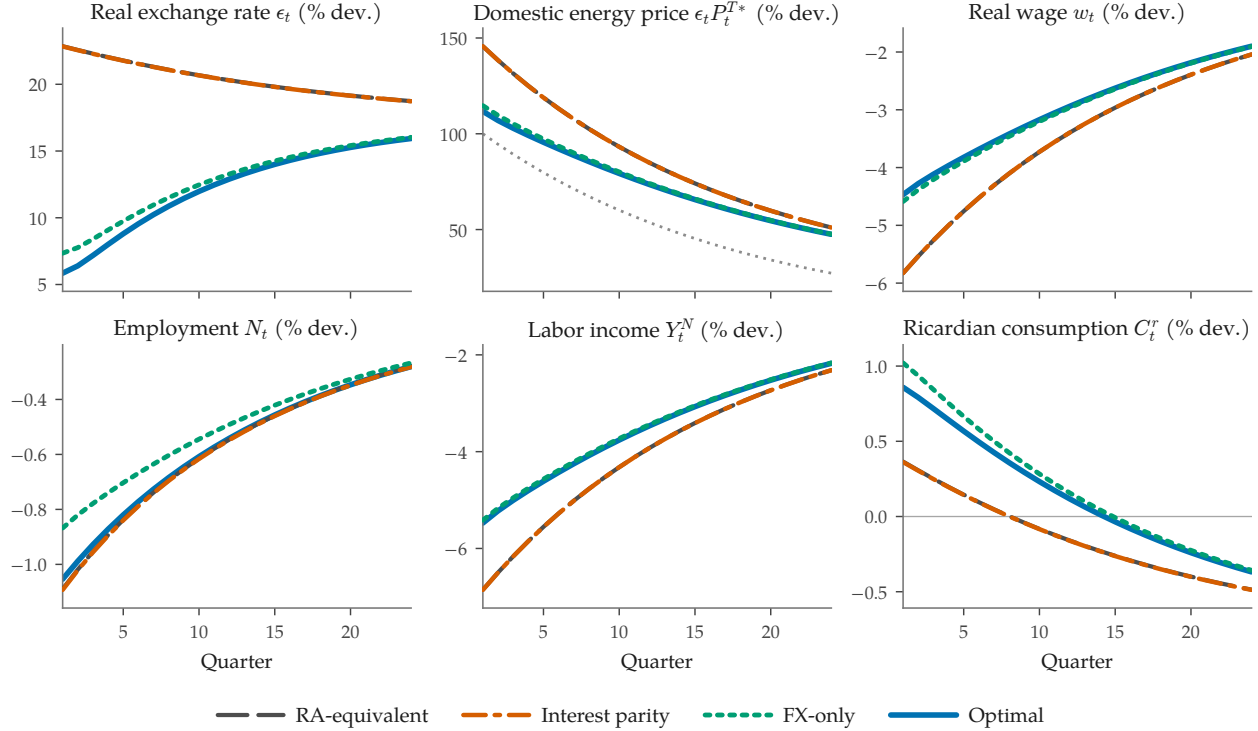


Figure 3. MIT shock: allocations and relative prices

Note: Percent deviations from the initial steady state. The dotted line is the world energy price.

intertemporal margin is left undistorted; monetary policy targets nominal rigidities and, in this economy, also corrects the static terms-of-trade distortion. With a mass of constrained households, optimal policy no longer uses the public balance sheet to undo the intermediation friction: it opens interest parity gaps deliberately, and pays the carry losses that the friction attaches to them, because the gap is what tilts the path of the real exchange rate and, through firms' costs and the zero-profit condition, the real wages and current incomes of constrained households.

The calibrated economy reproduces the amplification that the analytical framework establishes in closed form. Under the RA-equivalent benchmark the real exchange rate depreciates as the energy price rises, and the domestic real price of energy rises 1.46 percent for every percent rise in the world price (Figure 3). Real wages, employment, and labor income fall. Since HtM households consume their current labor income, the income loss passes directly into their consumption. Constrained households cannot borrow against future income, and the economy therefore relies more heavily on contemporaneous export revenue to finance the import bill. Expanding exports requires a real depreciation, which further raises the domestic energy price and compresses real wages. In the representative-agent economy of Appendix B.6, the same shock produces a smaller depreciation: 40 percent of the benchmark's impact depreciation reflects the borrowing that constrained households cannot undertake. Ricardian consumption instead rises on impact as households with access to asset markets borrow against the temporary loss, and then falls as the resulting foreign obligation is repaid.

The interest-parity Ramsey policy isolates what monetary policy achieves on its own. Suppressing the

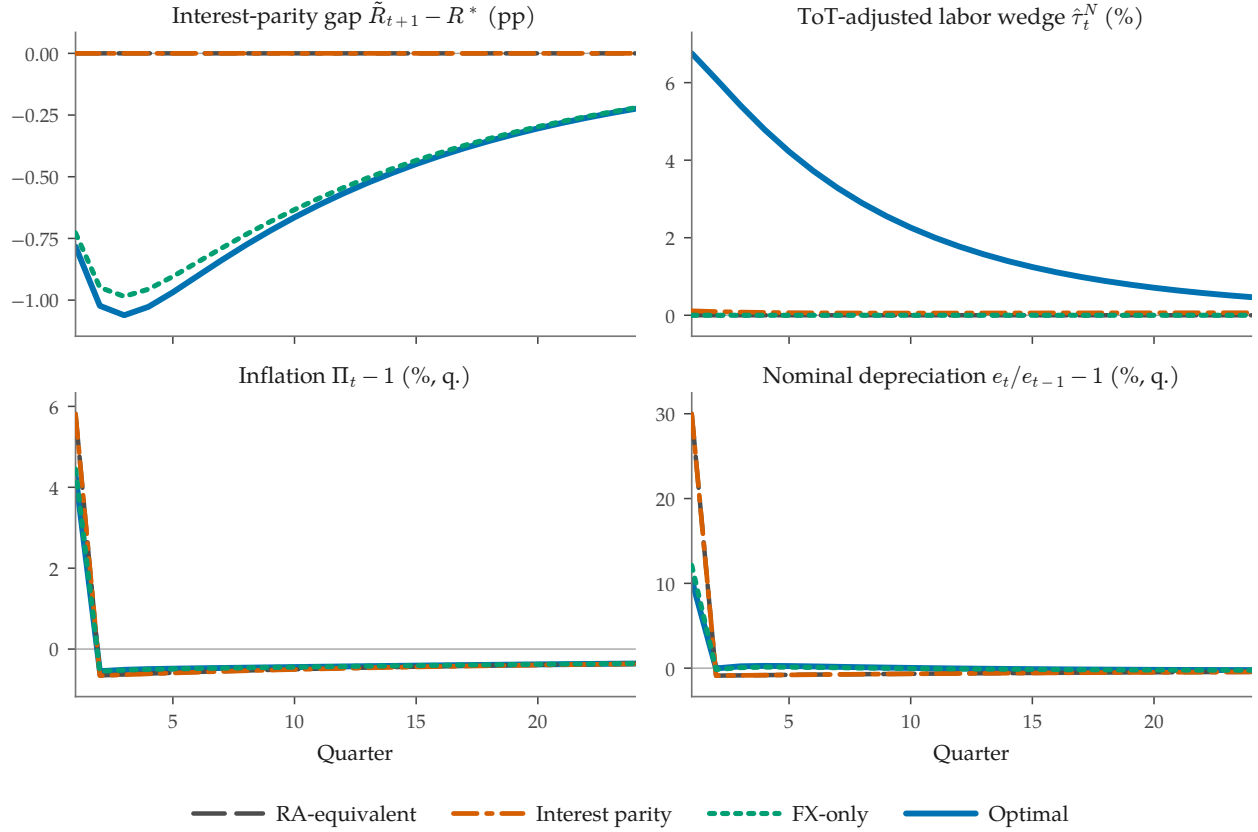


Figure 4. MIT shock: policy and implementation

intertemporal wedge leaves the Ricardian Euler equation undistorted, and it enters the Ramsey planner's problem as an additional constraint: the timing of the economy's foreign borrowing is then the one private agents choose at the world interest rate. Monetary policy keeps the labor wedge, and because the trade balance is nonzero while the energy price is high, the revaluation argument behind Proposition 4 continues to hold. The public agent sets a strictly positive terms-of-trade-adjusted labor wedge, restraining employment because an expansion would raise imported-input demand, depreciate the currency, and shift purchasing power toward Ricardian households through the revaluation of their financial flows. That restraint acts within the period. It cannot bring foreign purchasing power forward from the dates at which the necessity is cheap, and it is that re-timing which cuts the excess co-movement of the real exchange rate with the shock. The wedge the public agent selects is small, and both the inflation path that implements it and the allocation stay close to the RA-equivalent benchmark's. The separation of the two margins that Section 3 establishes in closed form reappears here at the calibrated parameters.

Exchange rate policy operates on the margin monetary policy cannot reach. The FX-only policy borrows abroad in the high-price quarters, and the borrowed foreign purchasing power replaces current export revenue; the real exchange rate appreciates relative to interest parity, the domestic cost of the imported input falls, and the real wage and labor income rise. This is most of the contemporaneous income stabilization that the joint optimum achieves. It is also most of the disinflation. Under the RA-equivalent benchmark,

the depreciation lowers the real wage through firms' costs, and with nominal wages costly for unions to move, the required fall arrives through a rising price level: the depreciation is imported as inflation. By holding the currency stronger, the intertemporal policy removes this imported component at its source. However, exchange rate policy alone does not determine employment or the division of the gain between wages and hours. With the labor wedge suppressed, the cheaper imported input induces firms to expand: employment and imports rise, financing the additional imports requires export revenue, and earning it gives back part of the appreciation. The FX-only policy therefore delivers slightly higher labor income than the joint optimum, through additional hours at a lower real wage. The resulting currency is weaker than under the joint optimum; the weaker currency revalues the borrowed foreign resources in favor of Ricardian households, and their consumption in Figure 3 is highest in this regime. These responses reverse as the foreign position is repaid. Exchange rate policy therefore changes the timing and incidence of the imported-resource loss; it does not remove that loss.

Monetary policy's role is defined conditional on the intertemporal policy: alongside it, the labor wedge determines how much of the appreciation survives and in what form its gains arrive. The appreciation lowers firms' marginal cost, and an accommodating monetary stance would let employment and imports expand in response; financing the additional imports requires export revenue, and earning it would give back part of the appreciation the intertemporal policy has bought. The expansion's depreciation would also revalue the borrowed positions in favor of Ricardian households in the quarters in which constrained households' marginal utility is highest. Optimal monetary policy therefore implements a lower inflation path than the FX-only policy: through the wage Phillips curve, lower inflation holds nominal wage growth and hours down, and employment stays below the FX-only path. Inflation enters the policy problem only through what it implements: the disinflationary stance is valuable for the employment path it delivers and for how that path distributes resources. Through the zero-profit condition, $w_t = A_t - \epsilon_t P_t^T \theta$, the lower domestic imported-input cost then appears as a higher real wage rather than primarily as additional hours. The intertemporal policy thus determines the size of the cost saving, while the monetary stance determines its composition and its recipients: wages rather than hours, and labor income for all households rather than revaluation gains for the unconstrained. Figure 4 displays the two policy margins. The optimal policy opens a negative interest parity gap, the decentralized form of the positive intertemporal wedge, and implements a terms-of-trade-adjusted labor wedge many times larger than the one the interest-parity policy selects, 6.8 percent against 0.11 percent on impact: chosen alongside the intertemporal policy, the same margin protects the appreciation and determines whether the cost saving arrives as wages or as hours. The nominal path that implements this allocation features lower inflation and less front-loaded nominal depreciation than the RA-equivalent benchmark.

Figure B.5 in Appendix B.7 reports the output reallocation under the MIT shock. Relative to the RA-equivalent benchmark, borrowed foreign purchasing power replaces export revenue in the high-price quarters, the export share falls, and more output remains for domestic consumption. The appreciation lowers the import-bill share, while the higher real wage raises the HtM consumption share. The Ricardian share also rises slightly, a response that Proposition 3 leaves ambiguous in the high-price period. As

Table 1: Analytical predictions under the MIT shock

Prediction	RA-equivalent	Optimal
<i>Intertemporal response (Prop. 2)</i>		
Real-exchange-rate volatility (%)	4.9	10.0
Real exchange rate, terminal steady state (% dev.)	17.09	17.61
Trade balance, high-price quarter (% of s.s. export revenue)	-106.5	-110.4
Trade balance, terminal steady state (% of s.s. export revenue)	4.06	4.18
<i>Domestic price and labor income (Prop. 2)</i>		
Volatility of domestic energy price (%)	109.8	80.0
Domestic energy price, terminal steady state (% dev.)	17.09	17.61
Volatility of labor income (%)	6.18	4.77
Labor income, terminal steady state (% dev.)	-0.72	-0.74
Volatility of the real wage (%)	5.18	3.79
<i>Output shares (Prop. 3)</i>		
Export share, high-price quarter (%)	4.40	3.65
Export share, terminal steady state (%)	4.10	4.12
HtM consumption share, high-price quarter (%)	31.69	32.15
HtM consumption share, terminal steady state (%)	33.424	33.417
Ricardian consumption share, terminal steady state (%)	62.48	62.46

the foreign position is repaid, the export and import-bill shares rise relative to the benchmark and both household consumption shares fall.

In Table 1 we set the analytical predictions beside the calibrated responses under the MIT shock, and each of them holds. Volatility is measured as the maximum absolute deviation from the terminal steady state, which keeps the permanent component that all regimes share out of the comparison. To stabilize the domestic price of energy and labor income, optimal policy makes the real exchange rate more volatile, while the domestic energy price, the real wage, and labor income all become less volatile: the exchange rate is the instrument of stabilization rather than its object. The high-price responses in Figure 3 and the terminal rows of the table show the same pattern in levels: the currency is stronger, the domestic energy price lower, and labor income higher while the import price is high, with each ranking reversed after the shock. The trade balance likewise moves from a larger deficit in the high-price quarter to a larger surplus later. The export and HtM shares move in the directions stated in Proposition 3, including the lower HtM and Ricardian shares after the shock. Finally, the negative interest parity gap and the positive terms-of-trade-adjusted labor wedge in Figure 4 verify the policy signs in Propositions 2, 4, and 6.

We run the same comparison along the realized 2021-2023 path in Figure 5, with the representative-agent reference paths for that price sequence in Figures B.3 and B.4 of Appendix B.6. Relative to that reference, the borrowing that constrained households cannot undertake accounts for 71 percent of the RA-equivalent benchmark's peak depreciation. Under the benchmark, every percent rise in the world price of energy raises its domestic real price by 1.30 percent at the peak, close to the 1.27 in the Japanese data of Figure 2. Under the optimal policy the factor falls to 0.92: exchange rate policy more than eliminates the depreciation-driven amplification, and at the peak the domestic real price of energy is 19.5 percent lower than under the benchmark. The prescribed path appreciates the currency through the run-up, places the strongest currency, above its pre-shock level, at the peak of the world price, and depreciates it afterward as the foreign position

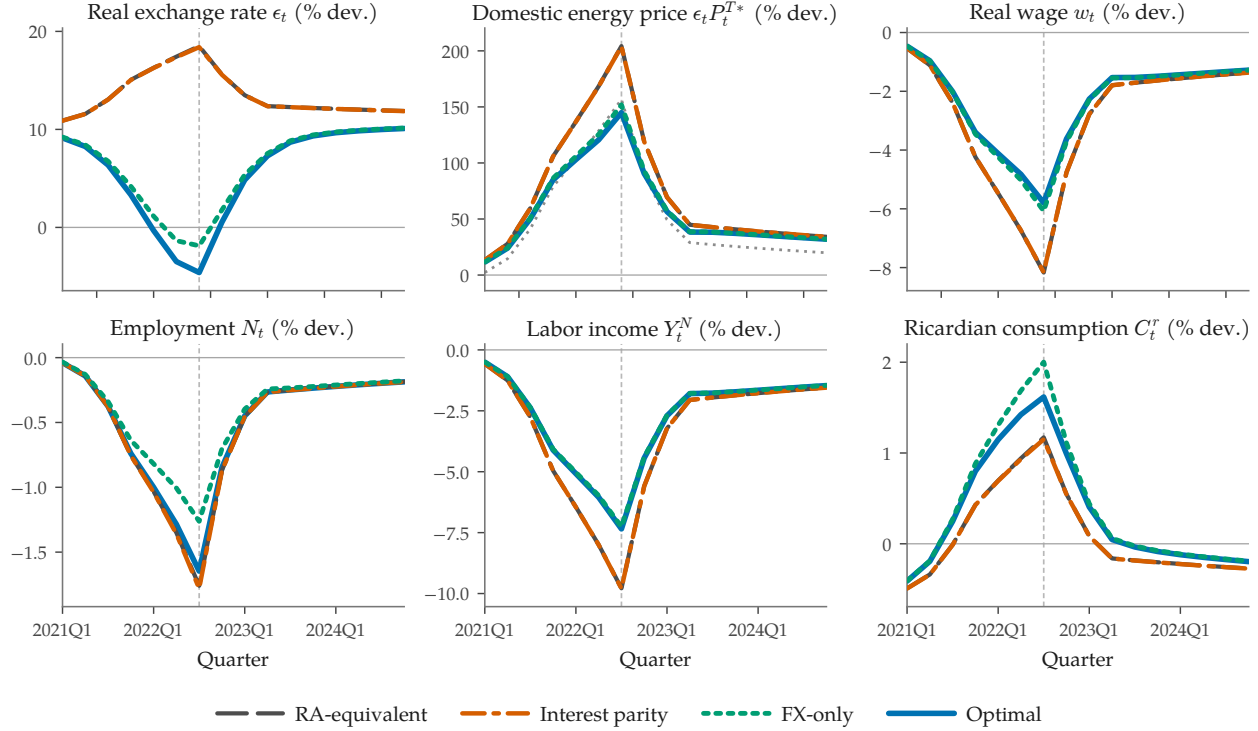


Figure 5. 2021-2023 shock: allocations and relative prices

Note: Percent deviations from the initial steady state. The dotted line is the world energy price; the vertical line marks its peak.

is unwound. The optimal policy undoes 29 percent of the benchmark's peak real-wage loss and 25 percent of its peak labor-income loss. Employment remains below the FX-only path, and the income protection therefore comes through a larger real wage rather than through additional hours. Ricardian consumption rises as foreign purchasing power is brought into the high-price quarters and falls as the position is repaid.

We connect the two policy margins to their observable implementation in Figure 6. Net capital inflows build as the energy price rises and recede after its peak: foreign purchasing power replaces export revenue when the import bill is high and the position is unwound afterward. The interest parity gap is positive early in the run-up and turns negative as the price approaches its peak, tilting the FX interventions toward the quarters in which foreign purchasing power is most valuable. The terms-of-trade-adjusted labor wedge rises with the import price and restrains employment and imported-input demand in the quarters when exchange rate policy holds the currency strong. Monetary policy is contractionary relative to the pre-shock steady state, as it is under every policy regime we consider: the nominal rate rises to 3.7 percent during the run-up, eases to 1.8 percent by the peak quarter, and stays above its pre-shock level throughout, falling further afterward as inflation recedes. Relative to the RA-equivalent benchmark, which sets the terms-of-trade-adjusted labor wedge to zero, the stance is tighter: inflation is lower through the run-up, peak-quarter inflation by 0.5 percentage points, and the domestic price level at the peak is 2.5 percent below the benchmark's. The peak-quarter rate nonetheless lies below the benchmark's 2.5 percent because it implements lower inflation; the labor wedge and inflation path identify the relative stance. Following a large initial depreciation, the nominal depreciation rate turns negative through the early run-up and positive again around the peak as

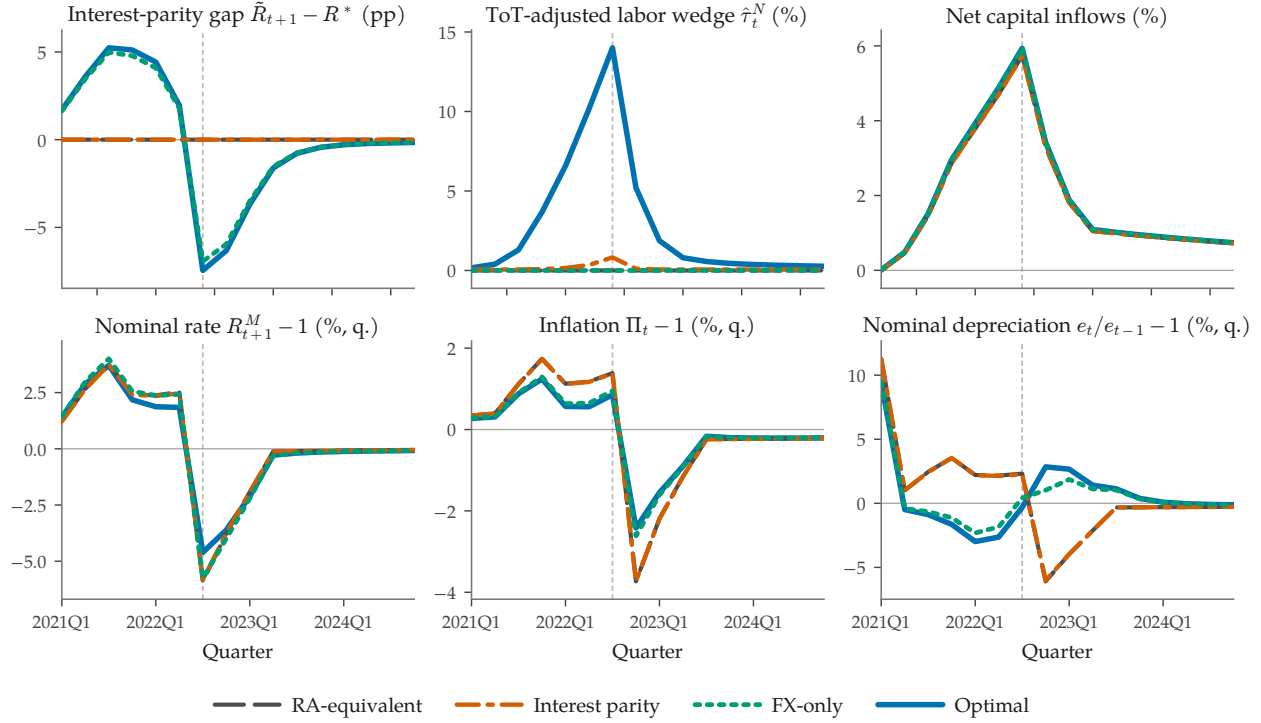


Figure 6. 2021-2023 shock: policy and implementation

Note: Net capital inflows are shown as a percent of initial steady-state gross output. The vertical line marks the energy-price peak.

the foreign position starts to unwind.

Figure B.6 of Appendix B.7 reports the corresponding output shares. Relative to the RA-equivalent benchmark, borrowed foreign purchasing power lowers the export share during the high-price quarters and leaves more output for domestic consumption. The appreciation lowers the import-bill share, while the higher real wage raises the HtM consumption share. The Ricardian share rises while foreign resources are brought forward. These differences recede as the energy price falls and the foreign position is unwound.

Table 2: Policy effectiveness and implementation

	RA-equivalent	Interest parity	FX-only	Optimal
<i>Panel A. MIT shock; impact quarter</i>				
Domestic energy-price amplification factor	1.46	1.46	1.15	1.12
Real-wage loss undone relative to RA-equivalent (%)	0	0	21	23
Labor-income loss undone relative to RA-equivalent (%)	0	0	21	20
Inflation reduction relative to RA-equivalent (pp)	0.00	0.01	1.37	1.51
Interest parity gap (pp)	0.00	0.00	-0.73	-0.78
Terms-of-trade-adjusted labor wedge $\hat{\tau}^N$ (%)	0.00	0.11	0.00	6.76
<i>Panel B. 2021–2023 energy-price path; peak quarter</i>				
Domestic energy-price amplification factor	1.30	1.30	0.97	0.92
Real-wage loss undone relative to RA-equivalent (%)	0	0	26	29
Labor-income loss undone relative to RA-equivalent (%)	0	0	26	25
Inflation reduction relative to RA-equivalent (pp)	0.00	0.01	0.43	0.53
Interest parity gap (pp)	0.00	0.00	-6.93	-7.45
Terms-of-trade-adjusted labor wedge $\hat{\tau}^N$ (%)	0.00	0.82	0.00	14.00

Table 2 collects the effectiveness measures and the policy wedges across regimes and both price paths. Under the MIT shock, the joint optimum lowers the amplification factor to 1.12 and undoes 23 percent of the RA-equivalent benchmark’s impact real-wage loss and 20 percent of its impact labor-income loss, with the domestic energy price 13.8 percent cheaper and the price level 1.4 percent lower than the benchmark at impact. The labor-income improvement is slightly smaller than under the FX-only policy because the positive labor wedge replaces part of the hours response with a larger real-wage response. Along the realized 2021-2023 path, these prescriptions constitute a full-commitment counterfactual: intervene to hold the currency stronger and restrain activity while the shock builds, then unwind as it fades.

The characterization of Section 3 therefore survives at a disciplined calibration and in an infinite-horizon economy whose instruments are the ones policy authorities operate: the signs of both wedges, the direction of the exchange rate path, and the division of the income gain between wages and hours all carry over. The magnitudes along a price path that an energy importer faced are large enough that the assignment of the two instruments carries consequences for domestic energy prices, real wages, and inflation. Both the analytical and the quantitative model rest on a two-agent structure, and Section 5 asks how far the results reach once that structure is relaxed.

5 Beyond the Two-Agent Model

The two-agent model delivers the real-income stabilization channel in its most transparent form: a fixed group of households consumes out of current labor income, and the public agent holds lump-sum instruments that finance its interventions without redistributing resources across households. In this section we remove both simplifications in turn. We first withdraw lump-sum transfers and taxes: all interventions must be financed through the public balance sheet at market prices. We then replace the fixed hand-to-mouth group with a continuum of households that self-insure against idiosyncratic expenditure risk subject to a borrowing limit: who is constrained, and for how long, becomes an equilibrium outcome. Once households differ across a full asset distribution, lump-sum taxes and transfers are redistributive instruments in their own right, and isolating what policy achieves through pecuniary general equilibrium effects alone requires their absence.

5.1 A Two-Agent Model without Lump-Sum Transfers

The two-agent model that we have considered so far assumes that the public agent is able to selectively allow Ricardian households to receive lump-sum transfers or be subject to lump-sum taxes. Lemma B.1 in Appendix B.4 shows that only the present discounted value of these transfers matters for implementing the Ramsey equilibrium, not their timing. This is a version of Ricardian equivalence, and it leaves the public agent’s and the Ricardian households’ positions in the domestic bond market undetermined. Removing lump-sum transfers and taxes eliminates this indeterminacy, and the Ricardian households’ present-value budget becomes an additional implementability constraint on the Ramsey planner. We couple this restriction with the assumption that assets in domestic markets are real rather than nominal claims, which removes

the time-inconsistency problem arising from the public agent's ex-post incentive to use inflation to revalue Ricardian claims, an incentive that was absent in the framework with lump-sum transfers. Monetary policy is then represented directly by a money-growth rule. In Appendix C.1 we present the framework and the Ramsey planner's problem under this restriction, show that its solution is implemented by the public agent's asset positions and the money-growth rule, and solve the problem at the quantitative calibration.

Removing transfers also moves the benchmark against which heterogeneity is measured. The Ricardian households' present-value budget prevents the labor wedge from carrying the whole terms-of-trade correction on its own. Consequently, the representative-agent economy under the same restriction distorts both margins as well. We measure the contribution of restricted asset market access as the difference between the $\lambda = 0$ and $\lambda > 0$ economies under the MIT shock of Section 4, and that difference reproduces the assignment of the baseline model (Figure C.1). The public agent holds the interest parity gap lower and the currency stronger while import prices are high, which lowers the domestic price of the import and raises the real wage in the quarters in which hand-to-mouth marginal utility is highest. Monetary policy sets a larger labor wedge and delivers lower inflation than in the $\lambda = 0$ economy, keeping an expansion of employment and imports from undoing the appreciation. The real-income stabilization channel therefore operates when all interventions must be financed through the public balance sheet at market prices.

5.2 A Heterogeneous-Agent Economy with Idiosyncratic Expenditure Risk

The two-agent framework fixes the boundary between constrained and unconstrained households: the hand-to-mouth share is a parameter, membership in the group is permanent, and no household can prepare for a shock by accumulating assets or finance it by drawing assets down. We ask whether the real-income stabilization motive survives once households can self-insure. We replace that structure with an incomplete-markets economy in the tradition of Bewley (1986), Huggett (1993), and Aiyagari (1994). A unit mass of households faces idiosyncratic risk and trades risk-free bonds subject to a common borrowing limit. Because saving choices determine who reaches the limit, the constrained share and the marginal utilities of households at the limit respond to the aggregate disturbance and to policy itself.

5.2.1 Environment and Equilibrium

There is a unit measure of households indexed by $i \in [0, 1]$. Household i faces an idiosyncratic autonomous consumption requirement z_t^i , which follows a Markov process

$$z_{t+1}^i \sim Z(\cdot \mid z_t^i)$$

with invariant distribution π_z satisfying $\int z d\pi_z = \bar{z} > 0$. The initial cross-sectional distribution of z_1^i is π_z : the idiosyncratic requirement therefore carries no aggregate risk, and its cross-sectional mean remains $\bar{z}$. The requirement z_t^i uses resources without entering utility, and discretionary consumption c_t^i remains the argument of utility.

All households participate in the domestic bond market, where the bond pays the gross real return R_{t+1} in

units of the domestic good. We write $b_{t+1}^i \geq 0$ for household i 's distance from the borrowing limit, i.e., its bond position is $b_{t+1}^i + \underline{b}$. As in the model without transfers, lump-sum taxes and transfers remain absent and domestic bonds are indexed to inflation: surprise inflation cannot revalue household claims. Unions set the nominal wage W_t as in the baseline quantitative model, and every household supplies the same N_t hours and earns the same real wage w_t per hour. A high realization of the requirement compresses discretionary consumption, and for a household at the borrowing limit it raises the marginal utility of current income, the state in which protection of the real wage is most valuable.

Household i solves

$$\begin{aligned} \max_{\{c_t^i, b_{t+1}^i\}_{t \geq 1}} \quad & \mathbb{E}_1 \sum_{t=1}^{\infty} \tilde{\beta}^{t-1} [u(c_t^i) - v(N_t)] \quad \text{s.t.} \\ c_t^i + b_{t+1}^i = & w_t N_t - z_t^i + R_t b_t^i + (R_t - 1)\underline{b} \\ b_{t+1}^i \geq & 0 \end{aligned} \quad (68)$$

The household first-order conditions are

$$u'(c_t^i) \geq \tilde{\beta} R_{t+1} \mathbb{E}_t u'(c_{t+1}^i), \quad b_{t+1}^i \left[u'(c_t^i) - \tilde{\beta} R_{t+1} \mathbb{E}_t u'(c_{t+1}^i) \right] = 0 \quad (69)$$

The Euler inequality holds with equality whenever the household is unconstrained. Households discount future utility at $\tilde{\beta} = s\beta$, where $s \in (0, 1)$ and $\beta = 1/R^*$ ⁹.

Policy follows the no-transfers economy of Section 5.1 and Appendix C.1: the monetary authority sets money growth g_t^M , and the public agent conducts foreign exchange intervention through its domestic and foreign asset positions $\{B_{t+1}^G, B_{t+1}^{G*}\}$, subject to its budget constraint and no-Ponzi conditions. Production, export demand, financial intermediation, labor aggregation, and nominal wage setting remain as in the baseline quantitative model, and Appendix C.2.1 states the complete decentralized equilibrium.

5.2.2 Optimal Policy under Timeless Commitment

The public agent chooses $\{g_t^M, B_{t+1}^G, B_{t+1}^{G*}\}$ under commitment to maximize utilitarian welfare subject to the decentralized equilibrium conditions in Appendix C.2.1. Incomplete markets add a distributional source of time inconsistency: once households have chosen their asset positions, a utilitarian public agent that re-optimizes would lower the previously announced real return in favor of indebted households with high marginal utility. [Dávila and Schaab \(2023\)](#) encode the corresponding promise in forward-looking individual Bellman equations. Our discrete-time sequence formulation carries it through household Euler constraints. In their closed economy monetary policy implements the desired movement in real returns, producing an inflationary bias under discretion. Their forward-looking Phillips curve supplies a second time-zero promise that remains with a representative household. Domestic claims are indexed, so Fisherian revaluation is

⁹We impose $\tilde{\beta} R^* < 1$ for a stationary household asset distribution and $\beta R^* = 1$ for an interior stationary public foreign asset position. A perpetual-youth structure provides a microfoundation. A household survives into the next period with probability s , and a newborn household replaces an exiting household and inherits its assets. Conditional on survival, households discount at β , and their sequence problem therefore has the effective discount factor $\tilde{\beta} = s\beta$. The utilitarian objective discounts at β and treats successive generations within a household lineage as one perpetually lived household.

absent from our Ramsey problem. The public balance sheet instead links the household return path to the real exchange rate and foreign exchange intervention.

In Appendix C.2.2 we derive both sources of time inconsistency and formulate commitment from a timeless perspective, following Marcet and Marimon (2019). The inherited household multipliers record the Euler conditions under which asset positions were chosen, and the inherited intermediary multiplier records the foreign exchange condition under which its position was chosen. Conditional on these promises, re-optimization reproduces the continuation of the earlier plan. We apply the inherited promises at the initial date as well, removing the one-time incentive to exploit predetermined positions before commitment begins. We show in Lemma C.2 that a solution satisfying the stated conditions is implemented by the monetary and foreign exchange instruments of the decentralized economy, with unique public asset positions.

5.2.3 Quantitative Results

With the timeless problem in place, we study the perfect-foresight response to an unanticipated 25 percent increase in the import price that subsequently decays at an AR(1) rate. The economy begins at the Ramsey steady state characterized in Appendix C.2.4, which supplies the initial distribution, the public balance-sheet positions, and the inherited promise multipliers that the timeless formulation requires. We choose the expenditure process to match a stationary constrained share of 35 percent and a one-quarter probability of remaining at the borrowing limit of 97 percent, the moments that discipline the size and persistence of the constrained region. Appendix C.2.5 reports the complete set of calibrated parameter values and targets.

The transition requires tracking the promises attached to households' Euler conditions alongside their asset positions, and those promises are history dependent: carried literally, they would enter as a third idiosyncratic state. In Appendix C.2.3 we show that they need not. Each household's promise is affine in the promise it inherits, with coefficients that depend only on its assets and its current expenditure requirement, and the public agent's optimality conditions involve promises only through their conditional mean on those two states. The policy problem therefore remains tractable in a state space with two idiosyncratic states, and we solve the resulting nonlinear system using the extended-path method described in Appendix C.2.6.

To isolate the real-income stabilization channel, we compare optimal policy with a counterfactual in which the public agent, when setting the real exchange rate, does not internalize the effect of the induced real wage on household consumption beyond its steady-state level, while households and the distribution respond in equilibrium throughout. We refer to this counterfactual as *policy without the real-income stabilization motive*. Appendix C.2.3 states its exchange-rate condition.

The real-income stabilization motive moves the exchange rate when households can self-insure and membership in the constrained group is an equilibrium outcome. A public agent without it lets the currency depreciate with the shock: every percent rise in the world import price raises its domestic price by 2.26 percent on impact. Optimal policy resists the depreciation while the import price is high: the same rise in the world price now raises its domestic price by 1.68 percent (Figure 7). The stronger currency lowers the unit cost of the imported input, and firms' zero-profit condition passes the saving into the real wage per

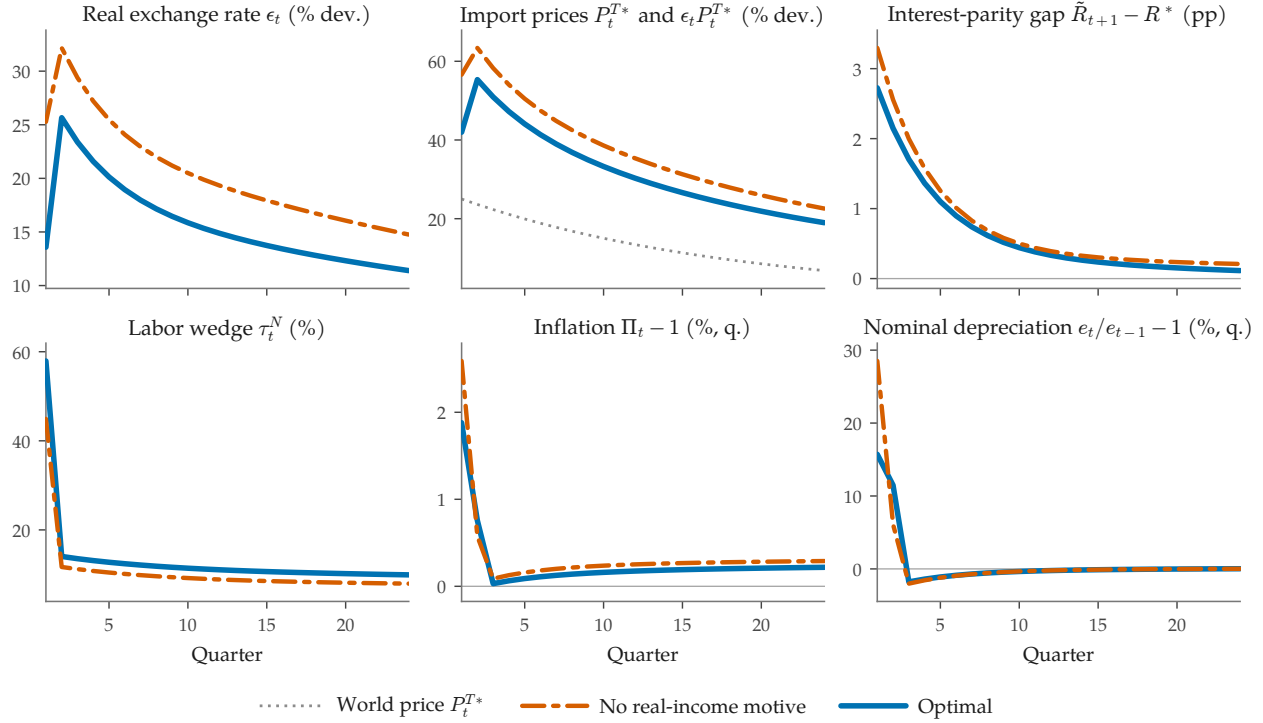


Figure 7. MIT shock in a heterogeneous-agent economy: relative prices and policy

hour, undoing 26 percent of the counterfactual's impact real-wage loss.

Holding the currency stronger while the import price is high requires additional external borrowing. Optimal policy finances the appreciation with a wider trade deficit during the high-price quarters. As the shock fades the pattern reverses: the currency ends more depreciated than under the counterfactual, and the additional exports repay 99.4 percent of the additional borrowing in present value, with the remainder the intermediation cost of the parity gaps that delivered it. The country therefore exports less while the necessity is expensive and more once its price has receded, the same reallocation of when it pays for its imports that the baseline model chose, now chosen against a full distribution of household positions.

Monetary policy supports exchange rate policy with a tighter stance: relative to the counterfactual it delivers lower impact inflation and implements a larger labor wedge. As in the two-agent model, much of the difference originates with the intertemporal policy itself. The counterfactual lets the currency depreciate with the shock; through firms' costs the depreciation lowers the real wage, and since nominal wages move slowly, the fall arrives as impact inflation. Optimal policy holds the currency stronger, and the smaller required fall in the real wage takes less inflation to deliver. The monetary stance adds the employment margin. The appreciation lowers the imported-input cost, and lower inflation, through the wage Phillips curve, keeps employment and imported-input demand from expanding in response; the cost reduction therefore passes into the real wage per hour rather than into an employment and import expansion. With inflation and the real depreciation both smaller on impact, the nominal depreciation is smaller on impact as well, and larger in the following quarter, when the deferred real depreciation arrives.

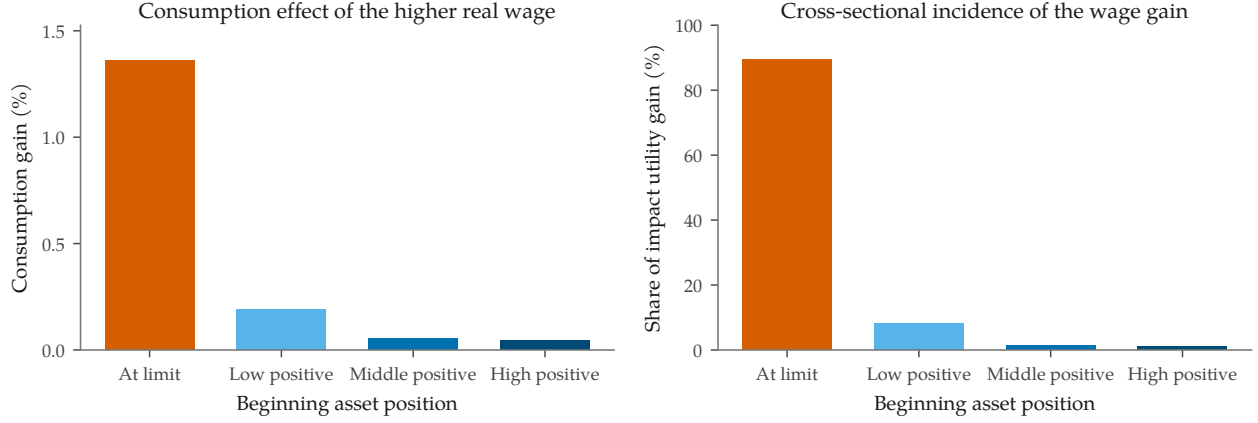


Figure 8. Heterogeneous-agent economy: cross-sectional incidence of the optimal-policy real wage

Note: The first group begins at the borrowing limit; households with positive assets are divided into three equal-population groups.

The aggregate responses do not identify who receives this protection: aggregate discretionary consumption falls by more on impact under optimal policy, because the tighter stance restrains hours, and the gains and losses fall on different parts of a distribution that the two-agent model divides between two types by assumption. Because the channel delivers its gains through the real wage, we isolate that margin directly with a wage-only incidence decomposition: starting from the counterfactual impact allocation, each household receives the optimal-policy real wage while employment, returns, and the distribution stay fixed, and the induced changes in consumption and current utility are attributed to households by their initial asset position (Appendix C.2.8).

A wage increase common to all households is most valuable to those who cannot borrow against the future. Households at the borrowing limit, roughly a third of the population, collect 83 percent of the consumption response and 90 percent of the utility gain, and the response fades with assets because households away from the limit smooth the same income through saving (Figure 8). Membership in the constrained region

Table 3: Heterogeneous-agent economy: impact responses

	No real-income motive	Optimal
<i>Panel A. Aggregate responses; impact quarter</i>		
Real exchange rate (% dev.)	25.3	13.6
Amplification factor (domestic per world price)	2.26	1.68
Real wage (% dev.)	-2.26	-1.68
Employment (% dev.)	-3.09	-5.06
Trade balance (% of s.s. export revenue)	-19.1	-22.9
Aggregate discretionary consumption (% dev.)	-7.5	-9.5
Interest-parity gap (pp)	3.30	2.73
Labor wedge τ_t^N (%)	44.9	57.9
Inflation (% , quarterly)	2.59	1.88
Nominal depreciation (% , quarterly)	28.5	15.7
Real-wage loss undone relative to counterfactual (%)	0	26
<i>Panel B. Distributional responses; impact quarter</i>		
Households choosing the borrowing limit (%)	35.6	35.9
Persistence conditional on beginning at the limit (%)	97.1	97.1

is an equilibrium outcome. On impact, however, the shock and the policy response leave its size and its persistence nearly unchanged, with 1.9 percent of households entering the region and 1.0 percent leaving it. The gains therefore operate through the marginal utilities of households already at the limit. Table 3 collects the impact responses. The intertemporal policy determines the size of the cost saving, and the monetary stance determines the form in which households receive it, as earnings per hour rather than longer hours. Asset positions then determine who values the gain most, with the largest gains accruing to households with the least scope for self-insurance.

6 Conclusion

This paper has studied how exchange rate and monetary policy should be jointly conducted by economies that depend on essential imports and face temporary shocks to their world prices. Conventional policy prescriptions, which rely on representative-agent models, assign monetary policy to target nominal distortions and argue for letting the exchange rate float freely with no distortions to private financial flows. When some households are borrowing constrained and transfers cannot reach them, that assignment need no longer hold: the economy borrows too little during the spike, the real exchange rate depreciates contemporaneously with import prices, and the amplified real income loss falls hardest on households who consume out of current incomes.

We have argued that exchange rate policy then takes on a role it does not possess in representative-agent economies: stabilizing the real incomes of borrowing-constrained households. A public agent that distorts the intertemporal margin to support stronger capital inflows while import prices are high, and unwinds them afterward, cuts the excess co-movement of the real exchange rate with the shock: less must be exported while the necessity is expensive, so the depreciation is smaller at the peak and larger once the shock has passed. The gains reach constrained households through pecuniary general equilibrium effects of exchange rate movements: the stronger currency lowers the domestic price of imports and raises real incomes in the periods in which constrained households' marginal utility is highest. We call this the open economy real-income stabilization channel of exchange rate policy, delivered through the policy's influence on the intertemporal margin. Monetary policy cannot perform this role in an open economy: it operates on the consumption-leisure margin. When imports are complementary to domestic employment, a monetary expansion to support real incomes of constrained workers is costly as it requires a depreciation that shifts purchasing power toward unconstrained households through revaluation effects; optimal monetary policy complements FX interventions with a tighter stance and lower inflation than the conventional prescription implies.

Calibrated to Japan and confronted with the realized 2022 path of world energy prices, our quantitative two-agent model compares jointly optimal policy with the representative-agent-equivalent policy and restricted optimal policies. The exercise isolates how heterogeneity in asset market access changes the policy prescription: optimal policy opens costly interest parity gaps to manage the real exchange rate and stabilize the incomes of the borrowing constrained, while monetary policy cuts inflation and restrains

employment relative to the representative-agent-equivalent stance. Together the two instruments dampen the transmission of the world price shock to the domestic economy. The policies work by re-timing the real income loss and changing its incidence across agents and time; they cannot eliminate the underlying resource loss in present-value terms. We extend our two-agent model to a heterogeneous-agent model with idiosyncratic risk and an endogenously determined, evolving mass of borrowing-constrained agents. A distributional time inconsistency arises in this framework, and we characterize the utilitarian-welfare-optimal policy problem under timeless commitment and solve for the joint exchange rate and monetary policy response to an unanticipated temporary shock. Comparing that response with the next best policy blind to the real-income motive shows that the exchange-rate-driven real-income stabilization motive materially changes how the exchange rate and inflation respond to the shock and how its incidence falls across the distribution. Optimal policy holds the currency stronger and inflation lower on impact, and the real wage it protects is worth most to households at the borrowing limit, who have the least scope to self-insure against the shock.

We have excluded three margins, and they mark the natural extensions for future research: fiscal transfers that are partial or delayed rather than absent, with the division of the stabilization task between the fiscal and exchange rate instruments; policy without commitment, where the promised unwinding of the FX interventions must be credible; and shocks that arrive as risk rather than as realized paths, under which the public foreign asset position acquires a precautionary role alongside the re-timing role isolated here.

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Appendix

A Analytical Model Appendix

A.1 Optimality Conditions

Consider the two-period constrained planning problem, (16). Form the Lagrangian:

$$\mathcal{L} = \sum_{t=1}^2 \beta^{t-1} \left[\lambda u(C_t^p) + (1-\lambda) u(C_t^r) - v(N_t) \right] + \mu \left[NX_1 + \frac{1}{R^*} NX_2 \right] \quad (70)$$

where μ is the Lagrange multiplier on the intertemporal budget constraint.

FOC with respect to ϵ_t . Differentiating (70) with respect to ϵ_t , using $C_t^p = Y_t^N$ and $C_t^r = Y_t^N - \frac{\epsilon_t}{1-\lambda} NX_t$:

$$\beta^{t-1} \left[\bar{u}'_t \frac{\partial Y_t^N}{\partial \epsilon_t} - u'(C_t^r) \left(NX_t + \epsilon_t \frac{\partial NX_t}{\partial \epsilon_t} \right) \right] + \mu (R^*)^{-(t-1)} \frac{\partial NX_t}{\partial \epsilon_t} = 0$$

where $\bar{u}'_t \equiv \lambda u'(C_t^p) + (1-\lambda) u'(C_t^r)$. Solving for μ :

$$\mu = (\beta R^*)^{t-1} \cdot \frac{u'(C_t^r) \left(NX_t + \epsilon_t \frac{\partial NX_t}{\partial \epsilon_t} \right) - \bar{u}'_t \frac{\partial Y_t^N}{\partial \epsilon_t}}{\frac{\partial NX_t}{\partial \epsilon_t}} \quad (71)$$

Using $NX_t = \zeta \epsilon_t^\chi - P_t^T \theta N_t$, rewrite Y_t^N as:

$$Y_t^N = A_t N_t - \zeta \epsilon_t^{1+\chi} + \epsilon_t NX_t$$

Differentiating with respect to ϵ_t and rearranging we get:

$$NX_t + \epsilon_t \frac{\partial NX_t}{\partial \epsilon_t} = \frac{\partial Y_t^N}{\partial \epsilon_t} + (1+\chi) \zeta \epsilon_t^\chi \quad (72)$$

Substitute (72) into the numerator of (71) and collect terms using $u'(C_t^r) - \bar{u}'_t = -\lambda[u'(C_t^p) - u'(C_t^r)]$:

$$\mu = (\beta R^*)^{t-1} \left[u'(C_t^r) \epsilon_t \left(1 + \frac{1}{\chi} \right) - \lambda [u'(C_t^p) - u'(C_t^r)] \frac{\frac{\partial Y_t^N}{\partial \epsilon_t}}{\frac{\partial NX_t}{\partial \epsilon_t}} \right] \quad (73)$$

Define:

$$\Omega_t \equiv \frac{\lambda [u'(C_t^p) - u'(C_t^r)] \left(-\frac{\partial Y_t^N / \partial \epsilon_t}{\partial NX_t / \partial \epsilon_t} \right)}{1 + \frac{1}{\chi}} \quad (74)$$

Setting $\tilde{\mu} \equiv \mu/(1 + 1/\chi)$:

$$(\beta R^*)^{t-1} \left[\epsilon_t u'(C_t^r) + \Omega_t \right] = \tilde{\mu} \quad (75)$$

Since $\tilde{\mu}$ is constant, equating (75) at $t = 1$ and $t = 2$ gives the intertemporal optimality condition:

$$\epsilon_1 u'(C_1^r) + \Omega_1 = \beta R^* \left[\epsilon_2 u'(C_2^r) + \Omega_2 \right] \quad (76)$$

FOC with respect to N_t . Differentiating (70) with respect to N_t and substituting μ from (71):

$$v'(N_t) = \bar{u}'_t \left[\frac{\partial Y_t^N}{\partial N_t} - \frac{\partial Y_t^N}{\partial \epsilon_t} \cdot \frac{\frac{\partial NX_t}{\partial N_t}}{\frac{\partial NX_t}{\partial \epsilon_t}} \right] + u'(C_t^r) NX_t \cdot \frac{\frac{\partial NX_t}{\partial N_t}}{\frac{\partial NX_t}{\partial \epsilon_t}} \quad (77)$$

Define:

$$\hat{w}_t \equiv \frac{\partial Y_t^N}{\partial N_t} + \left(NX_t - \frac{\partial Y_t^N}{\partial \epsilon_t} \right) \frac{\frac{\partial NX_t}{\partial N_t}}{\frac{\partial NX_t}{\partial \epsilon_t}} = w_t - \frac{1}{\chi} \epsilon_t P_t^{T*} \theta \quad (78)$$

Define:

$$\Lambda_t \equiv \lambda [u'(C_t^p) - u'(C_t^r)] NX_t \cdot \frac{\frac{\partial NX_t}{\partial N_t}}{\frac{\partial NX_t}{\partial \epsilon_t}} \quad (79)$$

Using $u'(C_t^r) = \bar{u}'_t - \lambda [u'(C_t^p) - u'(C_t^r)]$ we get:

$$v'(N_t) = \bar{u}'_t \hat{w}_t - \Lambda_t \quad (80)$$

The solution to the constrained planning problem, (16), $\{\epsilon_t, N_t\}_{t=1}^2$, is characterized by the intertemporal optimality condition, (76), and the intratemporal optimality conditions, (80), for $t = 1, 2$, together with the intertemporal budget constraint:

$$NX_1 + \frac{1}{R^*} NX_2 = 0 \quad (81)$$

Optimal Wedges Using the optimality conditions and the definitions of the labor wedge, (20), and the intertemporal wedge, (21), we can express the optimal wedges as:

$$\tau_t^N = \frac{1}{\chi} \frac{\epsilon_t P_t^{T*} \theta}{w_t} + \frac{\Lambda_t}{\bar{u}'_t w_t} \quad (82)$$

$$\tau = \frac{\Omega_1 - \beta R^* \Omega_2}{\epsilon_1 u'(C_1^r)} \quad (83)$$

A.2 Uniqueness of the Solution

The following Lemma establishes sufficient conditions for the uniqueness of the solution to the constrained planning problem, (16).

Lemma A.1. *Let $A_t > 0$ and $P_t^{T*} > 0$ for $t \in \{1, 2\}$. Let $R^* > 0$, $\beta > 0$, $\zeta > 0$, $\theta > 0$, $\chi > 0$, and $\lambda \in [0, 1)$. Let $u : \mathbb{R}_{++} \rightarrow \mathbb{R}$ be strictly increasing and strictly concave, and $v : \mathbb{R}_{++} \rightarrow \mathbb{R}$ be strictly increasing and strictly convex. Assume that there exists at least one solution, $\{N_t, \epsilon_t\}_{t=1}^2$, to equations (76), (80), and (81).*

For $\lambda \in (0, 1)$, assume that for each $t \in \{1, 2\}$ and each $\mu \in \mathbb{R}$, the function

$$\Phi_t(N_t, \epsilon_t; \mu) \equiv \beta^{t-1} \left[\lambda u(C_t^p(N_t, \epsilon_t)) + (1 - \lambda) u(C_t^r(N_t, \epsilon_t)) - v(N_t) \right] + \mu (R^*)^{-(t-1)} NX_t(N_t, \epsilon_t)$$

admits a unique interior maximizer $(N_t^(\mu), \epsilon_t^*(\mu))$ on its domain.*

Then the solution, $\{N_t, \epsilon_t\}_{t=1}^2$, is the unique maximizer of the constrained planning problem, (16).

Proof. Step 1: Uniqueness of the per-period maximizer.

For $\lambda \in (0, 1)$, uniqueness of $(N_t^*(\mu), \epsilon_t^*(\mu))$ is assumed directly.

For $\lambda = 0$, Φ_t simplifies to:

$$\Phi_t(N_t, \epsilon_t; \mu) = \beta^{t-1} \left[u(A_t N_t - \zeta \epsilon_t^{1+\chi}) - v(N_t) \right] + \mu (R^*)^{-(t-1)} (\zeta \epsilon_t^\chi - P_t^{T*} \theta N_t)$$

where $C_t^r = A_t N_t - \zeta \epsilon_t^{1+\chi}$ at $\lambda = 0$. Under the change of variables $z_t = \epsilon_t^\chi$, a bijection on $\mathbb{R}_{++}$ since $\chi > 0$, the exponent $(1 + \chi)/\chi > 1$ implies that $z_t \mapsto z_t^{(1+\chi)/\chi}$ is strictly convex. Since u is strictly increasing and strictly concave, $u(A_t N_t - \zeta z_t^{(1+\chi)/\chi})$ is strictly concave in (N_t, z_t) . The term $-v(N_t)$ is strictly concave in N_t , and $\mu (R^*)^{-(t-1)} (\zeta z_t - P_t^{T*} \theta N_t)$ is linear. Therefore Φ_t is strictly concave in (N_t, z_t) . Hence, whenever an interior maximizer exists, it is unique and maps back to a unique interior maximizer $(N_t^*(\mu), \epsilon_t^*(\mu))$ in the original variables.

Step 2: Strict monotonicity of $h_t(\mu) \equiv NX_t(N_t^*(\mu), \epsilon_t^*(\mu))$.

Let $\mu_2 > \mu_1$. We first show that

$$(N_t^*(\mu_2), \epsilon_t^*(\mu_2)) \neq (N_t^*(\mu_1), \epsilon_t^*(\mu_1))$$

Suppose for contradiction that both equal the same interior point (N_t, ϵ_t) . Since the maximizer is interior, it satisfies the first-order conditions of $\Phi_t(\cdot; \mu_i)$ for $i \in \{1, 2\}$:

$$\nabla_{(N_t, \epsilon_t)} U_t(N_t, \epsilon_t) + \mu_i (R^*)^{-(t-1)} \nabla_{(N_t, \epsilon_t)} NX_t(N_t, \epsilon_t) = 0, \quad i \in \{1, 2\}$$

where

$$U_t(N_t, \epsilon_t) \equiv \beta^{t-1} \left[\lambda u(C_t^p(N_t, \epsilon_t)) + (1 - \lambda) u(C_t^r(N_t, \epsilon_t)) - v(N_t) \right]$$

Subtracting the two conditions gives:

$$(\mu_2 - \mu_1)(R^*)^{-(t-1)} \nabla_{(N_t, \epsilon_t)} NX_t(N_t, \epsilon_t) = 0$$

Since $\mu_2 > \mu_1$ and $(R^*)^{-(t-1)} > 0$, this requires $\nabla_{(N_t, \epsilon_t)} NX_t(N_t, \epsilon_t) = 0$. But

$$\frac{\partial NX_t}{\partial N_t} = -P_t^T \theta < 0$$

at every interior point, so $\nabla_{(N_t, \epsilon_t)} NX_t \neq 0$, a contradiction. Hence the maximizers at μ_1 and μ_2 are distinct.

Denote the unique interior maximizers by (N_t^i, ϵ_t^i) for $i \in \{1, 2\}$, where

$$(N_t^i, \epsilon_t^i) = (N_t^*(\mu_i), \epsilon_t^*(\mu_i))$$

Set

$$U_t^i \equiv \beta^{t-1} [\lambda u(C_t^p(N_t^i, \epsilon_t^i)) + (1 - \lambda)u(C_t^r(N_t^i, \epsilon_t^i)) - v(N_t^i)], \quad H_t^i \equiv NX_t(N_t^i, \epsilon_t^i)$$

Since the maximizers are distinct, uniqueness at μ_2 and μ_1 respectively yields strict inequalities:

$$\begin{aligned} U_t^2 + \mu_2(R^*)^{-(t-1)} H_t^2 &> U_t^1 + \mu_2(R^*)^{-(t-1)} H_t^1 \\ U_t^1 + \mu_1(R^*)^{-(t-1)} H_t^1 &> U_t^2 + \mu_1(R^*)^{-(t-1)} H_t^2 \end{aligned}$$

Adding and cancelling $U_t^1 + U_t^2$ yields

$$(\mu_2 - \mu_1)(R^*)^{-(t-1)} (H_t^2 - H_t^1) > 0$$

Since $\mu_2 > \mu_1$ and $(R^*)^{-(t-1)} > 0$, it follows that

$$H_t^2 > H_t^1$$

Therefore $h_t(\mu)$ is strictly increasing in μ .

Step 3: Uniqueness of μ^* and of the maximizer.

Define

$$S(\mu) \equiv h_1(\mu) + R^{*-1} h_2(\mu)$$

Since h_1 and h_2 are strictly increasing and $R^* > 0$, S is strictly increasing. Hence there exists at most one μ^* satisfying

$$S(\mu^*) = 0$$

By assumption, there exists at least one solution $\{N_t, \epsilon_t\}_{t=1}^2$ to (76), (80), and (81). Any such solution defines a common multiplier μ through (71). Equation (76) implies that this multiplier is the same across the two periods, while (80) gives the corresponding first-order condition with respect to N_t . Hence the solution

coincides with

$$\{(N_t^*(\mu), \epsilon_t^*(\mu))\}_{t=1}^2$$

Since (81) holds, this multiplier satisfies $S(\mu) = 0$. Therefore at least one μ^* satisfying $S(\mu^*) = 0$ exists. By strict monotonicity of S , this μ^* is unique. Given this unique μ^* , Step 1 yields a unique pair

$$(N_t^*(\mu^*), \epsilon_t^*(\mu^*))$$

for each $t \in \{1, 2\}$.

It remains to verify that this allocation is the unique global maximizer of the constrained planning problem.

Let

$$x_t = (N_t, \epsilon_t), \quad x_t^* = (N_t^*(\mu^*), \epsilon_t^*(\mu^*))$$

Since x_t^* uniquely maximizes

$$\Phi_t(x_t; \mu^*) = U_t(x_t) + \mu^*(R^*)^{-(t-1)}NX_t(x_t)$$

for each $t \in \{1, 2\}$, any allocation (x_1, x_2) satisfies

$$\sum_{t=1}^2 [U_t(x_t) + \mu^*(R^*)^{-(t-1)}NX_t(x_t)] \leq \sum_{t=1}^2 [U_t(x_t^*) + \mu^*(R^*)^{-(t-1)}NX_t(x_t^*)]$$

with strict inequality unless $x_t = x_t^*$ for both $t = 1, 2$.

For any feasible allocation,

$$NX_1(x_1) + R^{*-1}NX_2(x_2) = 0$$

The allocation $\{x_t^*\}_{t=1}^2$ also satisfies this condition because $S(\mu^*) = 0$. Therefore the multiplier terms cancel, and

$$U_1(x_1) + U_2(x_2) \leq U_1(x_1^*) + U_2(x_2^*)$$

with strict inequality unless $(x_1, x_2) = (x_1^*, x_2^*)$. Hence $\{x_t^*\}_{t=1}^2$ is the unique global maximizer of problem (16). $\square$

Remark. Lemma A.1 requires an additional assumption on the uniqueness of the static optimizer for any $\lambda \in (0, 1)$ since uniqueness of the solution cannot be derived from strict concavity of u alone. This is because $C_t^p = (A_t - \epsilon_t P_t^T \theta)N_t$ and C_t^r are both bilinear in (N_t, ϵ_t) , and a composition with a strictly concave u does not yield a globally concave Φ_t for $\lambda \in (0, 1)$. The additional assumption rules out multiple local optima arising from this non-concavity. In practice, this assumption is satisfied for a large range of parameters under CRRA preferences, and verified numerically.

A.3 Proof of Lemma 1

Proof. Let $\{N_t, \epsilon_t\}_{t=1}^2$ solve (16). The intertemporal resource constraint gives

$$NX_1 + \frac{1}{R^*}NX_2 = 0$$

Define prices and quantities by

$$w_t = A_t - \epsilon_t P_t^{T*} \theta, \quad M_t = \theta N_t, \quad Y_t = A_t N_t, \quad X_t = \zeta \epsilon_t^{1+\chi}$$

and define household consumption by (17) and (18). Let aggregate net foreign assets be

$$F^* = NX_1$$

and set $f^* = F^*/(1 - \lambda)$. The intertemporal resource constraint implies $NX_2 = -R^*F^*$, so the two balance-of-payments equations in (13) hold.

Choose τ to satisfy the Ricardian Euler equation,

$$1 + \tau = \frac{\beta R^* u'(C_2^r) \epsilon_2}{u'(C_1^r) \epsilon_1}$$

and set $T = \tau \epsilon_1 f^*$. Then the fiscal budget constraint (12) holds. Moreover, using $F^* = NX_1$,

$$C_1^r = w_1 N_1 - \epsilon_1 f^*$$

so

$$C_1^r + \epsilon_1 f^* (1 + \tau) = w_1 N_1 + T$$

Similarly, using $NX_2 = -R^*F^*$,

$$C_2^r = w_2 N_2 + \epsilon_2 R^* f^*$$

Thus the Ricardian budget constraints hold. The HtM budget constraints hold by (17). Firms' optimality conditions and zero profits hold by the definitions of M_t , Y_t , and w_t above. Export demand holds by construction. Goods-market clearing follows from Walras' identity, since household budgets, firms' zero-profit conditions, the fiscal budget constraint, and the balance of payments all hold. Hence the allocation is implementable as an equilibrium under the policy $\{\tau, T\}$. $\square$

A.4 Proof of Proposition 1

Proof. Setting $\lambda = 0$ in (74) gives $\Omega_t = 0$. Equation (76) then coincides with the undistorted Ricardian Euler equation, so $\tau = 0$. Equation (80) becomes $v'(N_t) = u'(C_t^r) \hat{w}_t$. Therefore $1 - \tau_t^N = v'(N_t)/(u'(C_t^r) w_t) = \hat{w}_t/w_t$. Since $\hat{w}_t = w_t - \frac{1}{\chi} \epsilon_t P_t^{T*} \theta$, $\tau_t^N = \frac{1}{\chi} \epsilon_t P_t^{T*} \theta / w_t$. At an interior solution, $v'(N_t) > 0$ and $u'(C_t^r) > 0$ imply $\hat{w}_t > 0$, hence $w_t > 0$ and $\tau_t^N > 0$. $\square$

A.5 Proof of Lemma 2

Proof. Let $z_t \equiv e_t^X$ and $q \equiv (1 + \chi)/\chi > 1$, so that $C_t = AN_t - \zeta z_t^q$ and $NX_t = \zeta z_t - P_t^{T*} \theta N_t$.

Part 1. At $\lambda = 0$, the optimality system derived in the Appendix A reduces to

$$e_1 u'(C_1^r) = \beta R^* e_2 u'(C_2^r), \quad (84)$$

$$v'(N_t) = u'(C_t^r) \hat{w}_t, \quad t \in \{1, 2\}, \quad (85)$$

$$NX_1 + \frac{1}{R^*} NX_2 = 0 \quad (86)$$

Suppose $P_1^{T*} = P_2^{T*} = P^{T*}$. Let (N^0, z^0) solve the one-period system

$$\zeta z^0 - P^{T*} \theta N^0 = 0$$

and

$$v'(N^0) = u'(AN^0 - \zeta(z^0)^q) \left[A - q \zeta (z^0)^{q-1} \frac{P^{T*} \theta}{\zeta} \right]$$

This system is the first-order condition of the strictly concave one-period problem obtained after imposing $NX = 0$, and hence has at most one interior solution. Setting

$$N_1 = N_2 = N^0, \quad z_1 = z_2 = z^0$$

satisfies (85). It also satisfies (84) because $\beta R^* = 1$, and it satisfies (86) because $NX_1 = NX_2 = 0$. Hence it is a solution to the representative-agent optimality system. By Lemma A.1, the constrained efficient allocation is unique. Therefore

$$N_1 = N_2, \quad z_1 = z_2$$

It follows that $e_1 = e_2$ and $NX_1 = NX_2 = 0$.

Part 2. For $P > 0$ and $nx \in \mathbb{R}$, define the period value function:

$$V(P, nx) \equiv \max_{N, z} \{u(AN - \zeta z^q) - v(N)\} \quad \text{s.t.} \quad \zeta z - P \theta N = nx$$

Since the objective is concave in (N, z) and the constraint is affine, V is concave in nx . The two-period problem is equivalent to

$$\max_{nx_1} G(nx_1) \equiv V(P_1^{T*}, nx_1) + \beta V(P^{T*}, -R^* nx_1)$$

G is concave. Since $\beta R^* = 1$,

$$G'(0) = V_{nx}(P_1^{T*}, 0) - V_{nx}(P^{T*}, 0)$$

It suffices to show $V_{nx}(P, 0)$ is strictly decreasing in P .

At $nx = 0$ the constraint gives $\zeta z = P \theta N$. The envelope theorem yields $V_{nx}(P, 0) = -q C^{-\sigma} z^{q-1}$, so it suffices to show $C^{-\sigma} z^{q-1}$ is strictly increasing in P along the balanced-trade optimum.

Define $s \equiv \zeta z^q / (AN) \in (0, 1/q)$, so $C = AN(1 - s)$. The first-order condition with respect to N is:

$$A^{1-\sigma} N^{-\sigma} (1-s)^{-\sigma} (1-qs) = v'(N)$$

Implicit differentiation with respect to s gives:

$$\frac{d \log N}{ds} = \frac{\frac{\sigma}{1-s} - \frac{q}{1-qs}}{\sigma + \frac{Nv''(N)}{v'(N)}}$$

Since v is convex, $Nv''(N)/v'(N) \geq 0$, so

$$\frac{d \log N}{ds} < \frac{\sigma}{\sigma(1-s)} = \frac{1}{1-s} \quad (87)$$

From $s = (\theta^q / A \zeta^{q-1}) P^q N^{q-1}$:

$$\frac{d \log P}{ds} = \frac{1}{q} \left[\frac{1}{s} - (q-1) \frac{d \log N}{ds} \right]$$

Since $s < 1/q$ implies $(q-1)/(1-s) < 1/s$, combining with (87) gives $d \log P/ds > 0$: P is strictly increasing in s .

Up to a positive constant,

$$C^{-\sigma} z^{q-1} \propto s^{\frac{q-1}{q}} (1-s)^{-\sigma} N^{\frac{q-1}{q}-\sigma}$$

Its log-derivative with respect to s is:

$$\frac{q-1}{qs} + \frac{\sigma}{1-s} + \left(\frac{q-1}{q} - \sigma \right) \frac{d \log N}{ds}$$

Since $\sigma \geq (q-1)/q = 1/(1+\chi)$, the last coefficient is non-positive. Using the tighter bound $d \log N/ds < \sigma/[(\sigma + Nv''/v')(1-s)]$ from (87), the log-derivative is bounded below by:

$$\frac{q-1}{qs} + \frac{\sigma}{1-s} \left[1 - \frac{\sigma - (q-1)/q}{\sigma + Nv''(N)/v'(N)} \right] > 0$$

Hence $C^{-\sigma} z^{q-1}$ is strictly increasing in s , and since P is strictly increasing in s , it is strictly increasing in P . Therefore $V_{nx}(P, 0) = -q C^{-\sigma} z^{q-1}$ is strictly decreasing in P .

Since $P_1^{T*} > P^{T*}$, we have $G'(0) < 0$. Concavity of G then gives $NX_1 = nx_1^* < 0$, and (86) gives $NX_2 = -R^* NX_1 > 0$. $\square$

A.6 Proof of Lemma 4

Proof. With $\tau = 0$ and $\beta R^* = 1$, the laissez-faire equilibrium satisfies:

$$\epsilon_1 u'(C_1^r) = \epsilon_2 u'(C_2^r), \quad NX_1 + \frac{1}{R^*} NX_2 = 0$$

Suppose for contradiction $NX_1 \geq 0$, so $NX_2 \leq 0$. Since $P_1^{T*} > P^{T*}$:

$$\zeta \epsilon_1^X \geq P_1^{T*} \theta \bar{N} > P^{T*} \theta \bar{N} \geq \zeta \epsilon_2^X$$

giving $\epsilon_1 > \epsilon_2$. Since $NX_1 \geq 0$ and $NX_2 \leq 0$:

$$C_1^r \leq Y_1^N = (A - \epsilon_1 P_1^{T*} \theta) \bar{N} < (A - \epsilon_2 P^{T*} \theta) \bar{N} = Y_2^N \leq C_2^r$$

Therefore $u'(C_1^r) > u'(C_2^r)$, and with $\epsilon_1 > \epsilon_2$:

$$\epsilon_1 u'(C_1^r) > \epsilon_2 u'(C_2^r)$$

contradicting the Euler equation. Hence $NX_1 < 0$ and $NX_2 = -R^* NX_1 > 0$. Since $C_t^r - C_t^p = -\frac{\epsilon_t NX_t}{1-\lambda}$, it follows that $C_1^r > C_1^p$ and $C_2^r < C_2^p$. $\square$

A.7 Proof of Proposition 2

Proof. Strict concavity and gradient formula. Since C_t^p is affine and non-constant in ϵ_t and u is strictly concave, $W(\epsilon_1, \epsilon_2) \equiv \sum_{t=1}^2 \beta^{t-1} [\lambda u(C_t^p) + (1-\lambda)u(C_t^r)]$ is strictly concave. Direct differentiation of W , together with the definition of Ω_t , gives

$$\frac{\partial W}{\partial \epsilon_t} = -\beta^{t-1} \left(1 + \frac{1}{\chi} \right) \left[\epsilon_t u'(C_t^r) + \Omega_t \right] \frac{\partial NX_t}{\partial \epsilon_t} \quad (88)$$

From (74), (7), and (15):

$$|\Omega_t| = \frac{\lambda |u'(C_t^p) - u'(C_t^r)| P_t^{T*} \theta \bar{N}}{(1 + \chi^{-1}) \chi \zeta \epsilon_t^{X-1}}$$

When $NX_t > 0$, $C_t^r < C_t^p$, so $|u'(C_t^p) - u'(C_t^r)| < u'(C_t^r)$, and $P_t^{T*} \theta \bar{N} < \zeta \epsilon_t^X$. Hence

$$|\Omega_t| < \frac{\lambda u'(C_t^r) \zeta \epsilon_t^X}{(1 + \chi^{-1}) \chi \zeta \epsilon_t^{X-1}} = \frac{\lambda}{1 + \chi} \epsilon_t u'(C_t^r) < \epsilon_t u'(C_t^r) \quad (89)$$

where the last inequality uses $\lambda < 1 < 1 + \chi$. Hence $\epsilon_t u'(C_t^r) + \Omega_t > 0$ whenever $NX_t > 0$.

Proof of items ii and iii. By Lemma 4, $NX_1^{LF} < 0 < NX_2^{LF}$, so $\Omega_1^{LF} > 0 > \Omega_2^{LF}$. Setting $\tau = 0$ and $\beta R^* = 1$ in (11) gives $\epsilon_1^{LF} u'(C_1^{r,LF}) = \epsilon_2^{LF} u'(C_2^{r,LF})$. Hence, using $\Omega_1^{LF} > 0$, $\Omega_2^{LF} < 0$, and (89) at $t = 2$:

$$H_1^{LF} \equiv \epsilon_1^{LF} u'(C_1^{r,LF}) + \Omega_1^{LF} > \epsilon_2^{LF} u'(C_2^{r,LF}) > \epsilon_2^{LF} u'(C_2^{r,LF}) + \Omega_2^{LF} \equiv H_2^{LF} > 0 \quad (90)$$

For any feasible $(\epsilon_1, \epsilon_2) \neq (\epsilon_1^{LF}, \epsilon_2^{LF})$, strict concavity of W , (88), and $\beta R^* = 1$ give:

$$W(\epsilon_1, \epsilon_2) < W^{LF} - \underbrace{(1 + \chi^{-1}) \left[H_1^{LF} \frac{\partial NX_1^{LF}}{\partial \epsilon_1} \Delta \epsilon_1 + R^{*-1} H_2^{LF} \frac{\partial NX_2^{LF}}{\partial \epsilon_2} \Delta \epsilon_2 \right]}_{\equiv \mathcal{B}}$$

where $\Delta\epsilon_t = \epsilon_t - \epsilon_t^{LF}$. Decompose $\mathcal{B}$ as:

$$\mathcal{B} = H_2^{LF} \left[\frac{\partial NX_1^{LF}}{\partial \epsilon_1} \Delta\epsilon_1 + R^{*-1} \frac{\partial NX_2^{LF}}{\partial \epsilon_2} \Delta\epsilon_2 \right] + (H_1^{LF} - H_2^{LF}) \frac{\partial NX_1^{LF}}{\partial \epsilon_1} \Delta\epsilon_1$$

By concavity of $NX_1 + R^{*-1}NX_2$ and the equality constraint, the first bracket is non-negative. When $\Delta\epsilon_1 \geq 0$, both terms are non-negative by (90) and $\partial NX_t / \partial \epsilon_t > 0$, and at least one is strictly positive for $(\epsilon_1, \epsilon_2) \neq (\epsilon_1^{LF}, \epsilon_2^{LF})$. Hence

$$W(\epsilon_1, \epsilon_2) < W^{LF} \quad \text{for all feasible } (\epsilon_1, \epsilon_2) \text{ with } \epsilon_1 \geq \epsilon_1^{LF}$$

Along the constraint, $R^{*-1} \partial NX_2 / \partial \epsilon_2 d\epsilon_2 = -\partial NX_1 / \partial \epsilon_1 d\epsilon_1$, so by (88) and $\beta R^* = 1$:

$$dW^{LF} = -(1 + \chi^{-1}) (H_1^{LF} - H_2^{LF}) \frac{\partial NX_1^{LF}}{\partial \epsilon_1} d\epsilon_1 > 0 \quad \text{for } d\epsilon_1 < 0$$

so LF is suboptimal and there exist feasible allocations with $W > W^{LF}$. Since every feasible allocation with $\epsilon_1 \geq \epsilon_1^{LF}$ satisfies $W \leq W^{LF}$ and LF is suboptimal, any optimum satisfies $\epsilon_1^O < \epsilon_1^{LF}$. Since both allocations satisfy the constraint and NX_t is strictly increasing in ϵ_t ,

$$\epsilon_2^O > \epsilon_2^{LF}, \quad NX_1^O < NX_1^{LF} < 0 < NX_2^{LF} < NX_2^O$$

Proof of items iv and v. Item iv follows immediately from item ii. Since $Y_t^N = (A - \epsilon_t P_t^{T*} \theta) \bar{N}$ is strictly decreasing in $\epsilon_t P_t^{T*}$, item v follows from item iv.

Proof of item i. Since $NX_1^O < 0 < NX_2^O$, we have $\Omega_1^O > 0 > \Omega_2^O$. With $\beta R^* = 1$, (76) gives $\epsilon_1^O u'(C_1^{r,O}) + \Omega_1^O = \epsilon_2^O u'(C_2^{r,O}) + \Omega_2^O$, and (11) gives $\epsilon_1^O (1 + \tau) u'(C_1^{r,O}) = \epsilon_2^O u'(C_2^{r,O})$. Substituting the latter into the former gives

$$\tau = \frac{\Omega_1^O - \Omega_2^O}{\epsilon_1^O u'(C_1^{r,O})} > 0$$

Proof of item vi. This follows directly from Lemma A.2. □

Lemma A.2. Assume the conditions of Proposition 2 hold. Let $p \equiv \frac{P_1^{T*}}{P_2^{T*}} > 1$.

i. Define the allocation that ensures equal input prices across periods, $(\epsilon_1^E, \epsilon_2^E)$:

$$\epsilon_1^E P_1^{T*} = \epsilon_2^E P_2^{T*}$$

and satisfies the intertemporal budget constraint, (81). If such an allocation is infeasible or it is feasible and

$$\frac{\lambda}{1 + \chi} \leq \frac{(p + \beta)}{p(1 + \beta p \chi)}$$

then the constrained efficient allocation satisfies:

$$0 \leq \epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*} < \epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}$$

$$0 \leq Y_2^{N,O} - Y_1^{N,O} < Y_2^{N,LF} - Y_1^{N,LF}$$

ii. Define the mirror allocation as one that ensures input prices are as volatile as the laissez-faire allocation, $(\epsilon_1^M, \epsilon_2^M)$:

$$\epsilon_1^M P_1^{T*} - \epsilon_2^M P_2^{T*} = -(\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*})$$

with $\epsilon_1^M < \epsilon_1^{LF}$, $\epsilon_2^M > \epsilon_2^{LF}$, and satisfies the intertemporal budget constraint, (81). If such an allocation is infeasible or it is feasible and:

$$-NX_1^M \leq \left(1 - \frac{\lambda}{1 + \chi}\right) P_1^{T*} \theta \bar{N}$$

then the constrained efficient allocation satisfies:

$$|\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*}| < |\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}|$$

$$|Y_2^{N,O} - Y_1^{N,O}| < |Y_2^{N,LF} - Y_1^{N,LF}|$$

Proof. By Proposition 2,

$$\epsilon_1^O < \epsilon_1^{LF}, \quad \epsilon_2^O > \epsilon_2^{LF}$$

and

$$NX_1^O < NX_1^{LF} < 0 < NX_2^{LF} < NX_2^O$$

Also, from the laissez-faire lemma,

$$\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*} > 0$$

Since

$$Y_2^N - Y_1^N = \theta \bar{N} (\epsilon_1 P_1^{T*} - \epsilon_2 P_2^{T*})$$

it is enough to prove the corresponding claims for domestic input prices.

We first record an ordering property of the feasible frontier. Let $D(\epsilon_1, \epsilon_2) \equiv \epsilon_1 P_1^{T*} - \epsilon_2 P_2^{T*}$ and let (ϵ_1, ϵ_2) and $(\tilde{\epsilon}_1, \tilde{\epsilon}_2)$ both satisfy (81). If $\tilde{\epsilon}_1 > \epsilon_1$, then $NX_1(\tilde{\epsilon}_1) > NX_1(\epsilon_1)$, so (81) implies $NX_2(\tilde{\epsilon}_2) < NX_2(\epsilon_2)$. Since NX_2 is strictly increasing in ϵ_2 , $\tilde{\epsilon}_2 < \epsilon_2$. Hence

$$D(\tilde{\epsilon}_1, \tilde{\epsilon}_2) > D(\epsilon_1, \epsilon_2)$$

Along the feasible frontier, D is therefore strictly increasing in ϵ_1 . Since NX_2 is continuous and strictly increasing, the feasible frontier is continuous wherever it is defined.

We first prove the equal-input-price cutoff result. Suppose first that no feasible allocation satisfying (81) and

$$\epsilon_1 P_1^{T*} = \epsilon_2 P_2^{T*}$$

exists in the direction $\epsilon_1 < \epsilon_1^{LF}$. Since $D(\epsilon_1^{LF}, \epsilon_2^{LF}) > 0$ and D is strictly increasing along the feasible frontier, the constrained efficient allocation, which satisfies $\epsilon_1^O < \epsilon_1^{LF}$ by Proposition 2, must satisfy

$$\epsilon_1^O p_1^{T*} - \epsilon_2^O p_2^{T*} \geq 0$$

Together with

$$\epsilon_1^O < \epsilon_1^{LF}, \quad \epsilon_2^O > \epsilon_2^{LF}$$

this gives

$$0 \leq \epsilon_1^O p_1^{T*} - \epsilon_2^O p_2^{T*} < \epsilon_1^{LF} p_1^{T*} - \epsilon_2^{LF} p_2^{T*}$$

Now suppose the equal-input-price allocation $(\epsilon_1^E, \epsilon_2^E)$ is feasible. Since

$$\epsilon_1^E p_1^{T*} = \epsilon_2^E p_2^{T*}$$

and $p_1^{T*} > p_2^{T*}$, we have

$$\epsilon_1^E < \epsilon_2^E$$

Using (81), this allocation satisfies

$$NX_1^E < 0 < NX_2^E$$

Thus

$$C_1^{r,E} > C_1^{p,E} = C_2^{p,E} > C_2^{r,E}$$

Let

$$\kappa_t^E \equiv \frac{p_t^{T*} \theta \bar{N}}{(1 + \chi) \zeta(\epsilon_t^E)^{\chi-1}}$$

Expanding the two Ω terms at the equal-input-price allocation gives

$$\begin{aligned} & \epsilon_2^E u'(C_2^{r,E}) + \Omega_2^E - \epsilon_1^E u'(C_1^{r,E}) - \Omega_1^E \\ &= (\epsilon_2^E - \epsilon_1^E) u'(C_1^{p,E}) + (\epsilon_1^E - \lambda \kappa_1^E) \left[u'(C_1^{p,E}) - u'(C_1^{r,E}) \right] \\ & \quad + (\epsilon_2^E - \lambda \kappa_2^E) \left[u'(C_2^{r,E}) - u'(C_2^{p,E}) \right] \end{aligned}$$

The first term is strictly positive. Since $NX_2^E > 0$,

$$p_2^{T*} \theta \bar{N} < \zeta(\epsilon_2^E)^\chi$$

so

$$\lambda \kappa_2^E < \epsilon_2^E$$

Hence the third term is strictly positive. The condition

$$\frac{\lambda}{1 + \chi} \leq \frac{p + \beta}{p(1 + \beta p^\chi)}$$

is equivalent, using (81) and $\epsilon_2^E = p\epsilon_1^E$, to

$$\lambda P_1^{T*} \theta \bar{N} \leq (1 + \chi) \zeta(\epsilon_1^E)^\chi$$

and hence to

$$\lambda \kappa_1^E \leq \epsilon_1^E$$

Therefore the second term is non-negative, and so

$$\epsilon_2^E u'(C_2^{r,E}) + \Omega_2^E > \epsilon_1^E u'(C_1^{r,E}) + \Omega_1^E$$

We now use the strict concavity argument from Proposition 2. For any feasible allocation with $\epsilon_1 \leq \epsilon_1^E$, feasibility and concavity of the external resource constraint imply that the tangent feasibility term at $(\epsilon_1^E, \epsilon_2^E)$ is non-negative. Since the planner gradient at $(\epsilon_1^E, \epsilon_2^E)$ points strictly toward larger ϵ_1 along the feasible frontier, strict concavity implies that every feasible allocation with $\epsilon_1 \leq \epsilon_1^E$ is dominated. Hence the constrained efficient allocation must satisfy

$$\epsilon_1^O > \epsilon_1^E$$

Since both allocations satisfy (81) and D is strictly increasing along the feasible frontier, this implies

$$\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*} \geq 0$$

Together with the directional comparison in Proposition 2, this gives

$$0 \leq \epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*} < \epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}$$

Multiplying by $\theta \bar{N}$ gives

$$0 \leq Y_2^{N,O} - Y_1^{N,O} < Y_2^{N,LF} - Y_1^{N,LF}$$

We next prove the mirror cutoff result. Suppose first that no feasible allocation satisfying (81) and

$$\epsilon_1 P_1^{T*} - \epsilon_2 P_2^{T*} = -(\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*})$$

exists in the direction

$$\epsilon_1 < \epsilon_1^{LF}, \quad \epsilon_2 > \epsilon_2^{LF}$$

Since $D(\epsilon_1^{LF}, \epsilon_2^{LF}) > 0$ and D is strictly increasing along the feasible frontier, the constrained efficient allocation, which satisfies $\epsilon_1^O < \epsilon_1^{LF}$ by Proposition 2, must satisfy

$$\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*} > -(\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*})$$

Using again

$$\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*} < \epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}$$

we obtain

$$|\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*}| < |\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*}|$$

Now suppose the mirror allocation $(\epsilon_1^M, \epsilon_2^M)$ is feasible. By construction,

$$\epsilon_1^M P_1^{T*} - \epsilon_2^M P_2^{T*} = -(\epsilon_1^{LF} P_1^{T*} - \epsilon_2^{LF} P_2^{T*})$$

with

$$\epsilon_1^M < \epsilon_1^{LF}, \quad \epsilon_2^M > \epsilon_2^{LF}$$

Therefore

$$NX_1^M < NX_1^{LF} < 0 < NX_2^{LF} < NX_2^M$$

and

$$C_1^{r,M} > C_1^{p,M} > C_2^{p,M} > C_2^{r,M}$$

Let

$$\kappa_t^M \equiv \frac{P_t^{T*} \theta \bar{N}}{(1 + \chi) \zeta(\epsilon_t^M)^{\chi-1}}$$

Expanding the two Ω terms at the mirror allocation gives

$$\begin{aligned} & \epsilon_2^M u'(C_2^{r,M}) + \Omega_2^M - \epsilon_1^M u'(C_1^{r,M}) - \Omega_1^M \\ &= (\epsilon_2^M - \epsilon_1^M) u'(C_1^{p,M}) + \epsilon_2^M \left[u'(C_2^{p,M}) - u'(C_1^{p,M}) \right] \\ & \quad + (\epsilon_2^M - \lambda \kappa_2^M) \left[u'(C_2^{r,M}) - u'(C_2^{p,M}) \right] \\ & \quad + (\epsilon_1^M - \lambda \kappa_1^M) \left[u'(C_1^{p,M}) - u'(C_1^{r,M}) \right] \end{aligned}$$

The first two terms are strictly positive. Since $NX_2^M > 0$,

$$\lambda \kappa_2^M < \epsilon_2^M$$

so the third term is strictly positive. The condition

$$-NX_1^M \leq \left(1 - \frac{\lambda}{1 + \chi}\right) P_1^{T*} \theta \bar{N}$$

is equivalent to

$$\lambda P_1^{T*} \theta \bar{N} \leq (1 + \chi) \zeta(\epsilon_1^M)^\chi$$

and hence to

$$\lambda \kappa_1^M \leq \epsilon_1^M$$

Therefore the fourth term is non-negative, and so

$$\epsilon_2^M u'(C_2^{r,M}) + \Omega_2^M > \epsilon_1^M u'(C_1^{r,M}) + \Omega_1^M$$

The strict-concavity argument above implies that the constrained efficient allocation lies before the mirror

allocation:

$$\epsilon_1^O > \epsilon_1^M$$

Since D is strictly increasing along the feasible frontier,

$$\epsilon_1^O p_1^{T*} - \epsilon_2^O p_2^{T*} > -(\epsilon_1^{LF} p_1^{T*} - \epsilon_2^{LF} p_2^{T*})$$

Together with

$$\epsilon_1^O p_1^{T*} - \epsilon_2^O p_2^{T*} < \epsilon_1^{LF} p_1^{T*} - \epsilon_2^{LF} p_2^{T*}$$

this gives

$$|\epsilon_1^O p_1^{T*} - \epsilon_2^O p_2^{T*}| < |\epsilon_1^{LF} p_1^{T*} - \epsilon_2^{LF} p_2^{T*}|$$

Multiplying by $\theta \bar{N}$ gives

$$|Y_2^{N,O} - Y_1^{N,O}| < |Y_2^{N,LF} - Y_1^{N,LF}|$$

□

A.8 Proof of Proposition 3

Proof. Since $N_t = \bar{N}$ and $A_t = A$, output is $Y_t = A\bar{N}$ at all allocations and in both periods. All share comparisons therefore reduce to comparisons of levels. Proposition 2 gives $\epsilon_1^O < \epsilon_1^{LF}$, $\epsilon_2^O > \epsilon_2^{LF}$, $NX_1^O < NX_1^{LF} < 0 < NX_2^{LF} < NX_2^O$, $Y_1^{N,O} > Y_1^{N,LF}$, and $Y_2^{N,O} < Y_2^{N,LF}$.

Export shares. Export demand satisfies $X_t = \zeta \epsilon_t^{1+\chi}$, which is strictly increasing in ϵ_t . Hence $X_1^O < X_1^{LF}$ and $X_2^O > X_2^{LF}$. The export share of output is strictly lower in period 1 and strictly higher in period 2 under the constrained efficient allocation.

Domestic consumption shares. The resource constraint gives $C_t = Y_t - X_t = A\bar{N} - X_t$. Since output is the same across allocations, the domestic share moves opposite to the export share: $C_1^O > C_1^{LF}$ and $C_2^O < C_2^{LF}$. The domestic share is strictly higher in period 1 and strictly lower in period 2 under the constrained efficient allocation.

HtM consumption shares. HtM households consume their current labor income: $C_t^p = w_t \bar{N} = Y_t^N$. Hence $C_1^{p,O} > C_1^{p,LF}$ and $C_2^{p,O} < C_2^{p,LF}$. The HtM share is strictly higher in period 1 and strictly lower in period 2 under the constrained efficient allocation.

Ricardian consumption share in period 2. From the consumption function (18), with $NX_t > 0$ in period 2, $C_2^r = Y_2^N - \epsilon_2 NX_2 / (1 - \lambda)$. Since $\epsilon_2^O > \epsilon_2^{LF} > 0$ and $NX_2^O > NX_2^{LF} > 0$, $\epsilon_2^O NX_2^O > \epsilon_2^{LF} NX_2^{LF}$. Therefore

$$C_2^{r,O} = Y_2^{N,O} - \frac{\epsilon_2^O NX_2^O}{1 - \lambda} < Y_2^{N,LF} - \frac{\epsilon_2^{LF} NX_2^{LF}}{1 - \lambda} = C_2^{r,LF}$$

The Ricardian share is strictly lower in period 2 under the constrained efficient allocation.

Ricardian consumption share in period 1. From the consumption function (18), with $NX_1 < 0$, let

$B \equiv -NX_1 > 0$. Then $C_1^r = Y_1^N + \epsilon_1 B/(1 - \lambda)$, so

$$C_1^{r,O} - C_1^{r,LF} = \left(Y_1^{N,O} - Y_1^{N,LF} \right) + \frac{1}{1 - \lambda} (\epsilon_1^O B^O - \epsilon_1^{LF} B^{LF})$$

The first term is strictly positive. Since $\epsilon_1^O < \epsilon_1^{LF}$ and $B^O = -NX_1^O > -NX_1^{LF} = B^{LF}$, the sign of the second term is ambiguous. Hence the sign of $C_1^{r,O} - C_1^{r,LF}$ is ambiguous. $\square$

A.9 Proof of Proposition 4

Proof. From (80), $\hat{\tau}_t^N = 1 - v'(N_t)/(\hat{w}_t \bar{u}_t') = \Lambda_t/(\hat{w}_t \bar{u}_t')$. Since u is strictly increasing, $\bar{u}_t' > 0$, and by assumption $\hat{w}_t > 0$. Thus it suffices to show $\Lambda_t > 0$ whenever $NX_t \neq 0$.

From (15), $(\partial NX_t / \partial N_t) / (\partial NX_t / \partial \epsilon_t) < 0$. Moreover, (17) and (18) imply $C_t^r - C_t^p = -\epsilon_t NX_t / (1 - \lambda)$. If $NX_t > 0$, then $C_t^r < C_t^p$, so strict concavity of u gives $u'(C_t^p) - u'(C_t^r) < 0$. If $NX_t < 0$, then $C_t^r > C_t^p$, so $u'(C_t^p) - u'(C_t^r) > 0$. Hence $[u'(C_t^p) - u'(C_t^r)] NX_t < 0$ whenever $NX_t \neq 0$. Since $\lambda \in (0, 1)$, (79) gives $\Lambda_t > 0$, and therefore $\hat{\tau}_t^N > 0$. $\square$

A.10 Proof of Proposition 5

Proof. Since $A_t = A$, $P_t^{T*} = P^{T*}$, and $\beta R^* = 1$, we have $\Phi_2(\cdot; \mu) = \beta \Phi_1(\cdot; \mu)$ for all μ . Since $\beta > 0$, the maximizer is unchanged, so by Lemma A.1, $N_1^*(\mu) = N_2^*(\mu)$ and $\epsilon_1^*(\mu) = \epsilon_2^*(\mu)$ for all μ . Since the two periods share the same $(N^*, \epsilon^*, A, P^{T*})$, we have $NX_1 = NX_2$, so (81) gives $NX_1 = NX_2 = 0$. Substituting into the consumption functions yields $C_t^p = C_t^r$ for $t \in \{1, 2\}$, and hence $C_1^p = C_1^r = C_2^p = C_2^r$.

Since $C_t^p = C_t^r$, we have $\Omega_t = \Lambda_t = 0$. Equation (76) then reduces to the undistorted Ricardian Euler equation, giving $\tau = 0$. Equation (80) reduces to $v'(N_t) = \bar{u}_t' \hat{w}_t$, so $1 - \tau_t^N = v'(N_t)/(\bar{u}_t' \hat{w}_t) = \hat{w}_t / w_t$. Hence $\tau_t^N = \frac{1}{\chi} \epsilon_t P^{T*} \theta / w_t > 0$, since $v'(N_t) > 0$ and $\bar{u}_t' > 0$ imply $\hat{w}_t > 0$ and therefore $w_t > 0$. $\square$

A.11 Proof of Proposition 6

Proof. For a fixed import price P , let $h(P, \mu)$ denote the trade balance chosen by the unique interior maximizer of the static problem $\Phi(N, \epsilon; \mu)$ in Lemma A.1. Step 2 in the proof of that lemma implies that $h(P, \mu)$ is strictly increasing in μ .

Let $\mu^0(P)$ denote the multiplier associated with the balanced-trade static optimum, $h(P, \mu^0(P)) = 0$. At balanced trade, $C^p = C^r$, so the static problem coincides with the representative-agent static problem. The first-order condition with respect to ϵ gives $\mu^0(P) = \frac{1+\chi}{\chi} \epsilon(P) C(P)^{-\sigma}$. Along the balanced-trade optimum, the proof of Lemma 2 shows that $\epsilon(P) C(P)^{-\sigma}$ is strictly increasing in P when $0 < \chi \leq 1$ and $\sigma > 1/(1+\chi)$. Hence $\mu^0(P_1^{T*}) > \mu^0(P_2^{T*})$.

Let μ^* denote the multiplier for the constrained efficient allocation, and write $h_t(\mu) \equiv h(P_t^{T*}, \mu)$. Since

the allocation satisfies the intertemporal budget constraint,

$$h_1(\mu^*) + \frac{1}{R^*} h_2(\mu^*) = 0$$

Moreover, $h_1(\mu^0(P_2^{T*})) < h_1(\mu^0(P_1^{T*})) = 0$ and $h_2(\mu^0(P_1^{T*})) > h_2(\mu^0(P_2^{T*})) = 0$. Since $h_1(\mu) + R^{*-1}h_2(\mu)$ is strictly increasing in μ , it follows that

$$\mu^0(P_2^{T*}) < \mu^* < \mu^0(P_1^{T*})$$

Therefore

$$NX_1 = h_1(\mu^*) < 0 < NX_2 = h_2(\mu^*)$$

Since $C_t^r - C_t^p = -\epsilon_t NX_t / (1 - \lambda)$, we have $C_1^r > C_1^p$ and $C_2^r < C_2^p$. Strict concavity of u then gives $u'(C_1^p) > u'(C_1^r)$ and $u'(C_2^p) < u'(C_2^r)$. Since $-(\partial Y_t^N / \partial \epsilon_t) / (\partial NX_t / \partial \epsilon_t) > 0$, (74) gives $\Omega_1 > 0 > \Omega_2$. Using (83) and $\beta R^* = 1$, $\tau = (\Omega_1 - \Omega_2) / [\epsilon_1 u'(C_1^r)] > 0$.

Since $NX_t \neq 0$ for both periods and $\hat{w}_t > 0$, Proposition 4 implies $\hat{\tau}_t^N > 0$. $\square$

A.12 Imports with a CES Production Technology and Comparative Statics

This Appendix supports the discussion in Section 3.5.1. We state the constrained planning problem, describe the calibration used in the numerical simulations, and present the results of the comparative statics exercise.

A.12.1 The Planning Problem with CES Production Technology

A competitive firm in period t chooses labor and imported inputs to solve

$$\max_{N_t, M_t} A_t \left[(1 - \alpha) N_t^{\frac{\xi-1}{\xi}} + \alpha \left(\frac{M_t}{\theta} \right)^{\frac{\xi-1}{\xi}} \right]^{\frac{\xi}{\xi-1}} - w_t N_t - \epsilon_t P_t^{T*} M_t$$

For an interior allocation, the firm's first-order conditions are

$$w_t = A_t (1 - \alpha) N_t^{-1/\xi} \left[(1 - \alpha) N_t^{\frac{\xi-1}{\xi}} + \alpha \left(\frac{M_t}{\theta} \right)^{\frac{\xi-1}{\xi}} \right]^{\frac{1}{\xi-1}} \quad (91)$$

$$\epsilon_t P_t^{T*} = \frac{A_t \alpha}{\theta} \left(\frac{M_t}{\theta} \right)^{-1/\xi} \left[(1 - \alpha) N_t^{\frac{\xi-1}{\xi}} + \alpha \left(\frac{M_t}{\theta} \right)^{\frac{\xi-1}{\xi}} \right]^{\frac{1}{\xi-1}} \quad (92)$$

The utilitarian planner chooses employment, imported inputs, and the real exchange rate in both periods

subject to the firm's first-order conditions and the economy's intertemporal resource constraint:

$$\begin{aligned} \max_{\{N_t, M_t\}_{t=1}^2} \quad & \sum_{t=1}^2 \beta^{t-1} \left[\lambda \frac{(C_t^p)^{1-\sigma}}{1-\sigma} + (1-\lambda) \frac{(C_t^r)^{1-\sigma}}{1-\sigma} - \eta \frac{N_t^{1+\phi}}{1+\phi} \right] \quad \text{s.t.} \\ & NX_1 + \frac{1}{R^*} NX_2 = 0 \end{aligned}$$

where:

$$\begin{aligned} \epsilon_t &= \frac{A_t \alpha}{P_t^{T*} \theta} \left(\frac{M_t}{\theta} \right)^{-1/\xi} \left[(1-\alpha) N_t^{\frac{\xi-1}{\xi}} + \alpha \left(\frac{M_t}{\theta} \right)^{\frac{\xi-1}{\xi}} \right]^{\frac{1}{\xi-1}} \\ w_t &= A_t (1-\alpha) N_t^{-1/\xi} \left[(1-\alpha) N_t^{\frac{\xi-1}{\xi}} + \alpha \left(\frac{M_t}{\theta} \right)^{\frac{\xi-1}{\xi}} \right]^{\frac{1}{\xi-1}} \\ NX_t &= \zeta \epsilon_t^X - P_t^{T*} M_t \\ Y_t^N &= w_t N_t \\ C_t^p &= Y_t^N \\ C_t^r &= Y_t^N - \epsilon_t \frac{NX_t}{1-\lambda} \end{aligned}$$

The labor and intertemporal wedges are recovered from the constrained-efficient allocation using the definitions in (20) and (21):

$$1 - \tau_t^N = \frac{\eta N_t^\phi}{w_t [\lambda (C_t^p)^{-\sigma} + (1-\lambda) (C_t^r)^{-\sigma}]} \quad (93)$$

$$1 + \tau = \frac{\beta R^* (C_2^r)^{-\sigma} \epsilon_2}{(C_1^r)^{-\sigma} \epsilon_1} \quad (94)$$

The Leontief production technology with elastic labor supply is nested in the above specification as the limit $\xi \rightarrow 0$. The perfectly inelastic labor supply specification is nested as the limit $\phi \rightarrow \infty$.

A.12.2 The Necessity Nature of Imports

The elasticity of substitution ξ and the inverse Frisch elasticity ϕ are two important parameters that together govern how essential the imports are for the domestic economy. We illustrate this point with the help of a static representative agent economy and show how the price elasticity of import demand varies with these two parameters. Consider a static representative-agent set up with a CES production technology that combines imports and labor. Let $q \equiv e P^{T*}$ denote the relative price of the imported inputs in domestic units. Firms' first order conditions satisfy equations (91) and (92). The representative household's budget constraint is given by $C = wN$ and the household's optimal employment choice is given by $C^{-\sigma} w = \eta N^\phi$. Let $s_M \equiv \frac{qM}{Y}$, $s_N \equiv \frac{wN}{Y}$ with $s_M + s_N = 1$, denote the imported-input and labor shares. The firm's first-order conditions, combined with zero profits, which tie the real wage to the input price through

$s_N d \log w = -s_M d \log q$, imply that the input ratio responds to the import price according to

$$\frac{d \log(M/N)}{d \log q} = -\frac{\xi}{s_N}$$

Thus, ξ governs the extent to which firms can substitute away from imported inputs while holding the scale of production fixed. The household optimality condition implies

$$\frac{d \log N}{d \log q} = \frac{\sigma - 1}{\phi + \sigma} \frac{s_M}{s_N}$$

Thus, given σ , ϕ governs the response of employment, and hence the extent to which import demand can adjust through changes in the scale of domestic activity. Combining the substitution and scale responses gives the local price response of import demand:

$$\frac{d \log M}{d \log q} = -\frac{\xi}{s_N} + \frac{\sigma - 1}{\phi + \sigma} \frac{s_M}{s_N} \quad (95)$$

where the first term captures substitution effect, and the second captures the output or scale effect. Suppose $\sigma > 1$ is given. The parameters ξ and ϕ together govern the sensitivity of import demand to its price. As $\xi \rightarrow 0$, the substitution effect disappears and production approaches the Leontief case. As $\phi \rightarrow \infty$, employment becomes less responsive and the scale effect disappears. In the joint limit,

$$\lim_{\substack{\xi \rightarrow 0 \\ \phi \rightarrow \infty}} \frac{d \log M}{d \log q} = 0$$

so that the import demand becomes locally perfectly price inelastic. Varying ξ and ϕ therefore varies the price elasticity of import demand and hence the degree to which imported inputs are essential for the domestic economy, with the Leontief, fixed-employment allocation providing the limiting case.

A.12.3 Calibration

We use an illustrative calibration to present the comparative statics. We set $A_t = 1$ and $\theta = 1$. We conjecture (and verify) that in the ‘no-shock’ case, i.e., when import prices are equal across periods, the constrained-efficient allocation is also symmetric across periods: $N_1 = N_2$, $M_1 = M_2$, and $\epsilon_1 = \epsilon_2 \equiv \bar{\epsilon}$. We therefore calibrate this case with $P_t^{T*} = 1$ and normalize $N_t = M_t = 1$. The firms’ FOCs imply that $\bar{\epsilon} = \alpha$ and $\bar{w} = (1 - \alpha)$. α is chosen so that the domestic import bill is set equal to 4 percent of GDP:

$$\frac{\bar{\epsilon} \bar{P}^{T*} \bar{M}}{\bar{w} \bar{N}} = 0.04$$

The symmetric allocation involves $NX_1 = NX_2 = 0$. ζ is chosen so that the trade-balance condition is satisfied in the symmetric allocation:

$$\zeta \bar{\epsilon}^X = \bar{P}^{T*} \bar{M}$$

Given a value of ξ , the scale of labor disutility, η , is chosen so that the planner’s optimality condition for employment is satisfied with $N_t = 1$. Since $v'(1) = \eta$, this calibration is independent of ϕ . Moreover, when

import prices are identical across periods, the symmetric allocation satisfies $C_t^P = C_t^T$ so that heterogeneity is irrelevant and the allocation remains unchanged for any $\lambda \in [0, 1)$. The remaining parameters are set as follows: $\sigma = 2$, $\chi = 0.25$, $R^* = 1.0021$, and $\beta = \frac{1}{R^*}$. To isolate the role of heterogeneity, we compare how optimal wedges vary between the representative-agent economy where $\lambda = 0$ and a heterogeneous-agent economy where we set $\lambda = 0.35$. The import-price ‘shock’ case involves setting unequal import prices across periods for which we set $P_1^{T*} = 1.5$ and $P_2^{T*} = 1$.

The comparative statics exercise involves varying ξ and ϕ and computing the optimal wedges in each of four cases: (a) $\lambda = 0$ with equal import prices, (b) $\lambda = 0$ with unequal import prices, (c) $\lambda > 0$ with equal import prices, and (d) $\lambda > 0$ with unequal import prices. This allows us to compare how the optimal wedges vary with ξ , ϕ , and λ in the presence and absence of import-price shocks.

A.12.4 Comparative Statics Results

Figures A.1, A.2, and A.3 report the optimal intertemporal wedge and the labor wedges over the (ξ, ϕ) parameter space, one sub-panel for each of the four cases described above in each of the three figures.

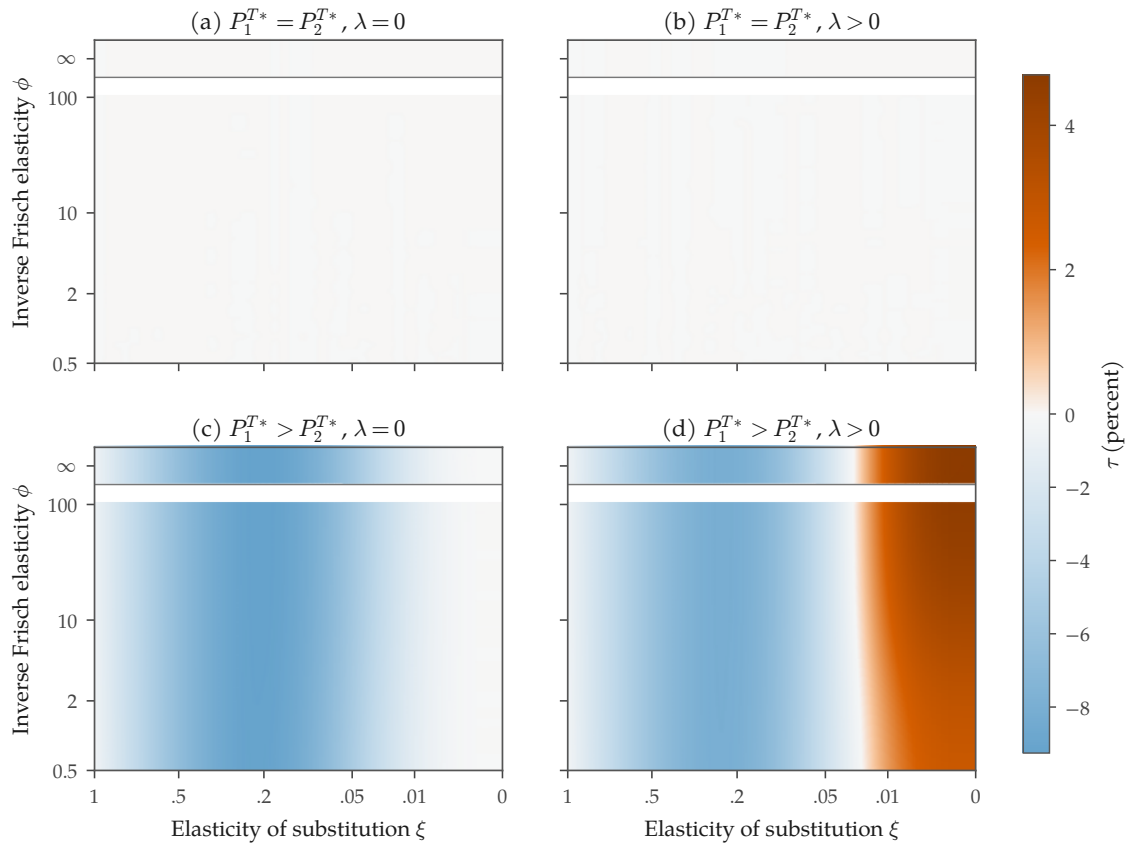


Figure A.1. Optimal intertemporal wedge τ

Note: The x-axis uses a cube-root scale.

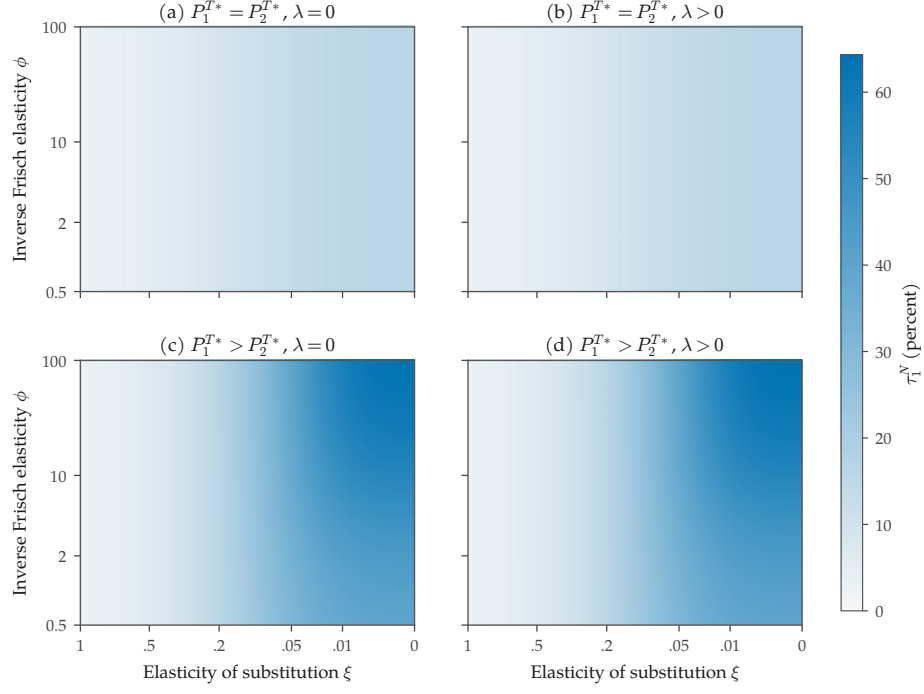


Figure A.2. Optimal period-1 labor wedge τ_1^N

Note: The x-axis uses a cube-root scale.

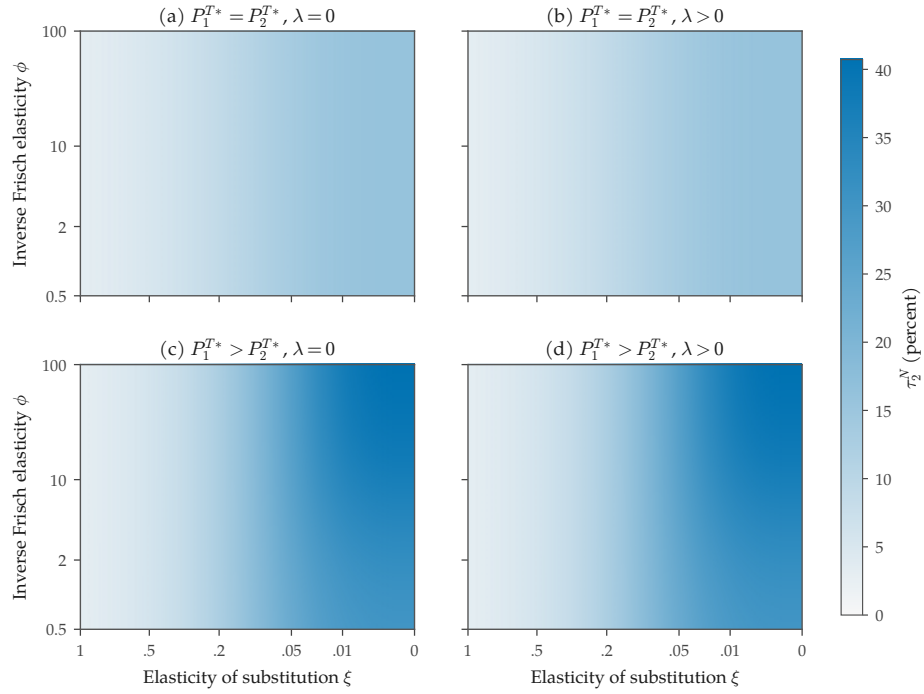


Figure A.3. Optimal period-2 labor wedge τ_2^N

Note: The x-axis uses a cube-root scale.

A.13 Imports as Final Goods and Comparative Statics

In this Appendix we consider an alternative model specification in which imports are utilized by consumers as final goods rather than being intermediate inputs in production. We continue to assume that the domestic tradeable good is produced only by the domestic economy and is either consumed domestically or exported: $Y_t = C_t + X_t$. The country faces a downward-sloping, price-elastic export demand for its own goods in the world market: $X_t = \zeta \epsilon_t^{1+\chi}$ where $P_t/e_t = 1/\epsilon_t$ is the world price of the domestic good in units of the world numeraire. The domestic economy continues to be small with respect to world asset markets where R^* is the world interest rate as well as small with respect to the market for imported goods with P_t^{T*} being the world price of imports in units of the world numeraire. By the law of one price, the domestic currency price of imports is $e_t P_t^{T*}$ and $\epsilon_t P_t^{T*}$ is the price in units of domestic output. While the typical price-level-based definition of the real exchange rate depends on both ϵ_t and P_t^{T*} , we continue to interpret ϵ_t as the real exchange rate, as it is the key endogenous variable that determines real outcomes.

Firms and Production Technology We assume that domestic tradeable output is produced by competitive firms using a linear production technology that uses just labor as the input: $Y_t = A_t N_t$. Firms' zero profit condition implies that the real wage is given by $w_t = A_t$.

Households Let C_t^j, M_t^j denote the consumption of domestic and imported goods respectively in period t by household j for $j \in \{p, r\}$. Households' period utility is given by:

$$u(C_t^j, M_t^j) - v(N_t)$$

HtM households face period-by-period static budget constraints:

$$C_t^p + \epsilon_t P_t^{T*} M_t^p = w_t N_t$$

Ricardian households are able to participate in international asset markets and are subject to capital controls:

$$C_t^r + \epsilon_t P_t^{T*} M_t^r + (1 + \tau_t) \epsilon_t f_{t+1}^* = w_t N_t + \epsilon_t R^* f_t^* + T_t$$

Household optimality conditions include an intratemporal condition that equates the marginal rate of substitution between domestic and imported goods to the terms of trade:

$$\frac{u_2(C_t^j, M_t^j)}{u_1(C_t^j, M_t^j)} = \epsilon_t P_t^{T*} \quad j \in \{p, r\}$$

and an Euler equation for Ricardian households:

$$\frac{u_1(C_t^r, M_t^r) \epsilon_t}{\beta u_1(C_{t+1}^r, M_{t+1}^r) \epsilon_{t+1}} = \frac{R^*}{1 + \tau_t}$$

Fiscal-Monetary Policy As in the main text, we assume that the exchange rate policy is chosen by a public agent who is able to distort the intertemporal margin by choosing capital controls with revenue rebated as lump-sum transfers to Ricardian households:

$$\tau_t \epsilon_t F_{t+1}^* = (1 - \lambda) T_t$$

Monetary policy continues to be reduced-form: the labor supply N_t is directly chosen by the public agent, reflecting the ability to distort the labor wedge. Tariffs and direct fiscal transfers to HtM households continue to remain absent.

Balance of Payments The balance of payments condition continues to be unchanged and is implied by all budget constraints and market clearing conditions:

$$NX_t = F_{t+1}^* - R^* F_t^*$$

where $NX_t = \frac{X_t}{\epsilon_t} - P_t^T M_t$ with $M_t = \lambda M_t^p + (1 - \lambda) M_t^r$.

Constrained Planning Problem For the purpose of numerical simulations and conducting the comparative statics exercise we consider a two-period framework. We also assume that Ricardian households begin with no initial assets. The intertemporal budget constraint is then given by $NX_1 + \frac{1}{R^*} NX_2 = 0$. The Ramsey planner's primal problem is given by:

$$\max_{\substack{\{N_t, \epsilon_t, C_t^p, C_t^r, \\ M_t^p, M_t^r\}_{t=1}^2}} \sum_{t=1}^2 \beta^{t-1} [\lambda u(C_t^p, M_t^p) + (1 - \lambda) u(C_t^r, M_t^r) - v(N_t)] \quad \text{s.t.}$$

$$NX_1 + \frac{1}{R^*} NX_2 = 0$$

$$C_t^p + \epsilon_t P_t^T M_t^p = A_t N_t$$

$$(1 - \lambda) C_t^r + \lambda C_t^p + \zeta \epsilon_t^{1+\chi} = A_t N_t$$

$$\frac{u_2(C_t^j, M_t^j)}{u_1(C_t^j, M_t^j)} = \epsilon_t P_t^T \quad j \in \{p, r\}$$

For the purpose of the comparative statics exercise, we continue to use the functional form $v(N) = \eta N^{1+\phi} / (1 + \phi)$ for labor disutility and for consumption we use a Stone-Geary specification:

$$u(C, M) = \frac{[C^{1-\omega} (M - \underline{M})^\omega]^{1-\sigma}}{1 - \sigma}, \quad M \geq \underline{M}, \quad (96)$$

where $\underline{M} \geq 0$ is a subsistence requirement for imported consumption and $\omega \in (0, 1)$ is the weight on imported consumption. Absent the subsistence term, the Cobb-Douglas aggregator keeps expenditure shares constant, so domestic and imported goods are neither gross substitutes nor gross complements. The

Stone-Geary specification over imports allows us to vary their degree of essentiality.

A.13.1 The Necessity Nature of Imports

The parameters $\underline{M}$ and ω together govern how essential the imports are for the domestic economy. We illustrate this point with the help of a static representative agent economy and show how the price elasticity of import demand varies with these two parameters. Consider a static representative-agent set up with Stone-Geary utility over domestic and imported goods. Let $q \equiv \epsilon P^{T*}$ denote the relative price of imported goods in domestic units and let I denote the income of the household. The household's budget constraint is given by $C + qM = I$. For an interior solution, the optimality condition is

$$\frac{\omega C}{(1 - \omega)(M - \underline{M})} = q$$

Combining this condition with the budget constraint gives the demand for imports:

$$M = \omega \frac{I}{q} + (1 - \omega)\underline{M} \quad (97)$$

Its absolute Marshallian price elasticity, holding income fixed, is given by

$$\varepsilon_M \equiv -\frac{\partial \log M}{\partial \log q} = \frac{\omega I}{\omega I + (1 - \omega)q\underline{M}} \quad (98)$$

When $\underline{M} = 0$, preferences reduce to Cobb–Douglas and import demand is unit elastic:

$$M = \omega \frac{I}{q}, \quad \varepsilon_M = 1$$

When $\underline{M} > 0$, equation (97) separates import demand into a price-responsive component and an inelastic subsistence component. A larger subsistence component lowers the price elasticity of import demand. In the limit in which the discretionary component vanishes,

$$\omega \rightarrow 0, \quad M \rightarrow \underline{M}, \quad \varepsilon_M \rightarrow 0$$

so that import demand becomes perfectly price inelastic. The parameters ω and $\underline{M}$ therefore jointly govern the price elasticity and hence the degree of essentiality of final-good imports.

A.13.2 Calibration

The comparative statics exercise involves jointly varying the parameters ω and $\underline{M}$ that allows us to vary the price elasticity of demand, ε_M , over $[0, 1]$. We use an illustrative calibration to examine how the optimal intertemporal and labor wedges vary with the price elasticity. We set $A_t = 1$ and conjecture (and verify) that in the ‘no-shock’ case, i.e., when import prices are equal across periods, the constrained-efficient allocation is also symmetric across periods: $N_1 = N_2 = \bar{N}$, $\epsilon_1 = \epsilon_2 = \bar{\epsilon}$, $C_1^p = C_1^r = C_2^p = C_2^r = \bar{C}$, $M_1^p = M_1^r = M_2^p = M_2^r = \bar{M}$. We therefore calibrate this case with the normalizations $P_t^{T*} = 1$ and $\bar{N} = \bar{M} = 1$.

The domestic import bill is calibrated to equal 4 percent of GDP:

$$\frac{\bar{\epsilon} \bar{P}^{T*} \bar{M}}{A \bar{N}} = 0.04 \quad (99)$$

The symmetric allocation involves $NX_1 = NX_2 = 0$. ζ is chosen so that the trade-balance condition is satisfied in the symmetric allocation: $\zeta \bar{\epsilon}^X = \bar{P}^{T*} \bar{M}$.

Let $\mathcal{E}_M \in [0, 1]$ denote the absolute price elasticity of import demand evaluated at the symmetric allocation. From (98) and (99), we have:

$$\mathcal{E}_M = \frac{\omega A \bar{N}}{\bar{\epsilon} \bar{P}^{T*} \bar{M}} = \frac{\omega}{0.04} \quad (100)$$

For each value of $\mathcal{E}_M$, we therefore set

$$\omega(\mathcal{E}_M) = 0.04 \mathcal{E}_M \quad (101)$$

The subsistence requirement is chosen so that import consumption remains equal to $\bar{M}$. Evaluating (97) at the symmetric allocation gives

$$\bar{M} = \omega(\mathcal{E}_M) \frac{A \bar{N}}{\bar{\epsilon} \bar{P}^{T*}} + [1 - \omega(\mathcal{E}_M)] \underline{M}(\mathcal{E}_M)$$

It follows that

$$\underline{M}(\mathcal{E}_M) = \bar{M} \frac{1 - \mathcal{E}_M}{1 - 0.04 \mathcal{E}_M} \quad (102)$$

When $\mathcal{E}_M = 1$, $\omega = 0.04$ and $\underline{M} = 0$, so that preferences reduce to the Cobb-Douglas case and import demand is unit elastic. As $\mathcal{E}_M \rightarrow 0$, $\omega \rightarrow 0$ and $\underline{M} \rightarrow \bar{M}$, and import demand approaches the perfectly price-inelastic case.

For every value of $\mathcal{E}_M \in [0, 1]$, the scale of labor disutility, η , is chosen so that the planner's optimality condition for employment is satisfied at $N_t = 1$ in the symmetric allocation. Since the symmetric allocation has $NX_t = 0$ and both household types consume the same bundle $(\bar{C}, \bar{M})$, household heterogeneity is irrelevant and the allocation remains unchanged for any $\lambda \in [0, 1]$. The remaining parameters are set as follows: $\sigma = 2$, $\phi = 29.5$, $\chi = 0.25$, $R^* = 1.0021$ and $\beta = \frac{1}{R^*}$. To isolate the role of heterogeneity, we compare how optimal wedges vary between the representative-agent economy where $\lambda = 0$ and a heterogeneous-agent economy where we set $\lambda = 0.35$. The import-price 'shock' case involves setting unequal import prices across periods for which we set $P_1^{T*} = 1.5$ and $P_2^{T*} = 1$.

The comparative statics exercise involves varying $\mathcal{E}_M$ and computing the optimal wedges in each of four cases: (a) $\lambda = 0$ with equal import prices, (b) $\lambda = 0$ with unequal import prices, (c) $\lambda > 0$ with equal import prices, and (d) $\lambda > 0$ with unequal import prices.

A.13.3 Comparative Statics Results

Figure A.4 reports the differences $\Delta\tau$ and $\Delta\tau_1^N$ between the optimal wedges of the heterogeneous and representative-agent economies facing the unequal import price path, plotted against the price elasticity of import demand. Figures A.5, A.6, and A.7 report the optimal intertemporal and labor wedges in levels, one panel for each of the four cases.

First, with equal import prices the optimal intertemporal wedge is zero at every elasticity, for $\lambda = 0$ and $\lambda > 0$ alike, and the labor wedge alone carries the static terms-of-trade correction. This correction now runs in the opposite direction from the production case: it is largest at Cobb-Douglas and vanishes at the perfectly inelastic corner, where subsistence pins imports, balanced trade pins the terms-of-trade path, and employment has no effect on it. With unequal prices and $\mathcal{E}_M > 0$, the planner distorts both margins against the terms-of-trade distortion absent tariffs, with $\lambda = 0$ and $\lambda > 0$ alike: the representative-agent intertemporal wedge is again the capital-control motive of Costinot, Lorenzoni and Werning (2014), and it vanishes at $\mathcal{E}_M = 0$, where the constrained-efficient allocation with $\lambda = 0$ coincides with laissez-faire. The perfectly inelastic subsistence limit is therefore the final-goods counterpart of the Leontief limit: any intertemporal distortion that arises there with $\lambda > 0$ reflects the role of heterogeneity alone.

Second, the intertemporal wedge difference rises with the degree of essentiality and is largest in the perfectly price inelastic demand limit. The real-income stabilization channel operates here directly, without any wage response: the real wage is fixed by the linear technology, and exchange rate policy reaches constrained households through the domestic price of their imported consumption.

Third, the labor wedge differences are negligible at every elasticity and approach zero as the degree of essentiality increases: heterogeneity generates no significant separate motive for monetary policy when imports are final goods. The complementarity between exchange rate policy and the monetary stance in the production economy runs through imported inputs that move with employment; with that link absent, the labor wedge is the same static correction in both economies, and exchange rate policy operates alone. While exchange rate policy continues to perform the real-income stabilization task by distorting the intertemporal margin, the scope and role of monetary policy are much weaker than in the production economy.

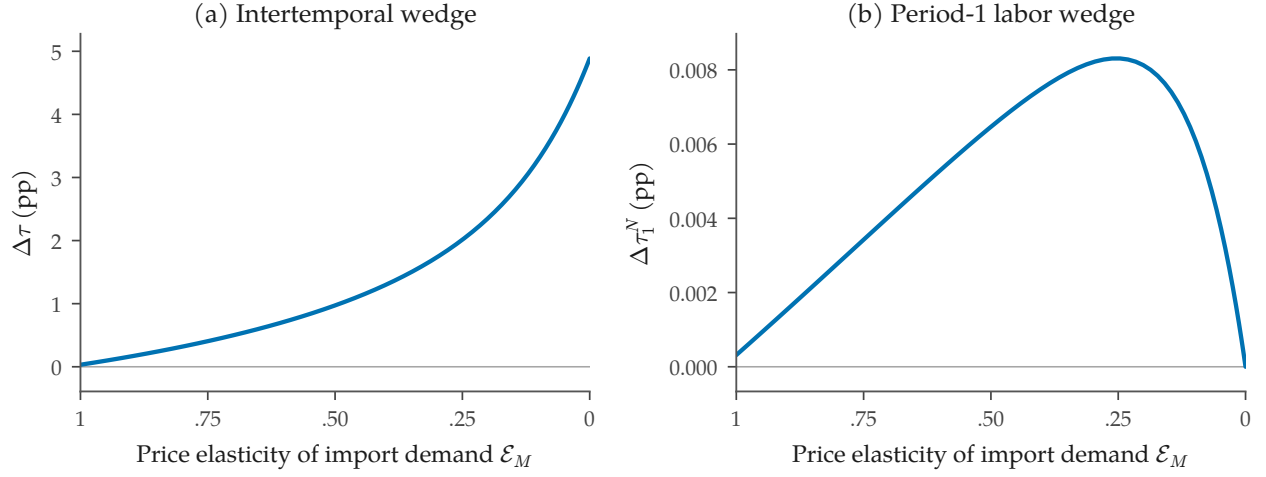


Figure A.4. Imports as final goods: optimal wedges, isolating the role of heterogeneity

Note: $\varepsilon_M = 1$ is the Cobb-Douglas case and $\varepsilon_M = 0$ the perfectly inelastic demand case.

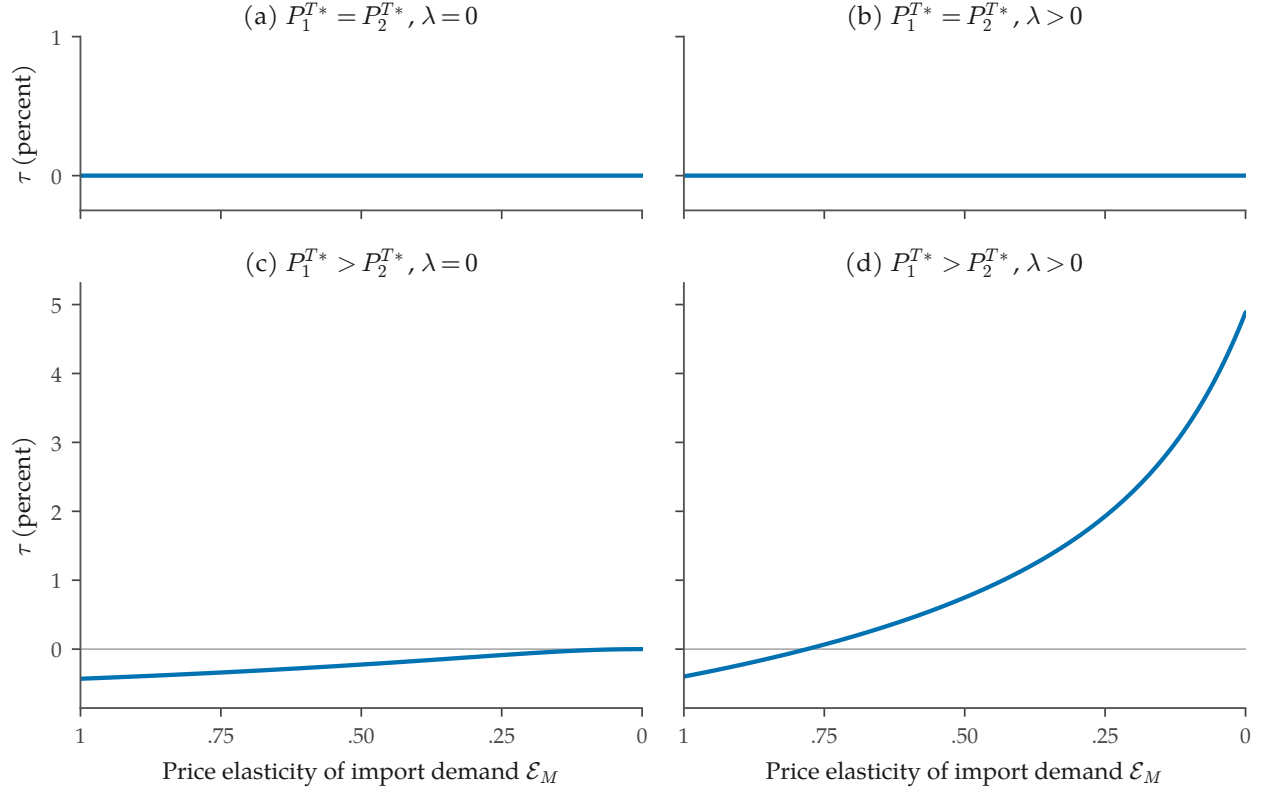


Figure A.5. Imports as final goods: optimal intertemporal wedge τ

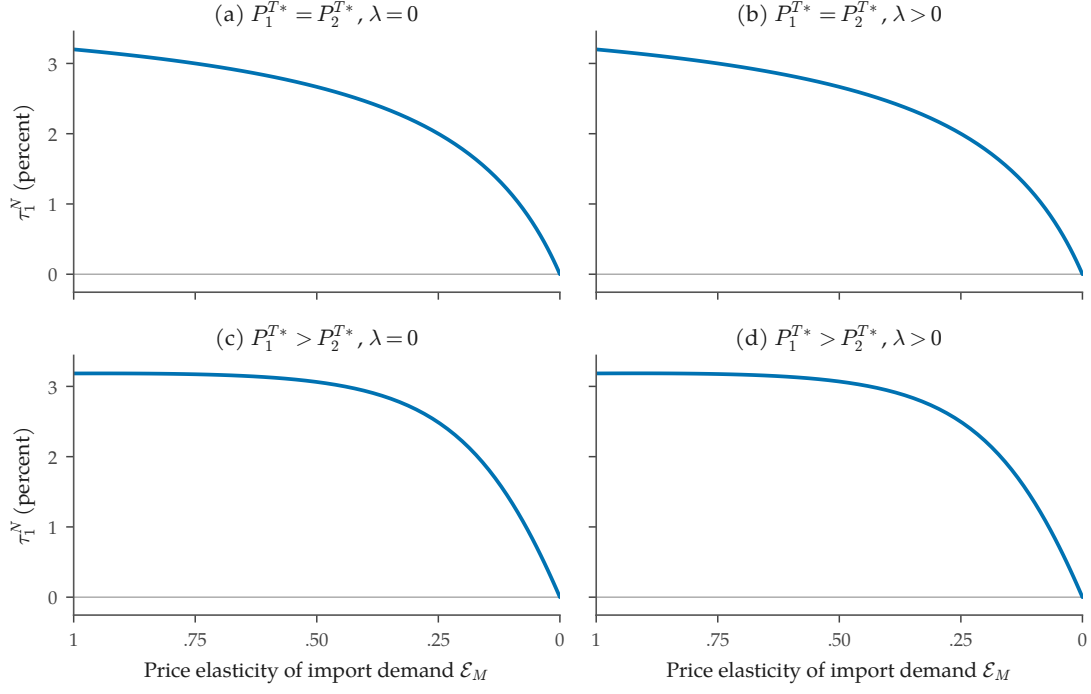


Figure A.6. Imports as final goods: optimal period-1 labor wedge τ_1^N

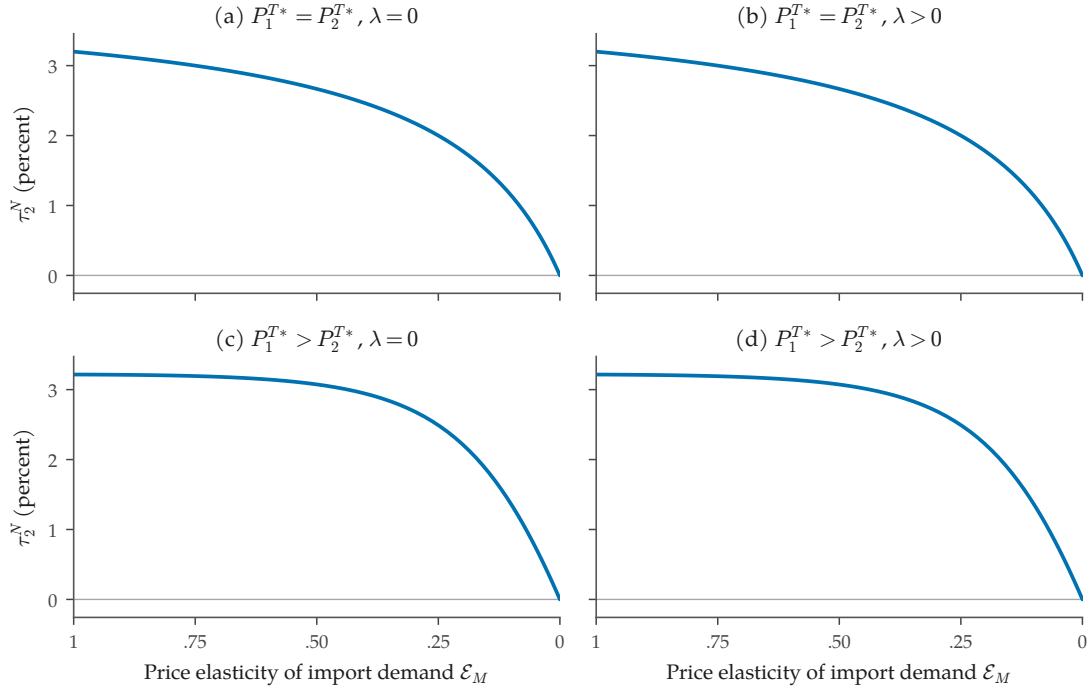


Figure A.7. Imports as final goods: optimal period-2 labor wedge τ_2^N

B Quantitative Model Appendix

B.1 Unions' Optimization Problem and the Wage Phillips Curve

At time t , union k sets the nominal wage $W_{k,t}$ to maximize (53) subject to the task demand schedule

$$N_{k,t} = \left(\frac{W_{k,t}}{W_t} \right)^{-\varepsilon_w} N_t$$

Each union is infinitesimal and takes W_t and N_t as given. Household real earnings are

$$Z_t = \frac{1}{P_t} \int_0^1 W_{k,t} \left(\frac{W_{k,t}}{W_t} \right)^{-\varepsilon_w} N_t dk$$

By the envelope theorem, we can evaluate how each household, j , values a marginal increase in labor incomes and hours worked. Then $\frac{\partial C_t^j}{\partial W_{k,t}} = \frac{\partial Z_t}{\partial W_{k,t}}$, where

$$\frac{\partial Z_t}{\partial W_{k,t}} = \frac{1}{P_t} N_{k,t} (1 - \varepsilon_w)$$

On the other hand, total hours worked by household j are $N_t^j \equiv \int_0^1 \left(\frac{W_{k,t}}{W_t} \right)^{-\varepsilon_w} N_t dk$ which falls when $W_{k,t}$ rises according to

$$\frac{\partial N_t^j}{\partial W_{k,t}} = -\varepsilon_w \frac{N_{k,t}}{W_{k,t}}$$

Therefore, the union's first order condition is given by:

$$\left(\frac{W_{k,t}}{W_{k,t-1}} - 1 \right) \frac{W_{k,t}}{W_{k,t-1}} = \frac{\varepsilon_w}{\psi_{nr}} \left[N_{k,t} v'(N_t) - \frac{\varepsilon_w - 1}{\varepsilon_w} \frac{W_{k,t}}{P_t} N_{k,t} \bar{u}_t' \right] + \beta \left(\frac{W_{k,t+1}}{W_{k,t}} - 1 \right) \frac{W_{k,t+1}}{W_{k,t}}$$

where $\bar{u}_t' \equiv \lambda u'(C_t^p) + (1 - \lambda) u'(C_t^r)$. In a symmetric equilibrium, $W_{k,t} = W_t$ and $N_{k,t} = N_t$ for all k . Let $\mu_w \equiv \varepsilon_w / (\varepsilon_w - 1)$. Then

$$\left(\frac{W_t}{W_{t-1}} - 1 \right) \frac{W_t}{W_{t-1}} = \frac{\varepsilon_w}{\psi_{nr}} \left[N_t v'(N_t) - \frac{1}{\mu_w} \frac{W_t}{P_t} N_t \bar{u}_t' \right] + \beta \left(\frac{W_{t+1}}{W_t} - 1 \right) \frac{W_{t+1}}{W_t}$$

Using $w_t = W_t/P_t$ and $\Pi_t = P_t/P_{t-1}$, this can be written as

$$\left(\frac{w_t}{w_{t-1}} \Pi_t - 1 \right) \frac{w_t}{w_{t-1}} \Pi_t = \frac{\varepsilon_w}{\psi_{nr}} \left[N_t v'(N_t) - \frac{1}{\mu_w} w_t N_t \bar{u}_t' \right] + \beta \left(\frac{w_{t+1}}{w_t} \Pi_{t+1} - 1 \right) \frac{w_{t+1}}{w_t} \Pi_{t+1}$$

Rearranging gives $v'(N_t) = (1 - \tau_t^N) w_t \bar{u}_t'$, where

$$1 - \tau_t^N = \frac{1}{\mu_w} + \frac{\left(\frac{w_t}{w_{t-1}} \Pi_t - 1 \right) \frac{w_t}{w_{t-1}} \Pi_t - \beta \left(\frac{w_{t+1}}{w_t} \Pi_{t+1} - 1 \right) \frac{w_{t+1}}{w_t} \Pi_{t+1}}{\frac{\varepsilon_w}{\psi_{nr}} w_t N_t \bar{u}_t'}$$

B.2 Balance of Payments and Intertemporal Budget Constraint

Aggregating the Ricardian and HtM household budget constraints gives the aggregate household budget constraint:

$$C_t + B_{t+1} = w_t N_t + R_t B_t + (1 - \lambda) \frac{T_t}{P_t} \quad (103)$$

Adding (103) to the government budget constraint (62) eliminates transfers and gives

$$C_t + B_{t+1} + B_{t+1}^G + \epsilon_t B_{t+1}^{G*} = w_t N_t + R_t (B_t + B_t^G) + \epsilon_t R^* B_t^{G*} \quad (104)$$

Firm zero profits and goods market clearing are given by

$$w_t N_t = A_t N_t - \epsilon_t P_t^T \theta N_t \quad C_t + \zeta \epsilon_t^{1+\chi} = A_t N_t \quad (105)$$

Substituting (105) into (104) yields

$$B_{t+1} + B_{t+1}^G + \epsilon_t B_{t+1}^{G*} = \zeta \epsilon_t^{1+\chi} - \epsilon_t P_t^T \theta N_t + R_t (B_t + B_t^G) + \epsilon_t R^* B_t^{G*} \quad (106)$$

Finally, using domestic bond market clearing (65), and dividing by ϵ_t , gives the period-by-period balance of payments condition (66).

To obtain the intertemporal budget constraint, use the intermediary supply rule (59) for the inherited position F_t^* when $t \geq 2$:

$$R_t \frac{\epsilon_{t-1}}{\epsilon_t} F_t^* = R^* F_t^* - \Gamma R^* (F_t^*)^2 \quad (107)$$

Substituting (107) into (66) gives

$$P_t^T \theta N_t + \Gamma R^* (F_t^*)^2 + F_{t+1}^* + B_{t+1}^{G*} = \zeta \epsilon_t^\chi + R^* (F_t^* + B_t^{G*}) \quad (108)$$

for all $t \geq 2$. The period-1 condition stays unchanged:

$$P_1^T \theta N_1 + F_2^* + B_2^{G*} = \zeta \epsilon_1^\chi + R_1 \frac{\epsilon_0}{\epsilon_1} F_1^* + R^* B_1^{G*} \quad (109)$$

The no-Ponzi condition on households, (50), and government's asset domestic position, (63), together with the domestic bonds market clearing, (65), imply:

$$\lim_{T \rightarrow \infty} \frac{\epsilon_T F_{T+1}^*}{\prod_{s=2}^T R_s} = 0 \quad (110)$$

Since $\prod_{s=2}^T \tilde{R}_s = (\prod_{s=2}^T R_s) \epsilon_1 / \epsilon_T$, (110) implies $F_{T+1}^* / \prod_{s=2}^T \tilde{R}_s \rightarrow 0$. The interest-rate consistency condition, (64), then gives

$$\lim_{T \rightarrow \infty} \frac{F_{T+1}^*}{R^{*T-1}} = 0 \quad (111)$$

Together with the government no-Ponzi condition on B_{T+1}^{G*} , (63), we have:

$$\lim_{T \rightarrow \infty} \frac{F_{T+1}^* + B_{T+1}^{G*}}{R^{*T-1}} = 0 \quad (112)$$

Combining (109) with (108), discounting by R^{*t-1} , summing from $t = 1$ to T , gives

$$\sum_{t=1}^T \frac{P_t^T \theta N_t}{R^{*t-1}} + \sum_{t=1}^{T-1} \frac{\Gamma(F_{t+1}^*)^2}{R^{*t-1}} + \frac{F_{T+1}^* + B_{T+1}^{G*}}{R^{*T-1}} = \sum_{t=1}^T \frac{\zeta \epsilon_t^X}{R^{*t-1}} + R^* B_1^{G*} + R_1 \frac{\epsilon_0}{\epsilon_1} F_1^* \quad (113)$$

Taking $T \rightarrow \infty$, using (112) and substituting $\Gamma(F_{t+1}^*)^2 = \Gamma^{-1}(1 - \tilde{R}_{t+1}/R^*)^2$ from (59) yields the intertemporal budget constraint, (67).

B.3 Decentralized Equilibrium Definition

Given:

- Initial states: $\{R_1^M, B_1, B_1^{G*}, \epsilon_0, F_1^*, w_0\}$,
- Exogenous sequence of import prices and productivity, $\{P_t^{T*}, A_t\}_{t=1}^\infty$,
- Fiscal-monetary policy: $\{B_{t+1}^G, B_{t+1}^{G*}, R_{t+1}^M, \frac{T_t}{P_t}\}_{t=1}^\infty, R_1$,

a decentralized equilibrium consists of an allocation: $\{C_t^r, C_t^p, B_{t+1}, N_t, M_t, Y_t, F_{t+1}^*, X_t\}_{t=1}^\infty$ and prices $\{\epsilon_t, w_t, \Pi_t, R_{t+1}\}_{t=1}^\infty$ such that:

- Given prices, Ricardian households solve (48) and HtM households satisfy their period-by-period budget constraints:

$$\begin{aligned} \frac{u'(C_t^r)}{\beta u'(C_{t+1}^r)} &= R_{t+1} \\ C_t^r + \frac{B_{t+1}}{1-\lambda} &= w_t N_t + R_t \frac{B_t}{1-\lambda} + \frac{T_t}{P_t} \\ \lim_{T \rightarrow \infty} \frac{B_{T+1}}{\prod_{s=2}^T R_s} &= 0 \\ C_t^p &= w_t N_t \end{aligned}$$

- Given prices, firms solve (3):

$$M_t = \theta N_t, \quad Y_t = A_t N_t$$

Zero profits imply:

$$w_t = A_t - \epsilon_t P_t^{T*} \theta$$

- Labor aggregator and unions' optimization gives the wage Phillips curve:

$$\left(\frac{w_t}{w_{t-1}} \Pi_t - 1 \right) \frac{w_t}{w_{t-1}} \Pi_t = \frac{\epsilon_w}{\psi_{nr}} \left[N_t v'(N_t) - \frac{1}{\mu_w} w_t N_t \tilde{u}'_t \right] + \beta \left[\left(\frac{w_{t+1}}{w_t} \Pi_{t+1} - 1 \right) \frac{w_{t+1}}{w_t} \Pi_{t+1} \right]$$

- Given prices, international financial intermediaries optimize:

$$F_{t+1}^* = \frac{1}{\bar{r}} \left[1 - \frac{R_{t+1}}{R^*} \left(\frac{\epsilon_t}{\epsilon_{t+1}} \right) \right]$$

- World demand for domestic output is given by:

$$X_t = \zeta \epsilon_t^{1+\chi}$$

- Fiscal policy satisfies the period-by-period budget constraints:

$$B_{t+1}^G + \epsilon_t B_{t+1}^{G*} + (1-\lambda) \frac{T_t}{P_t} = R_t B_t^G + \epsilon_t R^* B_t^{G*}$$

where $B_1^G = \epsilon_0 F_1^* - B_1$; satisfies no-Ponzi conditions:

$$\lim_{T \rightarrow \infty} \frac{B_{T+1}^G}{\prod_{s=2}^T R_s} = 0, \quad \lim_{T \rightarrow \infty} \frac{B_{T+1}^{G*}}{R^{*T-1}} = 0$$

together with the interest rate consistency condition:

$$0 < \inf_{T \geq 2} \prod_{s=2}^T \frac{\tilde{R}_s}{R^*} \leq \sup_{T \geq 2} \prod_{s=2}^T \frac{\tilde{R}_s}{R^*} < \infty$$

- Inflation is implied by monetary policy:

$$R_t^M = R_t \Pi_t, \quad t \geq 1$$

- Final goods market clears:

$$\lambda C_t^P + (1-\lambda) C_t^T + X_t = A_t N_t$$

- Labor market clears.

- Domestic bonds market clears:

$$B_{t+1} + B_{t+1}^G = \epsilon_t F_{t+1}^*$$

- Balance of payments is implied by all other budget constraints and market clearing conditions holding:

$$P_t^T \theta N_t + F_{t+1}^* + B_{t+1}^{G*} = \zeta \epsilon_t^\chi + R_t \frac{\epsilon_{t-1}}{\epsilon_t} F_t^* + R^* B_t^{G*}$$

B.4 Optimal Policy Problem

Given B_1^{G*} and under the assumption that $F_1^* = 0$, consider the constrained planning problem:

$$\begin{aligned}
& \max_{\{N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^r, C_t^p\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \beta^{t-1} [\lambda u(C_t^p) + (1-\lambda)u(C_t^r) - v(N_t)] \quad \text{s.t.} \\
& \sum_{t=1}^{\infty} \frac{\frac{1}{\Gamma} \left(1 - \frac{\tilde{R}_{t+1}}{R^*}\right)^2}{R^{*t-1}} + \sum_{t=1}^{\infty} \frac{P_t^{T*} \theta N_t}{R^{*t-1}} = \sum_{t=1}^{\infty} \frac{\zeta \epsilon_t^x}{R^{*t-1}} + R^* B_1^{G*} + R_1 \frac{\epsilon_0 F_1^*}{\epsilon_1} \\
& \tilde{R}_{t+1} = \frac{u'(C_t^r) \epsilon_t}{\beta u'(C_{t+1}^r) \epsilon_{t+1}} \\
& C_t^r = A_t N_t - \frac{\zeta \epsilon_t^{1+x}}{1-\lambda} + \frac{\lambda}{1-\lambda} \epsilon_t P_t^{T*} \theta N_t \\
& C_t^p = (A_t - \epsilon_t P_t^{T*} \theta) N_t \\
& N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^r, C_t^p \geq 0
\end{aligned} \tag{114}$$

The following Lemmas establish conditions under which a solution to (114) is implementable as a decentralized equilibrium defined in Section B.3.

Lemma B.1. *Given $F_1^* = 0$, B_1^{G*} , $\beta R^* = 1$, and positive sequences $\{A_t, P_t^{T*}\}_{t \geq 1}$ satisfying $A_t \leq \bar{A} < \infty$ and $P_t^{T*} \geq \underline{P} > 0$ for all $t \geq 1$, let $\{N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^r, C_t^p\}_{t \geq 1}$ solve (114). Suppose further that the solution satisfies for all $t \geq 1$:*

$$\epsilon_t \geq \underline{\epsilon}, \quad \underline{C}^r \leq C_t^r \leq \bar{C}^r$$

where $\underline{\epsilon}, \underline{C}^r, \bar{C}^r > 0$. In addition, assume the conditions of Lemma B.2 hold for $w_t = A_t - \epsilon_t P_t^{T*} \theta$ and $\bar{u}_t' = \lambda u'(C_t^p) + (1-\lambda)u'(C_t^r)$. Then there exists a non-unique sequence of transfers $\{\tau_t/P_t\}_{t \geq 1}$ such that the solution to (114) is implementable as a decentralized equilibrium defined in Section B.3.

Proof. Construct $w_t, M_t, Y_t, X_t, F_{t+1}^*$, and R_{t+1} directly:

$$\begin{aligned}
w_t &= A_t - \epsilon_t P_t^{T*} \theta, & M_t &= \theta N_t, & Y_t &= A_t N_t, & X_t &= \zeta \epsilon_t^{1+x} \\
F_{t+1}^* &= \frac{1}{\Gamma} \left(1 - \frac{\tilde{R}_{t+1}}{R^*}\right) \\
R_{t+1} &= \tilde{R}_{t+1} \frac{\epsilon_{t+1}}{\epsilon_t}
\end{aligned}$$

Since Ricardian Euler equation and HtM budget constraint are implementability conditions of (114), they are already satisfied. Using the wage determination equation, firms' zero profit condition: $A_t N_t = w_t N_t + \epsilon_t P_t^{T*} \theta N_t$ also holds. The condition $C_t^r = A_t N_t - \frac{\zeta \epsilon_t^{1+x}}{1-\lambda} + \frac{\lambda}{1-\lambda} \epsilon_t P_t^{T*} \theta N_t$ then implies that the goods market clearing condition holds: $(1-\lambda)C_t^r + \lambda C_t^p + X_t = A_t N_t$.

The Ricardian Euler equation and $\beta R^* = 1$ imply

$$\prod_{s=2}^T \frac{\tilde{R}_s}{R^*} = \frac{\epsilon_1 u'(C_1^r)}{\epsilon_T u'(C_T^r)}$$

Since $C_t^p = (A_t - \epsilon_t P_t^{T*} \theta) N_t > 0$ and $N_t > 0$,

$$\epsilon_t < \frac{A_t}{P_t^{T*} \theta} \leq \frac{\bar{A}}{\underline{P} \theta}$$

The lower bound on ϵ_t and the bounds assumptions on C_t^r therefore imply that there exist constants $0 < \underline{m} < \bar{m} < \infty$ such that $\underline{m} \leq \epsilon_t u'(C_t^r) \leq \bar{m}$ for all $t \geq 1$. Hence the interest rate consistency condition holds:

$$0 < \inf_{T \geq 2} \prod_{s=2}^T \frac{\tilde{R}_s}{R^*} \leq \sup_{T \geq 2} \prod_{s=2}^T \frac{\tilde{R}_s}{R^*} < \infty$$

Choose any sequence of transfers $\{T_t/P_t\}_{t \geq 1}$ satisfying

$$\sum_{t=1}^{\infty} \frac{(1-\lambda)T_t/P_t}{\prod_{s=2}^t R_s} = \sum_{t=1}^{\infty} \frac{C_t - w_t N_t}{\prod_{s=2}^t R_s} - R_1 B_1 \quad (115)$$

where $C_t = (1-\lambda)C_t^r + \lambda C_t^p$. This choice is non-unique. One admissible choice sets $(1-\lambda)T_1/P_1$ equal to the right-hand side of (115) and sets $T_t/P_t = 0$ for all $t \geq 2$. Using goods market clearing and zero profits, we have $C_t - w_t N_t = \epsilon_t (P_t^{T*} \theta N_t - \zeta \epsilon_t^x)$. Then

$$\frac{C_t - w_t N_t}{\prod_{s=2}^t R_s} = \epsilon_1 \frac{P_t^{T*} \theta N_t - \zeta \epsilon_t^x}{\prod_{s=2}^t \tilde{R}_s}$$

By the interest rate consistency condition, there exists $K < \infty$ such that $\frac{1}{\prod_{s=2}^t \tilde{R}_s} \leq \frac{K}{R^{*t-1}}$ for all $t \geq 1$. The intertemporal budget constraint in (114) implies

$$\sum_{t=1}^{\infty} \left| \frac{C_t - w_t N_t}{\prod_{s=2}^t R_s} \right| \leq K \epsilon_1 \sum_{t=1}^{\infty} \frac{P_t^{T*} \theta N_t + \zeta \epsilon_t^x}{R^{*t-1}} < \infty$$

Define $\{B_{t+1}\}_{t \geq 1}$ recursively by the consolidated household budget

$$B_{t+1} = w_t N_t + R_t B_t + (1-\lambda) \frac{T_t}{P_t} - C_t, \quad B_1 \text{ given} \quad (116)$$

Since $C_t^p = w_t N_t$ holds, subtracting $\lambda/(1-\lambda)$ times this from (116) and dividing by $(1-\lambda)$ yields the Ricardian budget:

$$C_t^r + \frac{B_{t+1}}{1-\lambda} = w_t N_t + R_t \frac{B_t}{1-\lambda} + \frac{T_t}{P_t}$$

Iterating (116) from $t = 1$ to T gives

$$\frac{B_{T+1}}{\prod_{s=2}^T R_s} = R_1 B_1 + \sum_{t=1}^T \frac{w_t N_t + (1-\lambda)T_t/P_t - C_t}{\prod_{s=2}^t R_s}$$

By (115), the right-hand side converges to zero as $T \rightarrow \infty$, so $\lim_{T \rightarrow \infty} \frac{B_{T+1}}{\prod_{s=2}^T R_s} = 0$.

Construct B_{t+1}^G from the bonds market clearing condition:

$$B_{t+1}^G = \epsilon_t F_{t+1}^* - B_{t+1}$$

Since $F_1^* = 0$, the initial condition $B_1^G = \epsilon_0 F_1^* - B_1 = -B_1$ holds.

Construct B_{t+1}^{G*} from the period-by-period BoP conditions:

$$B_{t+1}^{G*} = \zeta \epsilon_t^\chi + R_t \frac{\epsilon_{t-1}}{\epsilon_t} F_t^* + R^* B_t^{G*} - P_t^{T*} \theta N_t - F_{t+1}^*$$

with $F_1^* = 0, B_1^{G*}$ given. Since the solution satisfies the intertemporal budget constraint, it satisfies the no-Ponzi condition on the net foreign asset position:

$$\lim_{T \rightarrow \infty} \frac{B_{T+1}^{G*} + F_{T+1}^*}{R^{*T-1}} = 0$$

Moreover, since the solution satisfies:

$$\sum_{t=1}^{\infty} \frac{\frac{1}{\tilde{r}} \left(1 - \frac{\tilde{R}_{t+1}}{R^*}\right)^2}{R^{*t-1}} = \sum_{t=1}^{\infty} \frac{\Gamma(F_{t+1}^*)^2}{R^{*t-1}} < \infty$$

therefore $(F_{T+1}^*)^2 / R^{*T-1} \rightarrow 0$. Hence, the following hold separately

$$\lim_{T \rightarrow \infty} \frac{F_{T+1}^*}{R^{*T-1}} = 0, \quad \lim_{T \rightarrow \infty} \frac{B_{T+1}^{G*}}{R^{*T-1}} = 0$$

Using the bonds market clearing condition we have:

$$\frac{B_{T+1}^G}{\prod_{s=2}^T R_s} = \frac{\epsilon_1 F_{T+1}^*}{\prod_{s=2}^T \tilde{R}_s} - \frac{B_{T+1}}{\prod_{s=2}^T R_s}$$

Taking limits: $T \rightarrow \infty$, the second term on the RHS vanishes by Household no-Ponzi condition holding. For the first: since $F_{T+1}^* / R^{*T-1} \rightarrow 0$ and $\sup_T \prod_{s=2}^T (R^* / \tilde{R}_s) < \infty$, then $\frac{F_{T+1}^*}{\prod_{s=2}^T \tilde{R}_s} = \frac{F_{T+1}^*}{R^{*T-1}} \cdot \prod_{s=2}^T \frac{R^*}{\tilde{R}_s} \Rightarrow 0$.

Hence $\lim_{T \rightarrow \infty} \frac{B_{T+1}^G}{\prod_{s=2}^T R_s} = 0$.

The period-by-period government budget constraints hold by Walras' identity: multiplying period-by-period BoP conditions by ϵ_t and substituting $C_t + X_t = A_t N_t$ and $w_t N_t = A_t N_t - \epsilon_t P_t^{T*} \theta N_t$ gives

$$\epsilon_t F_{t+1}^* + \epsilon_t B_{t+1}^{G*} = w_t N_t - C_t + R_t \epsilon_{t-1} F_t^* + \epsilon_t R^* B_t^{G*}$$

Substituting $w_t N_t - C_t = B_{t+1} - R_t B_t - (1 - \lambda) T_t / P_t$ and $B_{t+1} + B_{t+1}^G = \epsilon_t F_{t+1}^*$ gives

$$B_{t+1}^G + \epsilon_t B_{t+1}^{G*} + (1 - \lambda) \frac{T_t}{P_t} = R_t B_t^G + \epsilon_t R^* B_t^{G*}$$

Finally, to construct the inflation path and the nominal interest rates, construct the labor wedge as:

$$\tau_t^N = 1 - \frac{v'(N_t)}{\bar{u}'_t w_t}$$

Then construct Π_t as a solution to:

$$1 - \tau_t^N = \frac{1}{\mu_w} + \left[\frac{\left(\frac{w_t}{w_{t-1}} \Pi_t - 1 \right) \frac{w_t}{w_{t-1}} \Pi_t - \beta \left[\left(\frac{w_{t+1}}{w_t} \Pi_{t+1} - 1 \right) \frac{w_{t+1}}{w_t} \Pi_{t+1} \right]}{\frac{\epsilon_w}{\psi_{nr}} w_t N_t \bar{u}'_t} \right]$$

By Lemma B.2, there exists $\{\Pi_t\}_{t \geq 1} \subset (0, \infty)$ satisfying the above equation.

Construct R_1, R_{t+1}^M using:

$$R_1 = \frac{R_1^M}{\Pi_1}, \quad R_{t+1}^M = R_{t+1} \Pi_{t+1}, \quad t \geq 1$$

All decentralized equilibrium conditions in Section B.3 are therefore satisfied. Non-uniqueness follows from the non-uniqueness of any transfer sequence satisfying (115). $\square$

Lemma B.2. *Let $\{N_t, w_t, \bar{u}'_t\}_{t \geq 1}$ be strictly positive sequences, with $w_0 > 0$ given. Suppose that, for every $t \geq 1$,*

$$\mathcal{X}_t \equiv \sum_{j=t}^{\infty} \beta^{j-t} \frac{\epsilon_w}{\psi_{nr}} \left[N_j v'(N_j) - \frac{1}{\mu_w} w_j N_j \bar{u}'_j \right] \text{ is finite and lies in } \left(-\frac{1}{4}, \infty \right) \quad (117)$$

Then there exists a sequence $\{\Pi_t\}_{t \geq 1} \subset (0, \infty)$ such that the wage Phillips curve (54) holds for all $t \geq 1$.

Proof. Define

$$\Pi_t = \frac{w_{t-1}}{w_t} \frac{1 + \sqrt{1 + 4\mathcal{X}_t}}{2}$$

By (117), $\Pi_t > 0$ for all $t \geq 1$ and

$$\mathcal{X}_t = \left(\frac{w_t}{w_{t-1}} \Pi_t - 1 \right) \frac{w_t}{w_{t-1}} \Pi_t$$

Moreover, the definition of $\mathcal{X}_t$ implies

$$\mathcal{X}_t = \frac{\epsilon_w}{\psi_{nr}} \left[N_t v'(N_t) - \frac{1}{\mu_w} w_t N_t \bar{u}'_t \right] + \beta \mathcal{X}_{t+1}$$

Therefore the wage Phillips curve holds for all $t \geq 1$. $\square$

Lemma B.3. *Given $F_1^* = 0$, B_1^{G*} , $\beta R^* = 1$, $w_0 > 0$, and positive sequences $\{A_t, P_t^{T*}\}_{t \geq 1}$ satisfying*

$A_t \rightarrow A_{ss} > 0$, and $P_t^{T*} \rightarrow P_{ss}^{T*} > 0$, let $\{N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^r, C_t^p\}_{t \geq 1}$ solve (114) with

$$N_t \rightarrow N_{ss} > 0, \quad \epsilon_t \rightarrow \epsilon_{ss} > 0, \quad C_t^r \rightarrow C_{ss}^r > 0, \quad C_t^p \rightarrow C_{ss}^p > 0, \quad \tilde{R}_{t+1} \rightarrow R^* > 0$$

Suppose also that condition (117) holds for every $t \geq 1$. Then this solution is implementable as a decentralized equilibrium. Moreover, under the larger-root convention, the inflation path $\{\Pi_t\}_{t \geq 1} \subset (0, \infty)$ satisfying the wage Phillips curve and converging to a positive finite limit is unique and converges to

$$\Pi_{ss} = \frac{1}{2} + \sqrt{\frac{1}{4} + \frac{1}{1-\beta} \frac{\epsilon_w}{\psi_{nr}} \left[N_{ss} v'(N_{ss}) - \frac{1}{\mu_w} w_{ss} N_{ss} \bar{u}'_{ss} \right]}$$

where $w_{ss} = A_{ss} - \epsilon_{ss} P_{ss}^{T*} \theta$.

Proof. Since $A_t \rightarrow A_{ss} > 0$ and $P_t^{T*} \rightarrow P_{ss}^{T*} > 0$, there exist $\bar{A} < \infty$ and $\underline{P} > 0$ such that $A_t \leq \bar{A}$ and $P_t^{T*} \geq \underline{P}$ for all $t \geq 1$. Since $\epsilon_t \rightarrow \epsilon_{ss} > 0$ and $C_t^r \rightarrow C_{ss}^r > 0$, there exist $\underline{\epsilon}, \underline{C}^r, \bar{C}^r > 0$ such that $\epsilon_t \geq \underline{\epsilon}$ and $\underline{C}^r \leq C_t^r \leq \bar{C}^r$ for all $t \geq 1$. Hence all conditions of Lemma B.1 are satisfied. Implementability follows as a corollary. Moreover by Lemma B.2, there exists an inflation path $\{\Pi_t\}_{t \geq 1} \subset (0, \infty)$ satisfying the wage Phillips curve. It remains to show that the inflation path is unique. Let

$$\mathcal{A}_t = \frac{\epsilon_w}{\psi_{nr}} \left[N_t v'(N_t) - \frac{1}{\mu_w} w_t N_t \bar{u}'_t \right]$$

Since $w_t = A_t - \epsilon_t P_t^{T*} \theta$, convergence gives $w_t \rightarrow w_{ss} > 0$ then

$$\mathcal{A}_t \rightarrow \mathcal{A}_{ss} = \frac{\epsilon_w}{\psi_{nr}} \left[N_{ss} v'(N_{ss}) - \frac{1}{\mu_w} w_{ss} N_{ss} \bar{u}'_{ss} \right]$$

For any inflation path satisfying the wage Phillips curve, the associated $\mathcal{X}_t$ satisfies

$$\mathcal{X}_t = \mathcal{A}_t + \beta \mathcal{X}_{t+1}$$

Thus

$$\mathcal{X}_t = \sum_{j=t}^{\infty} \beta^{j-t} \mathcal{A}_j + \kappa \beta^{-t}, \quad \kappa \in \mathbb{R}$$

Since $\mathcal{A}_t \rightarrow \mathcal{A}_{ss}$,

$$\sum_{j=t}^{\infty} \beta^{j-t} \mathcal{A}_j \rightarrow \frac{\mathcal{A}_{ss}}{1-\beta}$$

If $\kappa < 0$, then $\mathcal{X}_t \rightarrow -\infty$, so $\mathcal{X}_t < -1/4$ for all sufficiently large t , which is incompatible with $\Pi_t > 0$. If $\kappa > 0$, then $\mathcal{X}_t \rightarrow +\infty$, so the larger-root inflation path cannot converge to a finite limit. Hence any positive larger-root inflation path with a finite limit must have $\kappa = 0$. In that case,

$$\mathcal{X}_t = \sum_{j=t}^{\infty} \beta^{j-t} \mathcal{A}_j \rightarrow \frac{\mathcal{A}_{ss}}{1-\beta}$$

Under the larger-root convention,

$$\Pi_t = \frac{w_{t-1}}{w_t} \frac{1 + \sqrt{1 + 4\mathcal{X}_t}}{2}$$

Since $w_{t-1}/w_t \rightarrow 1$, it follows that

$$\Pi_t \rightarrow \frac{1}{2} + \sqrt{\frac{1}{4} + \frac{1}{1-\beta} \frac{\varepsilon_w}{\psi_{nr}} \left[N_{ss} v'(N_{ss}) - \frac{1}{\mu_w} w_{ss} N_{ss} \bar{u}'_{ss} \right]} \equiv \Pi_{ss} \in (0, \infty)$$

The case $\kappa = 0$ is the only case admitting a positive larger-root inflation path with a finite positive limit. Hence the convergent inflation path is unique. $\square$

B.5 Calibration

We calibrate the quantitative model to Japan and interpret the imported input as fossil fuel imports. A period is one quarter. The calibration is designed for MIT shocks to the imported-input price, $P_t^{\top*}$, and chooses the no-shock steady state as the initial state for all policy experiments. Table B.1 reports the parameter values used in the quantitative exercises.

The normalizations are $A_{ss} = 1$, $P_{ss}^{\top*} = 1$, $\theta = 1$, and $N_{ss} = 1$. The scale parameter ζ is chosen jointly with the steady real exchange rate to set the fossil-fuel import bill at 4 percent of GDP and net exports at -0.5 percent of GDP in the no-shock steady state. Since the no-shock steady state has $F^* = 0$, the public foreign asset position B^{G*} is calibrated to make the domestic value of net foreign assets equal to 60 percent of annual GDP. The world gross interest rate R^* is then chosen to satisfy the steady-state balance of payments at this net-export target.

Preferences use $u(C) = C^{1-\sigma}/(1-\sigma)$ and $v(N) = \eta N^{1+\phi}/(1+\phi)$. The household share λ is set to 0.35, in the range of estimates of hand-to-mouth households in [Kaplan, Violante and Weidner \(2014\)](#). The coefficient of relative risk aversion is $\sigma = 4$. In this model, χ is the Marshall-Lerner elasticity: it is the elasticity of foreign-currency export revenue, $X_t/\varepsilon_t = \zeta \varepsilon_t^\chi$, with respect to the real exchange rate. We set $\chi = 0.25$, so a one percent real depreciation raises the foreign purchasing power supplied by exports by 0.25 percent, equivalently the export-demand price elasticity is $1 + \chi = 1.25$. This imposes a positive but modest Marshall-Lerner response, consistent with empirical work that finds limited short-run trade-balance responses to exchange-rate movements; see [Rose \(1991\)](#) and the survey in [Bahmani-Oskooee and Ratha \(2004\)](#). The financial-intermediation parameter is set to $\Gamma = 9$, following [Fanelli and Straub \(2021\)](#). The wage elasticity is $\varepsilon_w = 6$, corresponding to a steady wage markup $\mu_w = 1.2$, as in the standard staggered-wage formulation of [Erceg, Henderson and Levin \(2000\)](#).

The labor-disutility curvature parameter ϕ and the nominal wage rigidity parameter ψ_{nr} discipline the response to the benchmark import-price shock. We set $\phi = 29.5$, at which a doubling of import prices lowers gross output, $A_t N_t$, by 1 percent in the representative-agent benchmark, where no household is borrowing constrained. We set $\psi_{nr} = 28000$, at which the same benchmark produces impact inflation of 5 percent after a doubling of import prices. The persistence of the imported-input price process is $\rho_{PT*} = 0.945$, estimated

Table B.1: Calibration

Parameter	Description	Value	Target or source
R^*	Gross world interest rate	1.0021	Net exports of -0.5 percent of GDP
β	Discount factor	0.9979	$1/R^*$
σ	Relative risk aversion	4.0	Standard value
η	Labor-disutility scale	0.9261	$N_{ss} = 1$ and steady labor FOC
ϕ	Inverse Frisch curvature	29.50	1 percent output decline after doubling P_t^{T*} in representative-agent benchmark
λ	HtM household share	0.35	Kaplan, Violante and Weidner (2014)
ζ	Export demand scale	1.9758	Fossil-fuel import bill of 4 percent of GDP and net exports of -0.5 percent of GDP
χ	Marshall-Lerner elasticity	0.25	Modest short-run response; Rose (1991) , Bahmani-Oskooee and Ratha (2004)
ρ_{PT^*}	Import-price persistence	0.945	AR(1) on detrended quarterly IMF energy-price index from FRED
Γ	Financial-intermediation friction	9.0	Fanelli and Straub (2021)
ε_w	Wage substitution elasticity	6.0	Wage markup $\mu_w = 1.2$; Erceg, Henderson and Levin (2000)
ψ_{nr}	Nominal wage rigidity	28000.0	5 percent impact inflation after doubling P_t^{T*} in representative-agent benchmark
B_{ss}^{G*}	Public foreign asset position	60.0	Net foreign assets of 60 percent of annual GDP

from an AR(1) regression on detrended quarterly observations of the IMF’s Global Price of Energy Index ([International Monetary Fund 2026](#)).

B.6 Representative-Agent Benchmark

This appendix reports the responses of the representative-agent economy, in which no household is borrowing constrained and every household participates in international asset markets and remains on its Euler equation, to the same two imported-energy price paths studied in the main text. By Proposition 1, optimal policy leaves the intertemporal margin undistorted and sets the terms-of-trade-adjusted labor wedge to zero, so the paths below coincide with the RA-equivalent policy applied in this economy. All deviations are relative to the representative-agent economy’s own no-shock steady state, which differs from the heterogeneous economy’s steady state; comparisons with the main-text figures should therefore be made in proportionate terms only.

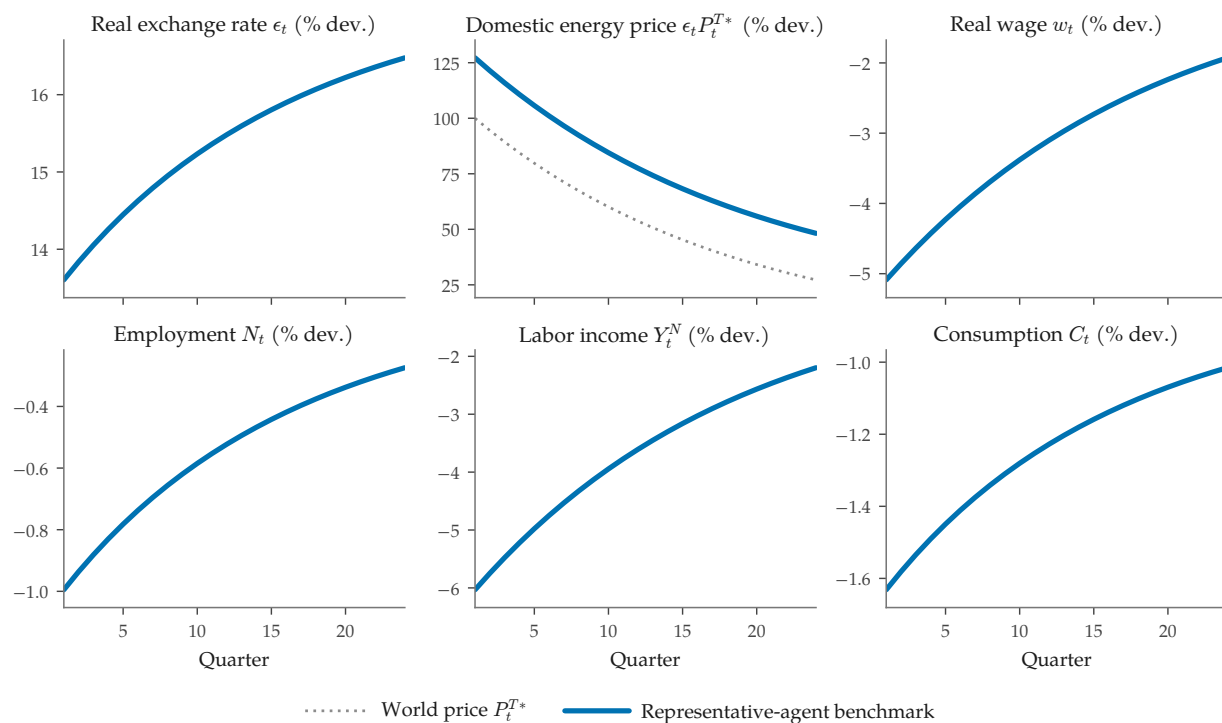


Figure B.1. Representative-agent benchmark: MIT-shock allocations

Note: The dotted line is the world energy price.

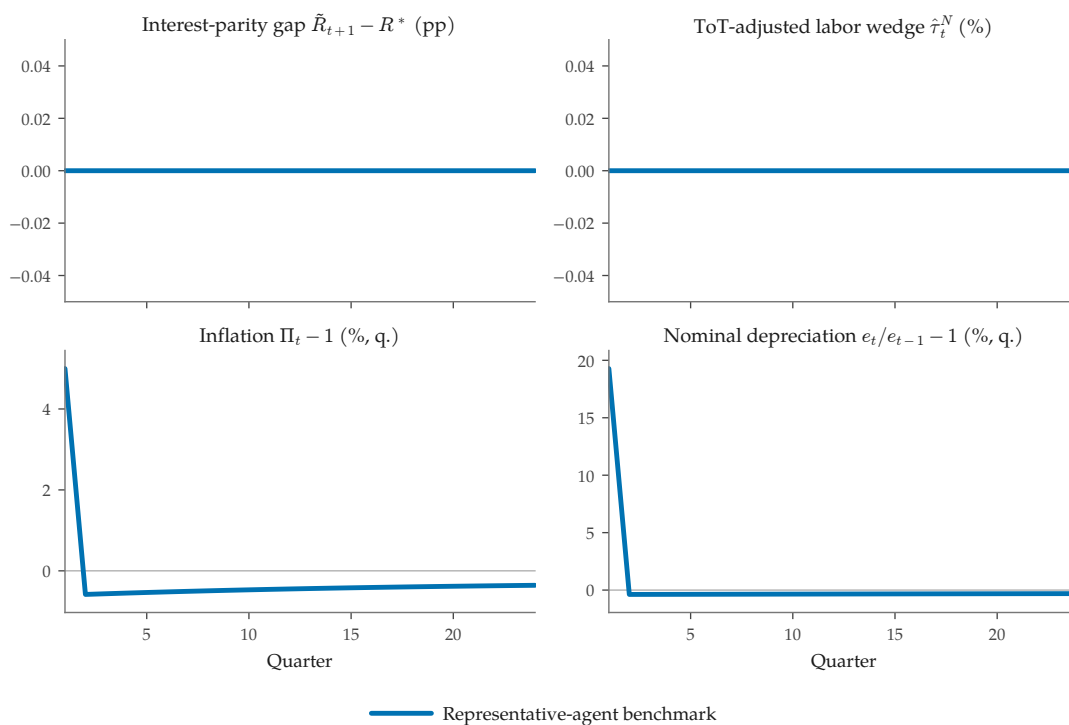


Figure B.2. Representative-agent benchmark: MIT-shock policy

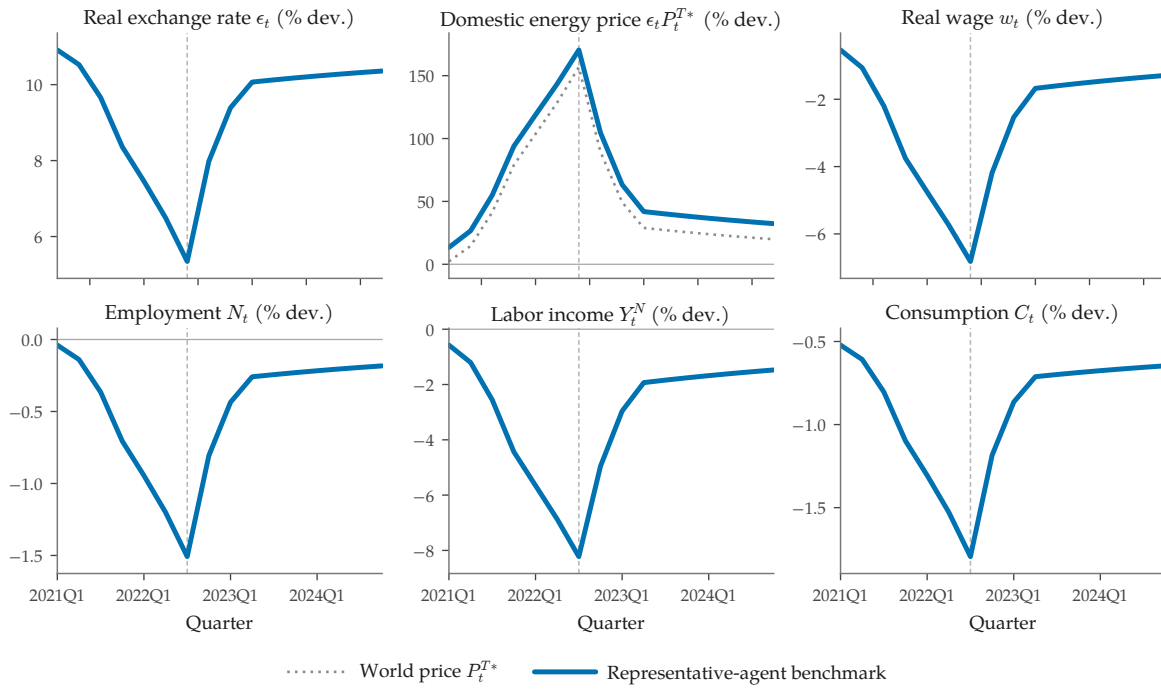


Figure B.3. Representative-agent benchmark: 2021-2023 allocations

Note: The dotted line is the world energy price; the vertical line marks its peak.

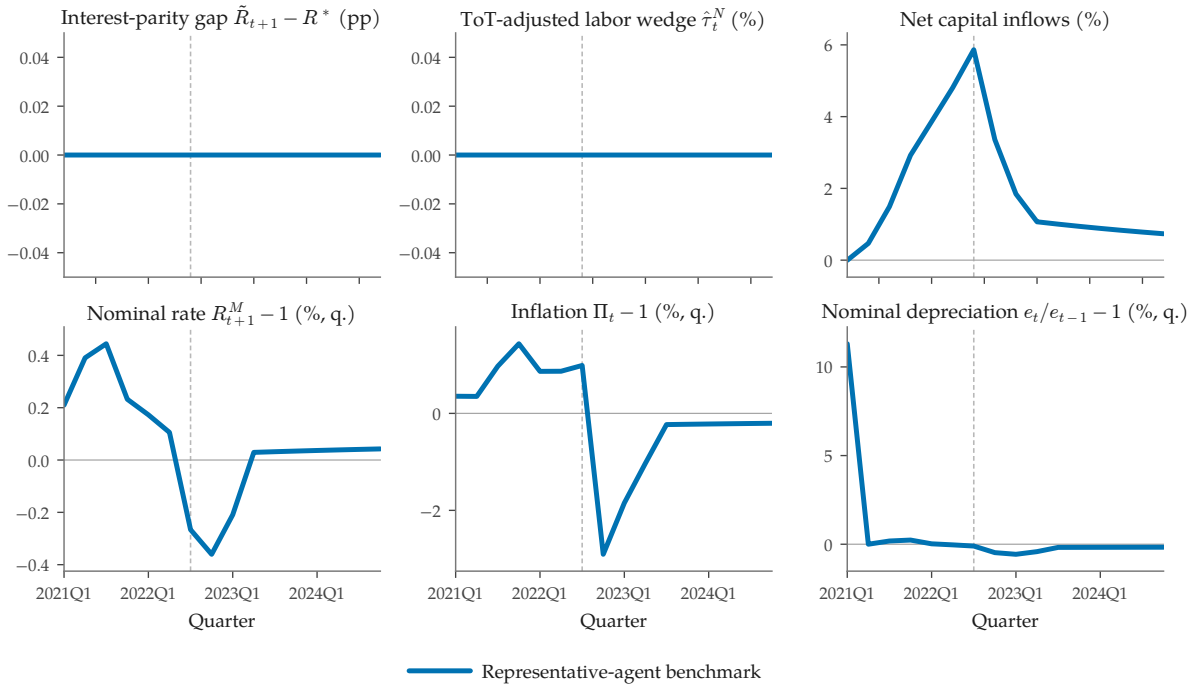


Figure B.4. Representative-agent benchmark: 2021-2023 policy

Note: Net capital inflows are shown as a percent of initial steady-state gross output. The vertical line marks the energy-price peak.

B.7 Resource Shares

The figures below report how gross output is divided between exports, the import bill, and the consumption of the two household types, as shares of gross output, under each policy regime and each of the two price paths. The main text reads the reallocation off these shares in Section 4.

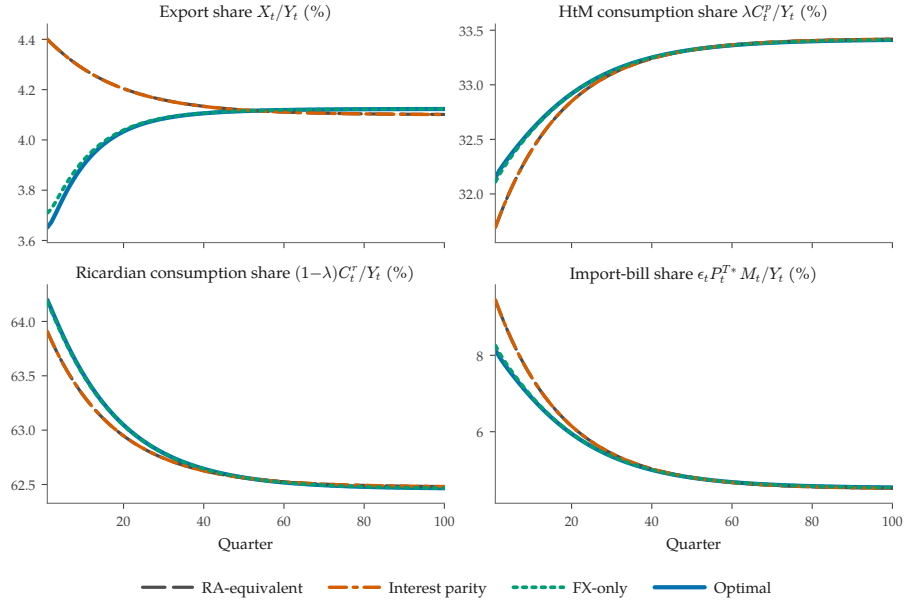


Figure B.5. MIT shock: resource shares

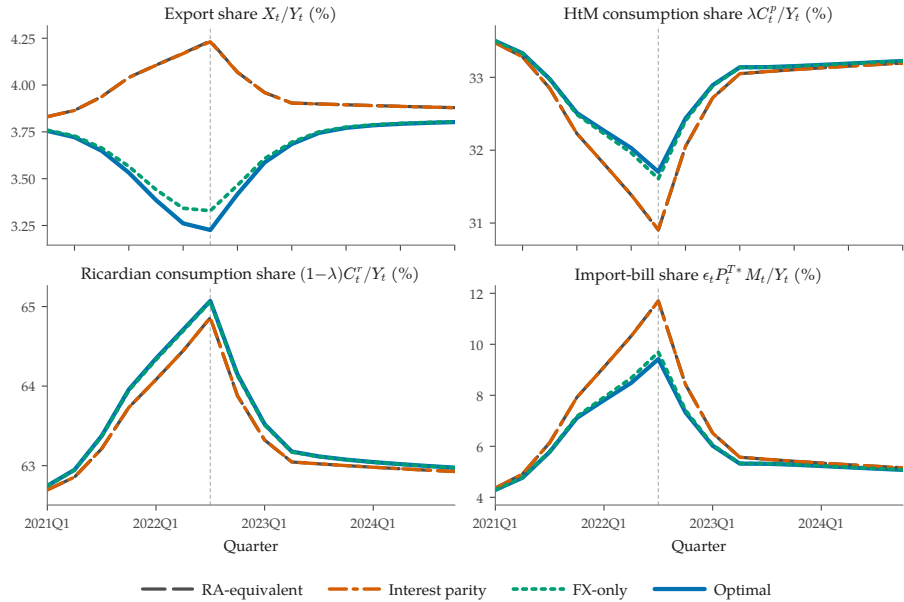


Figure B.6. 2021-2023 shock: resource shares

Note: The vertical line marks the energy-price peak.

C Beyond the Two-Agent Model

C.1 A Two-Agent Model without Lump-Sum Transfers

The main text's quantitative model assumes that the public agent can selectively make lump-sum transfers or impose lump-sum taxes on Ricardian households to finance carry trade losses or to distribute carry trade profits. In this appendix we withdraw this instrument entirely: lump-sum taxes and transfers are unavailable to the public agent. This restriction must be paired with a second one, that returns in domestic asset markets are inflation indexed, i.e., denominated in real domestic units. Without real indexation, the public agent would have an incentive to use inflation ex post to manipulate the real value of Ricardian asset holdings, a time-inconsistency that was absent in the framework with lump-sum transfers and taxes for Ricardian households.

The Ricardian households' budget constraint is now given by:

$$C_t^r + B_{t+1}^r = w_t N_t + R_t B_t^r \quad (118)$$

The Ricardian Euler equation and the no-Ponzi-transversality condition on the household budget, $\lim_{T \rightarrow \infty} \frac{B_{T+1}^r}{\prod_{s=2}^T R_s} = 0$, are unchanged. The period-by-period fiscal budget constraint is now given by:

$$B_{t+1}^G + \epsilon_t B_{t+1}^{G*} = R_t B_t^G + \epsilon_t R^* B_t^{G*} \quad (119)$$

together with unchanged no-Ponzi conditions:

$$\lim_{T \rightarrow \infty} \frac{B_{T+1}^G}{\prod_{s=2}^T R_s} = 0, \quad \lim_{T \rightarrow \infty} \frac{B_{T+1}^{G*}}{R^{*T-1}} = 0$$

and an unchanged consistency condition on interest rates which ensures that government asset positions satisfy no-Ponzi conditions in domestic as well as foreign units:

$$0 < \inf_{T \geq 2} \prod_{s=2}^T \frac{\tilde{R}_s}{R^*} \leq \sup_{T \geq 2} \prod_{s=2}^T \frac{\tilde{R}_s}{R^*} < \infty$$

The choice of monetary policy is now represented directly by a money-growth rule: money supply satisfies the standard exchange equation $\mathcal{M}_t = P_t Y_t$, and the monetary agent sets

$$g_t^{\mathcal{M}} = \Pi_t \frac{Y_t}{Y_{t-1}} \quad (120)$$

This rule was already implicitly present in the main text's quantitative model, but the choice of monetary policy was represented through the monetary agent choosing the ex-post real rate, R_1 , and the forward-looking nominal interest rates, R_{t+1}^M . Because bonds are denominated in real domestic units, the ex-post real rate, R_1 , is a state variable independent of the ex-post choice of inflation, while inflation itself remains under the monetary agent's control through the money-growth rule.

The balance of payments condition and the remaining equilibrium conditions of Appendix B.3 are unchanged. Given

- initial states $\{R_1, B_1, B_1^{G*}, \epsilon_0, F_1^*, w_0, Y_0\}$,
- an exogenous sequence of import prices and productivity $\{P_t^{T*}, A_t\}_{t=1}^\infty$, and
- fiscal-monetary policy $\{B_{t+1}^G, B_{t+1}^{G*}, g_t^M\}_{t=1}^\infty$,

a decentralized equilibrium is a sequence of allocations $\{C_t^r, C_t^p, B_{t+1}, N_t, M_t, Y_t, F_{t+1}^*, X_t\}_{t=1}^\infty$ and prices $\{\epsilon_t, w_t, \Pi_t, R_{t+1}\}_{t=1}^\infty$ that satisfy the conditions of Appendix B.3, with equations (118), (119), and (120) replacing the corresponding conditions.

C.1.1 Optimal Policy Problem

As in the main text, we restrict optimal policy to the case where $F_1^* = 0$ to avoid any time-inconsistency arising from the public agent's incentive to ex-post manipulate the foreign-unit value of the claims on financial intermediaries. With transfers absent, the Ricardian households' present-value budget constraint becomes an additional implementability constraint in the optimal policy problem, and the initial claims on Ricardian households, $R_1 B_1 / (1 - \lambda)$, become a state variable of the problem. Given B_1^{G*} , $R_1 B_1$, and $F_1^* = 0$, the Ramsey planner's optimization problem is to solve:

$$\begin{aligned}
& \max_{\{N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^r, C_t^p\}_{t=1}^\infty} \sum_{t=1}^\infty \beta^{t-1} [\lambda u(C_t^p) + (1 - \lambda)u(C_t^r) - v(N_t)] \quad \text{s.t.} \\
& \sum_{t=1}^\infty \frac{\frac{1}{r} \left(1 - \frac{\tilde{R}_{t+1}}{R^*}\right)^2}{R^{*t-1}} + \sum_{t=1}^\infty \frac{P_t^{T*} \theta N_t}{R^{*t-1}} = \sum_{t=1}^\infty \frac{\zeta \epsilon_t^x}{R^{*t-1}} + R^* B_1^{G*} + R_1 \frac{\epsilon_0 F_1^*}{\epsilon_1} \\
& \sum_{t=1}^\infty \beta^{t-1} u'(C_t^r) [C_t^r - (A_t - \epsilon_t P_t^{T*} \theta) N_t] = \frac{R_1 B_1}{1 - \lambda} u'(C_1^r) \\
& \tilde{R}_{t+1} = \frac{u'(C_t^r) \epsilon_t}{\beta u'(C_{t+1}^r) \epsilon_{t+1}} \\
& C_t^r = A_t N_t - \frac{\zeta \epsilon_t^{1+x}}{1 - \lambda} + \frac{\lambda}{1 - \lambda} \epsilon_t P_t^{T*} \theta N_t \\
& C_t^p = (A_t - \epsilon_t P_t^{T*} \theta) N_t \\
& N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^r, C_t^p \geq 0
\end{aligned} \tag{121}$$

The following lemma establishes conditions under which a solution to (121) is implementable as a decentralized equilibrium.

Lemma C.1. Given $F_1^* = 0$, B_1 , B_1^{G*} , $R_1 > 0$, $\beta R^* = 1$, and positive sequences $\{A_t, P_t^{T*}\}_{t \geq 1}$ satisfying $A_t \leq \bar{A} < \infty$ and $P_t^{T*} \geq \underline{P} > 0$ for all $t \geq 1$, let $\{N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^r, C_t^p\}_{t \geq 1}$ solve (121). Suppose further that the solution satisfies, for all $t \geq 1$,

$$\epsilon_t \geq \underline{\epsilon}, \quad \underline{C}^r \leq C_t^r \leq \bar{C}^r$$

where $\underline{\epsilon}, \underline{C}^r, \bar{C}^r > 0$, and that the conditions of Lemma B.2 hold for $w_t = A_t - \epsilon_t P_t^{T*} \theta$ and $\bar{u}'_t = \lambda u'(C_t^p) + (1 - \lambda) u'(C_t^r)$. Then the Ramsey allocation is implementable as a decentralized equilibrium of the economy without transfers. Conditional on the Ramsey allocation and the initial positions, the implementing paths $\{B_{t+1}, B_{t+1}^G, B_{t+1}^{G*}\}_{t \geq 1}$ are unique.

Proof. Construct $w_t, M_t, Y_t, X_t, F_{t+1}^*$, and R_{t+1} as in the proof of Lemma B.1:

$$w_t = A_t - \epsilon_t P_t^{T*} \theta, \quad M_t = \theta N_t, \quad Y_t = A_t N_t, \quad X_t = \zeta \epsilon_t^{1+\chi}$$

$$F_{t+1}^* = \frac{1}{\Gamma} \left(1 - \frac{\tilde{R}_{t+1}}{R^*} \right), \quad R_{t+1} = \tilde{R}_{t+1} \frac{\epsilon_{t+1}}{\epsilon_t}$$

The constraint $\tilde{R}_{t+1} = u'(C_t^r) \epsilon_t / [\beta u'(C_{t+1}^r) \epsilon_{t+1}]$ and the construction of R_{t+1} imply the Ricardian Euler equation for real-indexed bonds, $u'(C_t^r) = \beta R_{t+1} u'(C_{t+1}^r)$. The HtM budget constraint, firms' zero-profit condition, and goods market clearing hold as in the proof of Lemma B.1, and the interest-rate consistency condition follows by the same argument, which uses only the bounds on $\{A_t, P_t^{T*}, \epsilon_t, C_t^r\}$ assumed here.

The Euler equation and $\beta R^* = 1$ imply

$$\frac{1}{\prod_{s=2}^t R_s} = \beta^{t-1} \frac{u'(C_t^r)}{u'(C_1^r)}$$

Let $C_t = (1 - \lambda) C_t^r + \lambda C_t^p$. Since $C_t^p = w_t N_t$, we have $C_t - w_t N_t = (1 - \lambda) [C_t^r - w_t N_t]$. Dividing the household implementability constraint in (121) by $u'(C_1^r)$, multiplying by $(1 - \lambda)$, and substituting the two displays above shows that the constraint is equivalent to

$$\sum_{t=1}^{\infty} \frac{C_t - w_t N_t}{\prod_{s=2}^t R_s} = R_1 B_1 \quad (122)$$

The series converges absolutely: goods market clearing and zero profits give $C_t - w_t N_t = \epsilon_t (P_t^{T*} \theta N_t - \zeta \epsilon_t^{\chi})$, so

$$\frac{C_t - w_t N_t}{\prod_{s=2}^t R_s} = \epsilon_1 \frac{P_t^{T*} \theta N_t - \zeta \epsilon_t^{\chi}}{\prod_{s=2}^t \tilde{R}_s}$$

and, by the interest-rate consistency condition, there exists $K < \infty$ with $1 / \prod_{s=2}^t \tilde{R}_s \leq K / R^{*t-1}$, so the external intertemporal budget constraint in (121) bounds the series by $K \epsilon_1 \sum_t (P_t^{T*} \theta N_t + \zeta \epsilon_t^{\chi}) / R^{*t-1} < \infty$.

Define $\{B_{t+1}\}_{t \geq 1}$ recursively by the consolidated household budget

$$B_{t+1} = w_t N_t + R_t B_t - C_t, \quad B_1 \text{ given} \quad (123)$$

Since $C_t^p = w_t N_t$, subtracting $\lambda/(1 - \lambda)$ times the HtM budget from (123) and dividing by $(1 - \lambda)$ yields the Ricardian budget constraint (118) with $B_t^r = B_t/(1 - \lambda)$. Iterating (123) from $t = 1$ to T gives

$$\frac{B_{T+1}}{\prod_{s=2}^T R_s} = R_1 B_1 + \sum_{t=1}^T \frac{w_t N_t - C_t}{\prod_{s=2}^t R_s}$$

By (122) and absolute convergence, the right-hand side converges to zero as $T \rightarrow \infty$, so the household no-Ponzi condition $\lim_{T \rightarrow \infty} B_{T+1}/\prod_{s=2}^T R_s = 0$ holds.

Construct B_{t+1}^G from the bonds market clearing condition, $B_{t+1}^G = \epsilon_t F_{t+1}^* - B_{t+1}$; since $F_1^* = 0$, the initial condition $B_1^G = -B_1$ holds. Construct B_{t+1}^{G*} from the period-by-period balance of payments,

$$B_{t+1}^{G*} = \zeta \epsilon_t^X + R_t \frac{\epsilon_{t-1}}{\epsilon_t} F_t^* + R^* B_t^{G*} - p_t^{T*} \theta N_t - F_{t+1}^*, \quad B_1^{G*} \text{ given}$$

Since the solution satisfies the external intertemporal budget constraint, the no-Ponzi condition on the net foreign asset position, $\lim_{T \rightarrow \infty} (B_{T+1}^{G*} + F_{T+1}^*)/R^{*T-1} = 0$, holds; the carry-cost summability in (121) gives $(F_{T+1}^*)^2/R^{*T-1} \rightarrow 0$, so the two limits hold separately; and the bonds market clearing condition, the household no-Ponzi condition, and the interest-rate consistency condition give $\lim_{T \rightarrow \infty} B_{T+1}^G/\prod_{s=2}^T R_s = 0$, as in the proof of Lemma B.1.

The fiscal budget constraint (119) holds period by period by Walras' identity: multiplying the balance of payments by ϵ_t and substituting $C_t + X_t = A_t N_t$ and $w_t N_t = A_t N_t - \epsilon_t p_t^{T*} \theta N_t$ gives

$$\epsilon_t F_{t+1}^* + \epsilon_t B_{t+1}^{G*} = w_t N_t - C_t + R_t \epsilon_{t-1} F_t^* + \epsilon_t R^* B_t^{G*}$$

and substituting $w_t N_t - C_t = B_{t+1} - R_t B_t$ from (123) and $B_{t+1} + B_{t+1}^G = \epsilon_t F_{t+1}^*$ yields (119).

Finally, construct the labor wedge $\tau_t^N = 1 - v'(N_t)/(\bar{u}'_t w_t)$ and, by Lemma B.2, a sequence $\{\Pi_t\}_{t \geq 1} \subset (0, \infty)$ satisfying the wage Phillips curve. Set

$$g_t^{\mathcal{M}} = \Pi_t \frac{Y_t}{Y_{t-1}}$$

which implements this inflation path under $\mathcal{M}_t = p_t Y_t$. All decentralized-equilibrium conditions of the economy without transfers therefore hold.

For uniqueness, fix the Ramsey allocation and the initial positions (B_1, B_1^{G*}, R_1) . The recursion (123) determines $\{B_{t+1}\}_{t \geq 1}$ uniquely, the bonds market clearing condition then determines $\{B_{t+1}^G\}_{t \geq 1}$ uniquely, and the balance-of-payments recursion determines $\{B_{t+1}^{G*}\}_{t \geq 1}$ uniquely. Unlike Lemma B.1, no transfer sequence is available to reallocate claims across dates, and the implementing asset paths therefore admit no other solution. $\square$

C.1.2 Optimal Policy at the Quantitative Calibration

We solve the Ramsey problem (121) at the calibration of Appendix B.5, for the heterogeneous economy ($\lambda = 0.35$) and the representative-agent economy ($\lambda = 0$), under the MIT shock of Section 4. Under constant

import prices the computed solution reproduces the no-shock steady state at both values of λ : the multiplier on the household present-value constraint is zero, and the inherited claims $R_1 B_1$ create no incentive to distort any margin at date one. Real indexation of domestic bonds removes the incentive to revalue these claims through surprise inflation by construction. The constant-price solution shows that the inherited claims support no further distortion at this Ramsey steady state, and the removal of transfers therefore leaves steady-state policy unchanged.

Under the shock, however, this coincidence with the transfer economy ends, and it ends already in the representative-agent case: the planner opens interest parity gaps and moves the labor wedge (Figure C.1). With tariffs absent, the static terms-of-trade distortion, which arises because additional export revenue is earned only at a lower export price, falls on the wedges the planner retains, and the import price path makes the correction time varying. Under Leontief production this alone creates no intertemporal motive: the period-by-period labor wedge implements the entire correction, as in the comparative statics of Section 3.5, which is why the economy with transfers keeps interest parity at $\lambda = 0$. The household present-value constraint removes this assignment, since the allocation that corrects the distortion period by period need not satisfy the constraint at the inherited claims $R_1 B_1$; both margins share the correction, and neither wedge isolates it.

Heterogeneity enters on top of this correction, and its contribution is the difference between the two paths in Figure C.1. Relative to the $\lambda = 0$ economy, the planner holds the parity gap lower throughout: smaller at impact, and negative while the import price remains high. The real exchange rate barely depreciates on impact and approaches its long-run level only as the import price recedes. The domestic price of the import is therefore lower, and the real wage higher, in the quarters in which the marginal utility of hand-to-mouth households is highest. The nominal depreciation is deferred in the same way, while the foreign asset drawdown that finances the import bill is nearly identical across the two economies. Monetary policy retains its complementary role: relative to the $\lambda = 0$ path, impact inflation is lower and the labor wedge larger. The appreciation lowers the imported-input cost, the wedge keeps employment and imported-input demand from expanding in response, and the cost reduction passes into the real wage. Exchange rate policy sets the intertemporal distortion. The real-income stabilization channel of the main text therefore survives the removal of transfers: both wedges mix the terms-of-trade correction with the heterogeneity motive, and differencing the $\lambda = 0.35$ and $\lambda = 0$ paths isolates the part contributed by restricted asset market access.

C.2 A Heterogeneous-Agent Economy with Idiosyncratic Expenditure Risk

C.2.1 Environment and Decentralized Equilibrium

Production, export demand, financial intermediation, labor aggregation, and nominal wage rigidity follow the baseline framework. Section 5.2.1 describes the household sector. A union that sets the nominal wage for task k evaluates

$$\mathbb{E}_1 \sum_{t=1}^{\infty} \beta^{t-1} \left[\int_0^1 (u(c_t^i) - v(N_t)) di - \frac{\psi_{nr}}{2} \left(\frac{W_{k,t}}{W_{k,t-1}} - 1 \right)^2 \right]$$

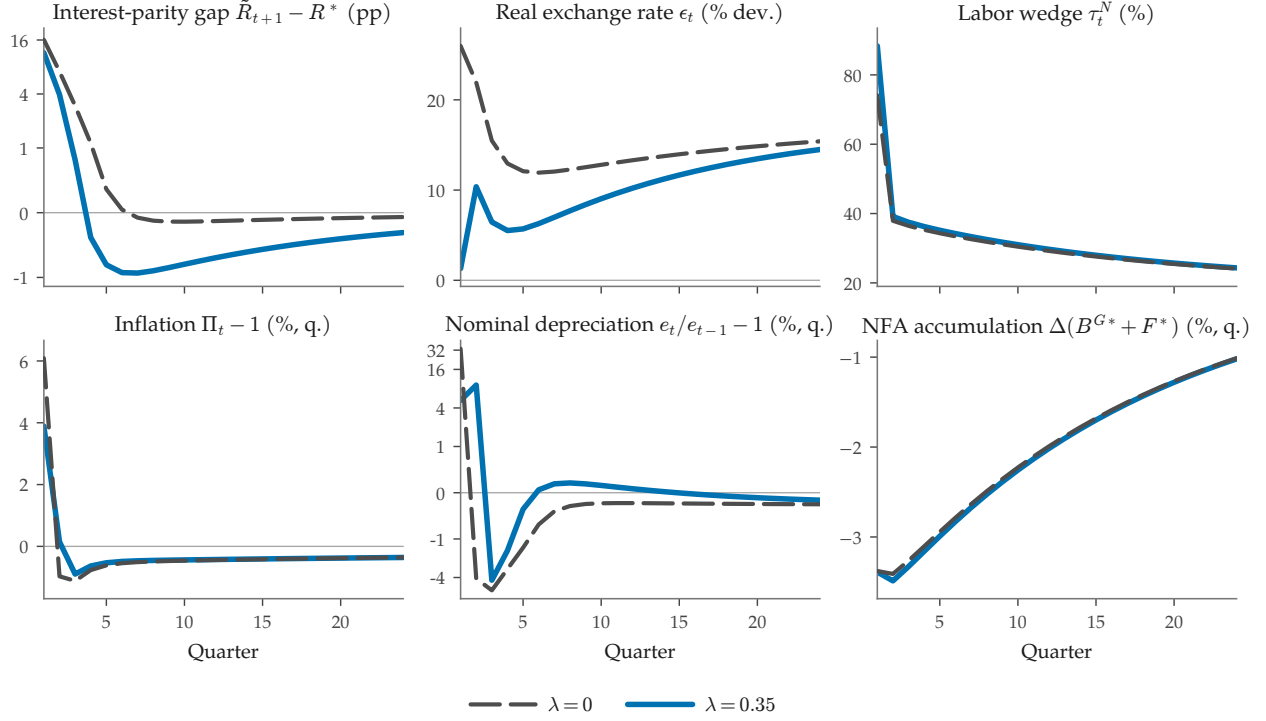


Figure C.1. Economy without lump-sum transfers: MIT-shock policy

Note: Deviations are relative to each economy's initial steady state; NFA accumulation is measured relative to steady-state output. The interest-parity-gap and nominal-depreciation panels use a symmetric log scale, linear within one percentage point of zero.

The wage Phillips curve remains (54), and the labor wedge remains (55), with average marginal utility $\bar{u}_t' = \int_0^1 u'(c_t^i) di$.

Fiscal and monetary policy. Policy follows the no-transfers economy of Section 5.1 and Appendix C.1. The money supply satisfies the exchange equation $\mathcal{M}_t = P_t Y_t$, and the monetary authority sets its growth rate according to

$$g_t^{\mathcal{M}} = \Pi_t \frac{Y_t}{Y_{t-1}} \quad (124)$$

Domestic bonds are indexed to inflation, with returns denominated in units of the domestic good. The ex-post real return is therefore independent of current inflation, eliminating the incentive to revalue inherited domestic claims. Nominal wage rigidity still gives monetary policy control over the labor wedge and employment.

Foreign exchange intervention is the only fiscal instrument. The public agent takes positions in domestic and foreign asset markets, lump-sum taxes and transfers are absent, and its positions satisfy the period budget constraint

$$B_{t+1}^G + \epsilon_t B_{t+1}^{G*} = R_t B_t^G + \epsilon_t R^* B_t^{G*} \quad (125)$$

The public agent satisfies no-Ponzi restrictions for both positions in domestic and foreign units

$$\begin{aligned} \lim_{T \rightarrow \infty} \frac{B_{T+1}^G}{\prod_{s=2}^T R_s} &= 0, & \lim_{T \rightarrow \infty} \frac{B_{T+1}^G}{\epsilon_T (R^*)^{T-1}} &= 0 \\ \lim_{T \rightarrow \infty} \frac{B_{T+1}^{G*}}{(R^*)^{T-1}} &= 0, & \lim_{T \rightarrow \infty} \frac{\epsilon_T B_{T+1}^{G*}}{\prod_{s=2}^T R_s} &= 0 \end{aligned} \quad (126)$$

Accounting conditions. Let $C_t \equiv \int_0^1 c_t^i di$ denote aggregate discretionary consumption. Goods-market clearing requires

$$C_t + \bar{z} + \zeta \epsilon_t^{1+\chi} = A_t N_t \quad (127)$$

Here $\bar{z}$ is the aggregate autonomous expenditure need. Domestic bond-market clearing remains (65), where $B_{t+1} \equiv \int_0^1 b_{t+1}^i di + \underline{b}$ is the aggregate household bond position.

Aggregating household budget constraints gives

$$\frac{B_{t+1}}{\epsilon_t} - \tilde{R}_t \frac{B_t}{\epsilon_{t-1}} = \frac{w_t N_t - C_t - \bar{z}}{\epsilon_t} \quad (128)$$

Aggregate household bond holdings satisfy transversality conditions in domestic and foreign units

$$\lim_{T \rightarrow \infty} \frac{B_{T+1}}{\prod_{s=2}^T R_s} = 0, \quad \lim_{T \rightarrow \infty} \frac{B_{T+1}}{\epsilon_T (R^*)^{T-1}} = 0 \quad (129)$$

The balance-of-payments condition (66) follows from household budget constraints, firms' zero-profit condition, the public budget constraint, and goods and bond-market clearing.

Recursive substitution for B_{t+1}^{G*} in (125), together with the public no-Ponzi restrictions (126), gives the public intertemporal budget constraint in foreign units

$$\sum_{t=1}^{\infty} \frac{\left(1 - \frac{\tilde{R}_{t+1}}{R^*}\right) \frac{B_{t+1}}{\epsilon_t}}{R^{*t-1}} = R_1 \frac{B_1^G}{\epsilon_1} + R^* B_1^{G*} \quad (130)$$

Goods-market clearing and firms' zero-profit condition imply $\frac{w_t N_t - C_t - \bar{z}}{\epsilon_t} = \zeta \epsilon_t^\chi - P_t^{T*} \theta N_t$. Iterating the aggregate household budget constraint (128) and applying (129) gives the aggregate household intertemporal budget constraint in foreign units

$$\sum_{t=1}^{\infty} \frac{\left(1 - \frac{\tilde{R}_{t+1}}{R^*}\right) \frac{B_{t+1}}{\epsilon_t}}{R^{*t-1}} = R_1 \frac{B_1}{\epsilon_1} + \sum_{t=1}^{\infty} \frac{\zeta \epsilon_t^\chi - P_t^{T*} \theta N_t}{R^{*t-1}} \quad (131)$$

Substituting bond-market clearing (65) and the household constraint (131) into (130), then applying intermediary optimality (59), gives the external intertemporal budget constraint

$$\sum_{t=1}^{\infty} \frac{\frac{1}{\bar{r}} \left(1 - \frac{\tilde{R}_{t+1}}{R^*}\right)^2}{R^{*t-1}} + \sum_{t=1}^{\infty} \frac{P_t^{T*} \theta N_t}{R^{*t-1}} = \sum_{t=1}^{\infty} \frac{\zeta \epsilon_t^\chi}{R^{*t-1}} + R^* B_1^{G*} + R_1 \frac{\epsilon_0 F_1^*}{\epsilon_1} \quad (132)$$

Decentralized equilibrium. Given

- an initial distribution over (b_1^i, z_1^i) ,
- aggregate states $\{R_1, w_0, \epsilon_0, F_1^*, B_1^{G*}, Y_0\}$,
- an exogenous sequence of import prices and productivity $\{P_t^{T*}, A_t\}_{t=1}^\infty$,
- fiscal-monetary policy $\{B_{t+1}^G, B_{t+1}^{G*}, g_t^M\}_{t=1}^\infty$,

a decentralized equilibrium consists of allocations $\{c_t^i, b_{t+1}^i, N_t, M_t, Y_t, X_t, F_{t+1}^*, \}_{t \geq 1}$ and prices $\{\epsilon_t, w_t, \Pi_t, R_{t+1}\}_{t=1}^\infty$ such that, for every $t \geq 1$:

- Given prices and policy, households satisfy their budget constraints and optimality conditions

$$c_t^i + b_{t+1}^i = R_t b_t^i + w_t N_t - z_t^i + (R_t - 1)\underline{b}, \quad b_{t+1}^i \geq 0$$

$$u'(c_t^i) \geq \tilde{\beta} R_{t+1} \mathbb{E}_t [u'(c_{t+1}^i)], \quad b_{t+1}^i \left[u'(c_t^i) - \tilde{\beta} R_{t+1} \mathbb{E}_t (u'(c_{t+1}^i)) \right] = 0$$

together with the aggregate household transversality conditions (129).

- Firms satisfy $Y_t = A_t N_t$, $M_t = \theta N_t$, and $w_t = A_t - \theta \epsilon_t P_t^{T*}$.
- The labor aggregator and unions satisfy

$$\left(\frac{w_t}{w_{t-1}} \Pi_t - 1 \right) \frac{w_t}{w_{t-1}} \Pi_t = \frac{\epsilon_w}{\psi_{nr}} \left[N_t v'(N_t) - \frac{1}{\mu_w} w_t N_t \bar{u}'_t \right] + \beta \left[\left(\frac{w_{t+1}}{w_t} \Pi_{t+1} - 1 \right) \frac{w_{t+1}}{w_t} \Pi_{t+1} \right]$$

- International financial intermediaries satisfy $F_{t+1}^* = \frac{1}{R^*} \left[1 - \frac{R_{t+1}}{R^*} \left(\frac{\epsilon_t}{\epsilon_{t+1}} \right) \right]$.
- The public budget constraint (125) and no-Ponzi conditions (126) hold, where $B_1^G = \epsilon_0 F_1^* - B_1$ and $B_1 = \int_0^1 b_1^i di + \underline{b}$.
- Monetary policy implies $g_t^M = \Pi_t \frac{Y_t}{Y_{t-1}}$.
- World demand for domestic output satisfies $X_t = \zeta \epsilon_t^{1+\chi}$.
- The final-goods market clears: $C_t + \bar{z} + X_t = A_t N_t$, where $C_t = \int_0^1 c_t^i di$.
- Labor market clears.
- The domestic bond market clears: $B_{t+1} + B_{t+1}^G = \epsilon_t F_{t+1}^*$, where $B_{t+1} = \int_0^1 b_{t+1}^i di + \underline{b}$.
- The other budget and market-clearing conditions imply the balance of payments $P_t^{T*} M_t + F_{t+1}^* + B_{t+1}^{G*} = \zeta \epsilon_t^\chi + R_t \frac{\epsilon_{t-1}}{\epsilon_t} F_t^* + R^* B_t^{G*}$.

C.2.2 Ramsey Problem and Distributional Commitments

The primal Ramsey problem represents the optimal policy of a utilitarian public agent subject to the decentralized equilibrium. The planner discounts future utility at $\beta = 1/R^*$, and households discount

at $\tilde{\beta} < \beta$. Fix an initial distribution over (b_1^i, z_1^i) , the Markov process for z_t^i , initial aggregate states $\{R_1, \epsilon_0, F_1^*, B_1^{G*}\}$, and a deterministic path of fundamentals $\{A_t, P_t^{T*}\}$. The planner solves

$$\begin{aligned}
& \max_{\{c_t^i, b_{t+1}^i, N_t, \epsilon_t, R_{t+1}, F_{t+1}^*, B_{t+1}^{G*}\}_{t=1}^{\infty}} \mathbb{E}_1 \sum_{t=1}^{\infty} \beta^{t-1} \left[\int u(c_t^i) di - v(N_t) \right] \quad \text{s.t.} \\
[\xi_t] & \quad \int c_t^i di + \bar{z} + \zeta \epsilon_t^{1+\chi} = A_t N_t \\
[\alpha_t^i] & \quad u'(c_t^i) \geq \tilde{\beta} R_{t+1} \mathbb{E}_t u'(c_{t+1}^i) \\
[\lambda_t^i] & \quad c_t^i + b_{t+1}^i = (A_t - \epsilon_t P_t^{T*} \theta) N_t - z_t^i + R_t b_t^i + (R_t - 1) \bar{b} \\
[\nu_t^i] & \quad b_{t+1}^i \geq 0 \\
[\rho_t^i] & \quad b_{t+1}^i \left[u'(c_t^i) - \tilde{\beta} R_{t+1} \mathbb{E}_t u'(c_{t+1}^i) \right] = 0 \\
[\varphi_t] & \quad P_t^{T*} \theta N_t + F_{t+1}^* + B_{t+1}^{G*} = \zeta \epsilon_t^{\chi} + R_t \frac{\epsilon_{t-1}}{\epsilon_t} F_t^* + R^* B_t^{G*} \\
[\psi_t] & \quad F_{t+1}^* = \frac{1}{\Gamma} \left[1 - \frac{R_{t+1}}{R^*} \frac{\epsilon_t}{\epsilon_{t+1}} \right]
\end{aligned} \tag{133}$$

A solution must, in addition, satisfy the no-Ponzi condition on the economy's net foreign asset position,

$$\lim_{T \rightarrow \infty} \frac{B_{T+1}^{G*} + F_{T+1}^*}{R^{*T-1}} = 0 \tag{134}$$

which rules out plans that finance the import bill through external borrowing without repayment. Each period- t constraint enters the planner's Lagrangian scaled by β^{t-1} , with the household-level constraints integrated over i . The first-order conditions for c_t^i and ϵ_t distinguish $t = 1$ from every later period. At $t = 1$, the FOC with respect to c_1^i is

$$u'(c_1^i) - \xi_1 - \lambda_1^i + (\alpha_1^i + \rho_1^i b_2^i) u''(c_1^i) = 0 \tag{135}$$

and for every $t \geq 2$ it is

$$u'(c_t^i) - \xi_t - \lambda_t^i + (\alpha_t^i + \rho_t^i b_{t+1}^i) u''(c_t^i) - \frac{\tilde{\beta}}{\beta} R_t (\alpha_{t-1}^i + \rho_{t-1}^i b_t^i) u''(c_t^i) = 0 \tag{136}$$

since period- t consumption enters the period- $(t-1)$ Euler constraint through its expected marginal utility. The FOC with respect to ϵ_t for $t \geq 2$ is:

$$\begin{aligned}
0 = & -\xi_t \zeta (1 + \chi) \epsilon_t^{\chi} - P_t^{T*} \theta N_t \int \lambda_t^i di + \varphi_t \zeta \chi \epsilon_t^{\chi-1} - \varphi_t R_t \frac{\epsilon_{t-1}}{\epsilon_t^2} F_t^* \\
& - \frac{\psi_t R_{t+1}}{\Gamma R^* \epsilon_{t+1}} + \beta \varphi_{t+1} \frac{R_{t+1} F_{t+1}^*}{\epsilon_{t+1}} + \frac{\psi_{t-1} R_t \epsilon_{t-1}}{\beta \Gamma R^* \epsilon_t^2}
\end{aligned} \tag{137}$$

At $t = 1$ the last term is absent, since the problem contains no $t = 0$ intermediary condition, and the initial intermediary position enters only through F_1^* . The remaining first-order conditions, for b_{t+1}^i, R_{t+1}, N_t ,

F_{t+1}^* , and B_{t+1}^{G*} , are symmetric across all $t \geq 1$:

$$0 = -\lambda_t^i + \rho_t^i \left[u'(c_t^i) - \tilde{\beta} R_{t+1} \mathbb{E}_t u'(c_{t+1}^i) \right] + v_t^i + \beta R_{t+1} \mathbb{E}_t \lambda_{t+1}^i \quad (138)$$

$$0 = -\tilde{\beta} \int (\alpha_t^i + \rho_t^i b_{t+1}^i) \mathbb{E}_t u'(c_{t+1}^i) di - \frac{\psi_t}{\Gamma R^*} \frac{\epsilon_t}{\epsilon_{t+1}} + \beta \int \mathbb{E}_t \lambda_{t+1}^i (b_{t+1}^i + \underline{b}) di \\ + \beta \varphi_{t+1} \frac{\epsilon_t}{\epsilon_{t+1}} F_{t+1}^* \quad (139)$$

$$0 = -v'(N_t) + \xi_t A_t + (A_t - \epsilon_t P_t^{T*} \theta) \int \lambda_t^i di - \varphi_t P_t^{T*} \theta \quad (140)$$

$$0 = -\varphi_t - \psi_t + \beta \varphi_{t+1} R_{t+1} \frac{\epsilon_t}{\epsilon_{t+1}} \quad (141)$$

$$0 = -\varphi_t + \beta R^* \varphi_{t+1} \quad (142)$$

together with the complementary slackness conditions

$$\alpha_t^i \geq 0, \quad \alpha_t^i \left[u'(c_t^i) - \tilde{\beta} R_{t+1} \mathbb{E}_t u'(c_{t+1}^i) \right] = 0, \quad v_t^i \geq 0, \quad v_t^i b_{t+1}^i = 0, \quad \rho_t^i \in \mathbb{R} \quad (143)$$

The asymmetry at $t = 1$ identifies two sources of time inconsistency. A problem beginning at $t = 1$ omits the household Euler constraints and intermediary optimality condition dated $t = 0$. The missing terms in (135) and in the $t = 1$ version of (137) are the derivatives of those conditions with respect to c_1^i and ϵ_1 . A planner re-optimizing at any later date t likewise drops the period- $(t - 1)$ conditions and generally chooses a different continuation.

The missing $t = 0$ Euler implementability constraints produce the real distributional time inconsistency studied by [Dávila and Schaab \(2023\)](#). Their recursive formulation records the corresponding promises through forward-looking individual Bellman equations; our sequence formulation records them through household Euler constraints. Once household asset positions are predetermined, a utilitarian planner that re-optimizes would lower real returns to favor indebted households with high marginal utility. Their forward-looking Phillips curve supplies a second time-zero promise that remains in the representative-agent case. That nominal promise is distinct from the household-Euler promise considered here. Domestic claims are indexed, so the Ramsey problem also excludes Fisherian revaluation of predetermined positions. Households chose their saving against the announced real-return path, and the lagged multipliers in (136) value the marginal utilities at which those choices were optimal. A planner re-optimizing at t treats past saving as predetermined. Because the Euler equalities of unconstrained households map consumption into real returns, dropping the inherited multipliers generally changes the announced return path and redistributes across the asset distribution. Appendix C.2.7 illustrates this incentive under constant fundamentals.

The missing $t = 0$ intermediary condition produces a separate external-balance-sheet source of time inconsistency. Intermediaries choose F_t^* at $t - 1$ given the announced ϵ_t . When the planner re-optimizes at t , F_t^* is predetermined and the period- $(t - 1)$ intermediary condition is absent, which changes the incentive governing ϵ_t . This source is present in the representative-agent and two-agent models as well.

Ramsey problem under timeless commitment. Following [Marcet and Marimon \(2019\)](#), the planner carries the multipliers on forward-looking implementability conditions across periods as state variables. A problem beginning in period τ honors the promises inherited from period $\tau - 1$. Define $\Theta_t^i \equiv \alpha_t^i + \rho_t^i b_{t+1}^i$, and let $(\{\Theta_{\tau-1}^i\}_{i \in [0,1]}, \psi_{\tau-1})$ denote the inherited states. Rearranging the Lagrangian associated with (133) gives the timeless problem in Proposition C.1.

Proposition C.1. *Given a period $\tau \geq 1$, inherited states $(\{\Theta_{\tau-1}^i\}_{i \in [0,1]}, \psi_{\tau-1})$, an initial distribution over idiosyncratic states (b_τ^i, z_τ^i) , and initial aggregate states $(R_\tau, \epsilon_{\tau-1}, F_\tau^*, B_\tau^{G*})$, define*

$$\tilde{W}_\tau = \mathbb{E}_\tau \sum_{t=\tau}^{\infty} \beta^{t-\tau} \left[\int u(c_t^i) di - v(N_t) \right] - \frac{\tilde{\beta}}{\beta} R_\tau \int \Theta_{\tau-1}^i u'(c_\tau^i) di - \frac{\psi_{\tau-1}}{\beta \Gamma R^*} \frac{R_\tau \epsilon_{\tau-1}}{\epsilon_\tau} \quad (144)$$

The Ramsey problem under timeless commitment maximizes $\tilde{W}_\tau$ over $\{c_t^i, b_{t+1}^i, N_t, \epsilon_t, R_{t+1}, F_{t+1}^, B_{t+1}^{G*}\}_{t \geq \tau}$ subject to the constraints of (133) for $t \geq \tau$. For every $\tau' > \tau$, the continuation from τ' onward of the solution of the period- τ problem solves the period- τ' problem whose inherited states $(\{\Theta_{\tau'-1}^i\}_i, \psi_{\tau'-1})$ are the corresponding multipliers of that solution: the planner has no incentive to re-optimize at any period.*

Proof. The inherited terms in (144) involve only c_τ^i and ϵ_τ , so no optimality condition at $t > \tau$ changes. Differentiation with respect to c_τ^i adds $-\frac{\tilde{\beta}}{\beta} R_\tau \Theta_{\tau-1}^i u''(c_\tau^i)$ to the initial-period consumption condition, which turns it into (136) at $t = \tau$ with $\alpha_{\tau-1}^i + \rho_{\tau-1}^i b_\tau^i$ replaced by $\Theta_{\tau-1}^i$. Differentiation with respect to ϵ_τ adds $\frac{\psi_{\tau-1}}{\beta \Gamma R^*} \frac{R_\tau \epsilon_{\tau-1}}{\epsilon_\tau^2}$, the term of (137) that the initial-period condition otherwise lacks. The conditions for b_{t+1}^i , R_{t+1} , N_t , F_{t+1}^* , and B_{t+1}^{G*} in equations (138) through (143) are unaffected. The optimality system of the period- τ problem therefore takes the same form at every $t \geq \tau$, and for $t \geq \tau'$ it coincides with the optimality system of the period- τ' problem whose inherited states are the multipliers $(\{\Theta_{\tau'-1}^i\}_i, \psi_{\tau'-1})$ of the period- τ solution. The two problems also share their constraints for $t \geq \tau'$, so the continuation of the period- τ solution satisfies the optimality conditions of the period- τ' problem, and re-optimization at τ' selects the same plan. $\square$

Implementability. The following lemma establishes that a solution to the timeless Ramsey problem can be decentralized by the policy instruments of Appendix C.2.1.

Lemma C.2. *Given $F_1^*, B_1^{G*}, R_1 > 0, w_0 > 0, \beta R^* = 1$, an initial distribution over (b_1^i, z_1^i) whose z -marginal is π_z , and positive sequences $\{A_t, P_t^{T*}\}_{t \geq 1}$ satisfying $A_t \leq \bar{A} < \infty$ and $\underline{P} \leq P_t^{T*} \leq \bar{P}$ with $\underline{P} > 0$ and $\bar{P} < \infty$ for all $t \geq 1$, let $\{c_t^i, b_{t+1}^i, N_t, \epsilon_t, R_{t+1}, F_{t+1}^*, B_{t+1}^{G*}\}_{t \geq 1}$ solve the Ramsey problem of Proposition C.1 with $\tau = 1$. Suppose further that the solution satisfies for all $t \geq 1$:*

$$\epsilon_t \geq \underline{\epsilon}, \quad N_t \leq \bar{N}, \quad 0 < R_{t+1} \leq \bar{R}, \quad B_{t+1} \leq \bar{B}, \quad \lim_{T \rightarrow \infty} \prod_{s=2}^T R_s = \infty$$

where $\underline{\epsilon} > 0$ and $\bar{N}, \bar{R}, \bar{B} < \infty$, and that the conditions of Lemma B.2 hold for $w_t = A_t - \epsilon_t P_t^{T} \theta$ and $\bar{u}_t' = \int u'(c_t^i) di$. Then the Ramsey allocation is implementable as a decentralized equilibrium of*

Appendix C.2.1. Conditional on the Ramsey allocation and the initial positions, the implementing paths $\{B_{t+1}^G, B_{t+1}^{G}\}_{t \geq 1}$ are unique.*

Proof. Construct w_t , M_t , Y_t , and X_t as in the proof of Lemma B.1:

$$w_t = A_t - \epsilon_t P_t^{T*} \theta, \quad M_t = \theta N_t, \quad Y_t = A_t N_t, \quad X_t = \zeta \epsilon_t^{1+\chi}$$

These paths satisfy firm optimality and export demand. The household budget constraints, Euler inequalities, and complementarity conditions, goods market clearing, the balance of payments, and intermediary optimality are constraints of the Ramsey problem and hold at the solution, with $\{R_{t+1}, F_{t+1}^*, B_{t+1}^{G*}\}_{t \geq 1}$ the solution's own paths. Since $w_t > 0$ and $P_t^{T*} \geq \underline{P}$,

$$\epsilon_t < \frac{A_t}{P_t^{T*} \theta} \leq \frac{\bar{A}}{\underline{P} \theta} \equiv \bar{\epsilon}$$

Hence $\tilde{R}_{t+1} \equiv R_{t+1} \epsilon_t / \epsilon_{t+1} \in (0, \bar{R} \bar{\epsilon} / \underline{\epsilon}]$, so $F_{t+1}^* = \frac{1}{\bar{r}} \left(1 - \tilde{R}_{t+1} / R^*\right)$ is bounded, and since $b_{t+1}^i \geq 0$, the aggregate position satisfies $\underline{b} \leq B_{t+1} \leq \bar{B}$.

Aggregating the household budget constraints over i and using $\int z_t^i di = \bar{z}$ gives

$$C_t + B_{t+1} = w_t N_t - \bar{z} + R_t B_t \quad (145)$$

Construct B_{t+1}^G from the bonds market clearing condition, $B_{t+1}^G = \epsilon_t F_{t+1}^* - B_{t+1}$; the initial condition $B_1^G = \epsilon_0 F_1^* - B_1$ of Appendix C.2.1 holds by construction. The fiscal budget constraint (125) then holds period by period by Walras' identity: multiplying the balance of payments by ϵ_t and substituting goods market clearing (127) and firms' zero profits gives

$$\epsilon_t F_{t+1}^* + \epsilon_t B_{t+1}^{G*} = w_t N_t - C_t - \bar{z} + R_t \epsilon_{t-1} F_t^* + \epsilon_t R^* B_t^{G*}$$

and substituting $w_t N_t - C_t - \bar{z} = B_{t+1} - R_t B_t$ from (145) and the bonds market clearing condition yields (125).

For the no-Ponzi conditions (126), write the balance of payments in terms of the net foreign asset position $W_{t+1} \equiv B_{t+1}^{G*} + F_{t+1}^*$, using intermediary optimality to substitute $\tilde{R}_t F_t^* = R^* F_t^* - \Gamma R^* (F_t^*)^2$:

$$W_{t+1} = R^* W_t + d_t, \quad d_t \equiv \zeta \epsilon_t^\chi - P_t^{T*} \theta N_t - \Gamma R^* (F_t^*)^2$$

The sequence d_t is bounded, $|d_t| \leq \zeta \bar{\epsilon}^\chi + \bar{P} \theta \bar{N} + \Gamma R^* \sup_t (F_t^*)^2$. Iterating forward gives $W_{T+1} / R^{*T} = W_1 + \sum_{t=1}^T d_t / R^{*t}$, so the no-Ponzi condition (134) selects the forward solution

$$W_T = - \sum_{j=0}^{\infty} \frac{d_{T+j}}{R^{*j+1}}, \quad |W_T| \leq \frac{\sup_t |d_t|}{R^* - 1}$$

Hence W_t , and with it $B_{t+1}^{G*} = W_{t+1} - F_{t+1}^*$ and $B_{t+1}^G = \epsilon_t F_{t+1}^* - B_{t+1}$, are bounded. All four conditions in (126) and the aggregate household transversality conditions (129) follow: B_{T+1} , B_{T+1}^G , $\epsilon_T B_{T+1}^{G*}$, and

B_{T+1}^{G*} are bounded, $\epsilon_T R^{*T-1} \geq \underline{\epsilon} R^{*T-1} \rightarrow \infty$, and $\prod_{s=2}^T R_s \rightarrow \infty$ by hypothesis, so each limit is zero.

Finally, by Lemma B.2 applied with w_t and $\bar{u}'_t = \int u'(c_t^i) di$, there exists $\{\Pi_t\}_{t \geq 1} \subset (0, \infty)$ satisfying the wage Phillips curve, and the money-growth rule (124) implements this inflation path. All conditions of the decentralized equilibrium in Appendix C.2.1 therefore hold.

For uniqueness, fix the Ramsey allocation and the initial positions. Aggregating the household policies determines $\{B_{t+1}\}_{t \geq 1}$, the bonds market clearing condition then determines $\{B_{t+1}^G\}_{t \geq 1}$ uniquely, and the balance of payments determines $\{B_{t+1}^{G*}\}_{t \geq 1}$ uniquely given $\{F_{t+1}^*\}_{t \geq 1}$ and the initial positions. $\square$

C.2.3 The Ramsey System in State Space

The promise multiplier Θ_t^i depends on individual history. For a given deterministic sequence $\{A_t, P_t^{T*}\}_{t \geq 1}$, current assets and the autonomous consumption state are sufficient for household choices. We write the policy functions as

$$c_t^i = g_{ct}(b_t^i, z_t^i), \quad b_{t+1}^i = g_{bt}(b_t^i, z_t^i) \quad (146)$$

where $b \in \mathbb{R}_+$ and z belongs to the support $\mathcal{Z}$ of the Markov process Z . Let μ_t denote the distribution of (b_t, z_t) , with μ_1 given. Household optimality requires

$$g_{ct}(b, z) + g_{bt}(b, z) = (A_t - \epsilon_t P_t^{T*} \theta) N_t - z + R_t b + (R_t - 1) \underline{b} \quad (147)$$

$$u'(g_{ct}(b, z)) \geq \tilde{\beta} R_{t+1} \mathbb{E}_z[u'(g_{c,t+1}(g_{bt}(b, z), z'))] \quad (148)$$

$$g_{bt}(b, z) \geq 0 \quad (149)$$

$$g_{bt}(b, z) \left\{ u'(g_{ct}(b, z)) - \tilde{\beta} R_{t+1} \mathbb{E}_z[u'(g_{c,t+1}(g_{bt}(b, z), z'))] \right\} = 0 \quad (150)$$

The policies induce a sequence of distributions satisfying, for every bounded continuous function h ,

$$\int h(b', z') d\mu_{t+1}(b', z') = \int \mathbb{E}_z[h(g_{bt}(b, z), z')] d\mu_t(b, z) \quad (151)$$

Aggregate consumption and household bond holdings are

$$C_t = \int g_{ct}(b, z) d\mu_t(b, z), \quad B_{t+1} = \int [g_{bt}(b, z) + \underline{b}] d\mu_t(b, z) \quad (152)$$

The external intertemporal budget constraint and intermediary optimality eliminate B_{t+1}^{G*} and F_{t+1}^* from the Ramsey system

$$\sum_{t=1}^{\infty} \frac{1}{(R^*)^{t-1}} \left[\frac{1}{\Gamma} \left(1 - \frac{R_{t+1}}{R^*} \frac{\epsilon_t}{\epsilon_{t+1}} \right)^2 + P_t^{T*} \theta N_t - \zeta \epsilon_t^x \right] = R^* B_1^{G*} + R_1 \frac{\epsilon_0 F_1^*}{\epsilon_1} \quad (153)$$

Let φ denote the multiplier on (153). The individual consumption and asset conditions are

$$\lambda_t^i + \xi_t = u'(c_t^i) + \left(\Theta_t^i - \frac{\tilde{\beta}}{\beta} R_t \Theta_{t-1}^i \right) u''(c_t^i) \quad (154)$$

$$\lambda_t^i = \beta R_{t+1} \mathbb{E}_t \lambda_{t+1}^i + \rho_t^i \left[u'(c_t^i) - \tilde{\beta} R_{t+1} \mathbb{E}_t u'(c_{t+1}^i) \right] + \nu_t^i \quad (155)$$

Given the household policies, these equations are linear in the current and inherited promises. Hence

$$\Theta_t^i = a_t(b_t^i, z_t^i) + q_t(b_t^i, z_t^i) \Theta_{t-1}^i \quad (156)$$

At a state where the borrowing constraint binds strictly,

$$g_{bt}(b, z) = 0, \quad u'(g_{ct}(b, z)) > \tilde{\beta} R_{t+1} \mathbb{E}_z [u'(g_{c,t+1}(g_{bt}(b, z), z')))] \Rightarrow a_t(b, z) = q_t(b, z) = 0 \quad (157)$$

Because ρ_t^i is unrestricted, equation (155) can be satisfied at such a state without imposing another equation on a_t or q_t . We assume that states at the Euler kink have zero probability. At an unconstrained state, the coefficient of Θ_{t-1}^i satisfies

$$\begin{aligned} & \left[q_t(b, z) - \frac{\tilde{\beta}}{\beta} R_t \right] u''(g_{ct}(b, z)) \\ &= \beta R_{t+1} q_t(b, z) \mathbb{E}_z \left[\left(q_{t+1}(g_{bt}(b, z), z') - \frac{\tilde{\beta}}{\beta} R_{t+1} \right) u''(g_{c,t+1}(g_{bt}(b, z), z')) \right] \end{aligned} \quad (158)$$

The constant terms satisfy

$$\begin{aligned} & u'(g_{ct}(b, z)) - \xi_t + a_t(b, z) u''(g_{ct}(b, z)) \\ &= \beta R_{t+1} \mathbb{E}_z [u'(g_{c,t+1}(g_{bt}(b, z), z')) - \xi_{t+1} + a_{t+1}(g_{bt}(b, z), z') u''(g_{c,t+1}(g_{bt}(b, z), z')))] \\ & \quad + \beta R_{t+1} a_t(b, z) \mathbb{E}_z \left[\left(q_{t+1}(g_{bt}(b, z), z') - \frac{\tilde{\beta}}{\beta} R_{t+1} \right) u''(g_{c,t+1}(g_{bt}(b, z), z')) \right] \end{aligned} \quad (159)$$

We select the bounded solution to equations (158) and (159), with the reset in (157) whenever the borrowing constraint binds strictly.

Define the conditional promise functions by

$$\bar{\Theta}_t^-(b, z) = \mathbb{E}[\Theta_{t-1}^i \mid b_t^i = b, z_t^i = z], \quad \bar{\Theta}_t(b, z) = a_t(b, z) + q_t(b, z) \bar{\Theta}_t^-(b, z) \quad (160)$$

At $t = 1$, the function $\bar{\Theta}_1^-$ is given. For every bounded continuous function h , its subsequent values satisfy

$$\int h(b', z') \bar{\Theta}_{t+1}^-(b', z') d\mu_{t+1}(b', z') = \int \mathbb{E}_z [h(g_{bt}(b, z), z')] \bar{\Theta}_t(b, z) d\mu_t(b, z) \quad (161)$$

Taking conditional expectations in (154) gives

$$\begin{aligned}\bar{\lambda}_t(b, z) &\equiv \mathbb{E}[\lambda_t^i \mid b_t^i = b, z_t^i = z] \\ &= u'(g_{ct}(b, z)) - \xi_t + \left[a_t(b, z) + \left(q_t(b, z) - \frac{\tilde{\beta}}{\beta} R_t \right) \bar{\Theta}_t^-(b, z) \right] u''(g_{ct}(b, z))\end{aligned}\quad (162)$$

When the Euler inequality is strict, complementary slackness gives $\alpha_t^i = 0$ and $b_{t+1}^i = 0$, so $\Theta_t^i = 0$. On the remaining states the Euler equation holds with equality, and therefore

$$\tilde{\beta} \int \Theta_t^i \mathbb{E}_t u'(c_{t+1}^i) di = \frac{1}{R_{t+1}} \int \bar{\Theta}_t(b, z) u'(g_{ct}(b, z)) d\mu_t(b, z) \quad (163)$$

Goods market clearing and the aggregate first-order conditions require

$$0 = \int g_{ct}(b, z) d\mu_t(b, z) + \bar{z} + \zeta \epsilon_t^{1+\chi} - A_t N_t \quad (164)$$

$$0 = v'(N_t) + \varphi P_t^{T*} \theta - \xi_t A_t - (A_t - \epsilon_t P_t^{T*} \theta) \int \bar{\lambda}_t(b, z) d\mu_t(b, z) \quad (165)$$

The first-order conditions (141) and (142) give $\varphi_t = \varphi$ and $\psi_t = \beta \varphi (\tilde{R}_{t+1} - R^*)$. Substituting these and (163) into (139) and multiplying by R^* gives

$$\begin{aligned}0 = & \int [g_{bt}(b, z) + \underline{b}] \mathbb{E}_z \left[u'(g_{c,t+1}(g_{bt}(b, z), z')) - \xi_{t+1} \right. \\ & \quad \left. + a_{t+1}(g_{bt}(b, z), z') u''(g_{c,t+1}(g_{bt}(b, z), z')) \right. \\ & \quad \left. + \left(q_{t+1}(g_{bt}(b, z), z') - \frac{\tilde{\beta}}{\beta} R_{t+1} \right) \bar{\Theta}_t(b, z) \times u''(g_{c,t+1}(g_{bt}(b, z), z')) \right] d\mu_t(b, z) \\ & - \frac{R^*}{R_{t+1}} \int \bar{\Theta}_t(b, z) u'(g_{ct}(b, z)) d\mu_t(b, z) + \frac{2\varphi}{\Gamma} \left(1 - \frac{R_{t+1}}{R^*} \frac{\epsilon_t}{\epsilon_{t+1}} \right) \frac{\epsilon_t}{\epsilon_{t+1}}\end{aligned}\quad (166)$$

The same substitutions in (137) give the real-exchange-rate conditions. At $t = 1$,

$$\begin{aligned}0 = & -P_1^{T*} \theta N_1 \int \bar{\lambda}_1(b, z) d\mu_1(b, z) - \xi_1 \zeta (1 + \chi) \epsilon_1^\chi + \varphi \zeta \chi \epsilon_1^{\chi-1} \\ & + \frac{2\varphi \beta R_2}{\Gamma \epsilon_2} \left(1 - \frac{R_2}{R^*} \frac{\epsilon_1}{\epsilon_2} \right) + \frac{\psi_0 R_1 \epsilon_0}{\Gamma \epsilon_1^2} - \frac{\varphi R_1 \epsilon_0 F_1^*}{\epsilon_1^2}\end{aligned}\quad (167)$$

For every $t \geq 2$,

$$\begin{aligned}0 = & -P_t^{T*} \theta N_t \int \bar{\lambda}_t(b, z) d\mu_t(b, z) - \xi_t \zeta (1 + \chi) \epsilon_t^\chi + \varphi \zeta \chi \epsilon_t^{\chi-1} + \frac{2\varphi \beta R_{t+1}}{\Gamma \epsilon_{t+1}} \left(1 - \frac{R_{t+1}}{R^*} \frac{\epsilon_t}{\epsilon_{t+1}} \right) \\ & - \frac{2\varphi R_t \epsilon_{t-1}}{\Gamma \epsilon_t^2} \left(1 - \frac{R_t}{R^*} \frac{\epsilon_{t-1}}{\epsilon_t} \right)\end{aligned}\quad (168)$$

Equations (147) through (161), (164) through (168), and (153) form the infinite-horizon Ramsey system for a

given path of fundamentals.

Policy without the real-income stabilization motive. The counterfactual in Section 5.2.3 removes the cyclical component of the real-income stabilization motive from the exchange-rate choice. In conditions (167) and (168), the term

$$-P_t^T \theta N_t \int \bar{\lambda}_t(b, z) d\mu_t(b, z) \quad (169)$$

is the planner's marginal valuation of the effect of the real exchange rate on household real incomes. The counterfactual replaces this term with its Ramsey steady-state value and solves the system of the preceding paragraphs with the modified conditions in place of (167) and (168); household policies, distributions, and aggregate quantities are evaluated at their realized values.

C.2.4 The Ramsey Steady State

The quantitative transition begins at a stationary Ramsey allocation under constant fundamentals. Set $A_t = A$ and $P_t^T = P^T$, impose $\beta R^* = 1$ and $\tilde{\beta} R^* < 1$, and select the stationary allocation with $F^* = 0$. Intermediary optimality then implies $R = R^*$. For a given preference parameter η , employment N and the real exchange rate ϵ are endogenous. Let $s = (b, z)$, and let h be any bounded continuous function. The stationary household policies and distribution satisfy

$$g_c(b, z) + g_b(b, z) = (A - \epsilon P^T \theta)N - z + R^*b + (R^* - 1)\underline{b} \quad (170)$$

$$u'(g_c(b, z)) \geq \tilde{\beta} R^* \mathbb{E}_z u'(g_c(g_b(b, z), z')) \quad (171)$$

$$g_b(b, z) \geq 0 \quad (172)$$

$$g_b(b, z) \left[u'(g_c(b, z)) - \tilde{\beta} R^* \mathbb{E}_z u'(g_c(g_b(b, z), z')) \right] = 0 \quad (173)$$

$$\int h(b', z') d\mu(b', z') = \int \mathbb{E}_z [h(g_b(b, z), z')] d\mu(b, z) \quad (174)$$

$$\int g_c(b, z) d\mu + \bar{z} + \zeta \epsilon^{1+\chi} = AN \quad (175)$$

$$B^{G*} = \frac{P^T \theta N - \zeta \epsilon^\chi}{R^* - 1} \quad (176)$$

In the stationary allocation, the affine promise equations satisfy the reset

$$g_b(s) = 0, \quad u'(g_c(s)) > \tilde{\beta} R^* \mathbb{E}[u'(g_c(s')) | s] \Rightarrow a(s) = q(s) = 0 \quad (177)$$

At an unconstrained state, equations (158) and (159) become

$$\left[q(s) - \tilde{\beta} (R^*)^2 \right] u''(g_c(s)) = q(s) \mathbb{E} \left[\left(q(s') - \tilde{\beta} (R^*)^2 \right) u''(g_c(s')) | s \right] \quad (178)$$

$$\begin{aligned} u'(g_c(s)) + a(s)u''(g_c(s)) &= \mathbb{E}[u'(g_c(s')) + a(s')u''(g_c(s')) \mid s] \\ &\quad + a(s)\mathbb{E}\left[\left(q(s') - \tilde{\beta}(R^*)^2\right) u''(g_c(s')) \mid s\right] \end{aligned} \quad (179)$$

Let $\bar{\Theta}^-(s) = \mathbb{E}[\Theta_{t-1}^i \mid s_t^i = s]$ and $\bar{\Theta}(s) = \mathbb{E}[\Theta_t^i \mid s_t^i = s]$. Their stationary conditional means satisfy

$$\begin{aligned} \bar{\Theta}(s) &= a(s) + q(s)\bar{\Theta}^-(s) \\ \bar{\Theta}^-(s') &= \mathbb{E}[\bar{\Theta}(s_t) \mid s_{t+1} = s'] \end{aligned} \quad (180)$$

Define $\mathcal{M}$ and B by

$$\mathcal{M}(s) = u'(g_c(s)) + [\bar{\Theta}(s) - \tilde{\beta}(R^*)^2 \bar{\Theta}^-(s)] u''(g_c(s)) \quad (181)$$

$$B = \int [g_b(s) + \underline{b}] d\mu \quad (182)$$

Since $g_b(s) + \underline{b}$ is measurable with respect to the arrival state s' , iterated expectations replace $\bar{\Theta}(s)$ in the integrated return condition with $\mathbb{E}[\bar{\Theta}(s) \mid s'] = \bar{\Theta}^-(s')$. The stationary return and real exchange rate conditions determine ξ and φ . The stationary conditions of the full balance-sheet formulation imply $\psi = 0$

$$\xi = \frac{\int [g_b(s) + \underline{b}] \mathbb{E}[\mathcal{M}(s') \mid s] d\mu - \int \bar{\Theta}(s) u'(g_c(s)) d\mu}{B} \quad (183)$$

$$\varphi = \frac{\xi [\zeta(1 + \chi)\epsilon^\chi - P^{T*}\theta N] + P^{T*}\theta N \int \mathcal{M}(s) d\mu}{\zeta\chi\epsilon^{\chi-1}} \quad (184)$$

$$\psi = 0 \quad (185)$$

The stationary labor condition is

$$\mathcal{L}(N, \epsilon) \equiv \eta N^\phi + \varphi P^{T*}\theta - \xi \epsilon P^{T*}\theta - (A - \epsilon P^{T*}\theta) \int \mathcal{M}(s) d\mu = 0 \quad (186)$$

The Ramsey steady state supplies the initial distribution μ , the aggregate state $(R, \epsilon, F^*, B^{G*})$, and the inherited promise states $(\bar{\Theta}^-, \psi_-)$. Stationarity gives $\psi_- = \psi = 0$, and B together with bond-market clearing determines the domestic bond positions.

Let the domain for (N, ϵ) be

$$\mathcal{D} = [N_L, N_H] \times [\epsilon_L, \epsilon_H] \subset (0, \infty) \times \left(0, \frac{A}{P^{T*}\theta}\right) \quad (187)$$

For each $(N, \epsilon) \in \mathcal{D}$, define

$$\mathcal{G}(N, \epsilon) = AN - \bar{z} - \zeta\epsilon^{1+\chi} - \int g_{c,N,\epsilon}(s) d\mu_{N,\epsilon} \quad (188)$$

and let $\mathcal{L}(N, \epsilon)$ be given by (186).

Lemma C.3. Suppose $R^* > 1$, $\beta R^* = 1$, $\tilde{\beta} R^* < 1$, and $\eta > 0$. For every $(N, \epsilon) \in \mathcal{D}$, suppose equations (170)

through (174) and (177) through (180) admit bounded solutions with finite integrals and $B \neq 0$, and suppose $\mathcal{G}$ and $\mathcal{L}$ are continuous. If, possibly after reversing both inequalities in either line,

$$\mathcal{G}(N_L, \epsilon) \leq 0 \leq \mathcal{G}(N_H, \epsilon), \quad \text{for every } \epsilon \in [\epsilon_L, \epsilon_H] \quad (189)$$

$$\mathcal{L}(N, \epsilon_L) \leq 0 \leq \mathcal{L}(N, \epsilon_H), \quad \text{for every } N \in [N_L, N_H] \quad (190)$$

then a Ramsey steady state exists. It is unique if $(\mathcal{G}, \mathcal{L}) = (0, 0)$ has a unique solution in $\mathcal{D}$ and the equations in the statement have a unique solution at that point.

Proof. The Poincaré-Miranda theorem and equations (189) and (190) give $(N, \epsilon) \in \mathcal{D}$ such that $\mathcal{G}(N, \epsilon) = \mathcal{L}(N, \epsilon) = 0$. These equalities are equations (175) and (186). The assumed solutions satisfy equations (170) through (174) and (177) through (180). Equations (176) and (181) through (185) define the remaining quantities. The stated conditions imply uniqueness. $\square$

C.2.5 Calibration

Parameters shared with the baseline model retain the values and normalizations in Table B.1. The heterogeneous-agent economy's calibration sets $R^* = 1.01$, $\tilde{\beta} = 0.97$, $\sigma = 2$, $\phi = 10$, $\chi = 0.60$, $\Gamma = 0.25$, and $\underline{b} = 0$. Mean autonomous expenditure is 35 percent of steady-state output. The labor-disutility scale sets steady-state employment to one, and the export-demand scale preserves the baseline import bill relative to wage income. Net foreign assets and the export share are equilibrium implications.

The autonomous-expenditure process is an ergodic five-state Markov chain. We choose its support and transition probabilities to match a stationary constrained share of 35 percent and a one-quarter probability of remaining constrained, conditional on beginning at the limit, of 97 percent. Households enter and leave the constrained region as their expenditure requirements and saving evolve. With the remaining parameters inherited from the baseline calibration, the exercise does not target a broader set of Japanese moments.

C.2.6 Computational Algorithm

We discretize the asset and expenditure states and replace the integrals in the Ramsey conditions by probability-weighted sums. For a horizon T , the nonlinear unknowns are

$$\{\epsilon_t, N_t, R_{t+1}, \xi_t\}_{t=1}^T \quad \text{and} \quad \varphi \quad (191)$$

The vector contains $4T + 1$ unknowns. Conditional on these paths, household policies, distributions, and promise functions are computed outside the nonlinear unknown vector.

We approximate the infinite-horizon system by the extended-path method. The continuation after the import-price disturbance is endogenous, and the terminal elements of the paths in (191) represent candidate limiting values. We impose a flat continuation at those values, after deriving the infinite-horizon first-order conditions, and the terminal allocation therefore remains free to differ from the initial Ramsey steady state. We choose a horizon over which the transition approaches its limit. Each evaluation of the nonlinear system

proceeds as follows.

- i. Hold the fundamentals constant after date T and extend $(\epsilon_T, N_T, R_{T+1}, \xi_T)$ as a flat tail. Solve the stationary household policies, invariant distribution, and bounded affine promise functions associated with this continuation. These objects provide the forward values used in the date- T conditions.
- ii. Given the candidate paths of employment, the real exchange rate, and the return, compute wage income at every date. Starting from the terminal consumption policy, solve equations (147) through (150) backward by the endogenous-grid method to obtain g_{ct} and g_{bt} .
- iii. Starting from the given initial distribution, apply equation (151) forward under the asset policies. At each date, aggregate consumption and saving over the resulting distribution.
- iv. Given the household policies, solve equation (158) backward for q_t and equation (159) backward for α_t , imposing the reset in equation (157). Carry the conditional inherited promises forward using equation (161), and construct $\bar{\lambda}_t$ from equation (162).
- v. Form a residual vector from the T goods-market-clearing conditions (164), the T labor conditions (165), the T return conditions (166), the T real-exchange-rate conditions (167) through (168), and the external intertemporal budget constraint (153). This gives $4T + 1$ equations for the unknowns in (191).
- vi. Update the nonlinear unknowns with a nonlinear root solver and repeat the preceding steps until the residual system is solved.

C.2.7 A Quantitative Illustration of Distributional Time Inconsistency

We start from the same Ramsey steady state under constant fundamentals. Timeless commitment honors the inherited household Euler promises and leaves the allocation unchanged, whereas reoptimization removes only those inherited promises at the initial date.

Figure C.2 shows that the reoptimizing planner lowers both the domestic real return and its foreign-unit counterpart at date 1, raising current consumption and employment while accepting higher inflation and a real depreciation. The inherited Euler multipliers prevent this reweighting of current marginal utilities by preserving the promises associated with the initial asset distribution. Because the public balance sheet determines the household return together with the real exchange rate, the distributional incentive studied by Dávila and Schaab (2023) also changes the intervention path in the open economy. Indexed domestic claims remove Fisherian revaluation, leaving the real-return promises embodied in earlier household Euler equations as the source of the deviation.

C.2.8 Cross-Sectional Incidence

Figure 8 isolates the impact of the real-wage difference between the two policies. Superscripts opt and cf denote optimal policy and policy without the real-income stabilization motive. The calculation starts from

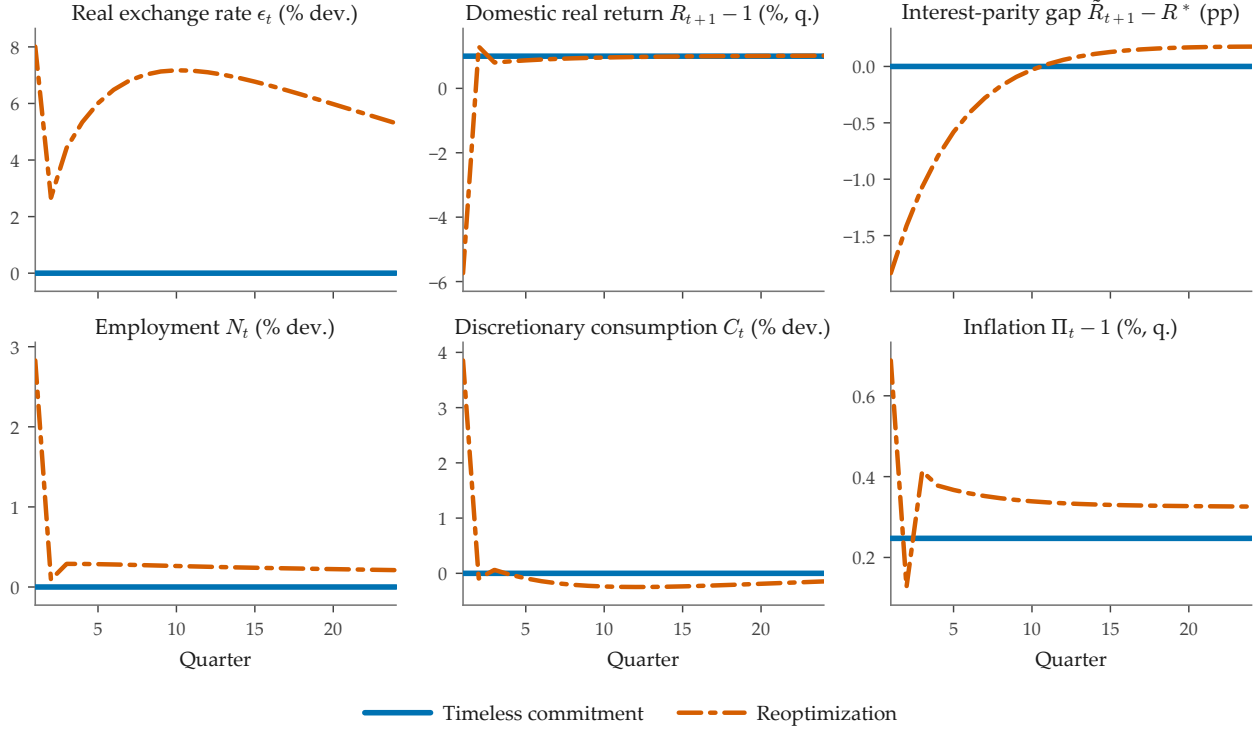


Figure C.2. Distributional time inconsistency under constant fundamentals

the counterfactual impact allocation and holds N_1^{cf} , the initial distribution, returns, and all continuation paths fixed. Replacing w_1^{cf} with w_1^{opt} increases each household's cash on hand by $(w_1^{\text{opt}} - w_1^{\text{cf}})N_1^{\text{cf}}$. The corresponding discretionary consumption is

$$g_{c1}^w(b, z) = g_{c1}^{\text{cf}}\left(b + \frac{(w_1^{\text{opt}} - w_1^{\text{cf}})N_1^{\text{cf}}}{R_1}, z\right) \quad (192)$$

For a set of impact states $\mathcal{S}$, define its contributions to the change in consumption and current utility by

$$\Delta C_1(\mathcal{S}) = \int_{\mathcal{S}} [g_{c1}^w(b, z) - g_{c1}^{\text{cf}}(b, z)] d\mu_1(b, z) \quad (193)$$

$$\Delta U_1(\mathcal{S}) = \int_{\mathcal{S}} [u(g_{c1}^w(b, z)) - u(g_{c1}^{\text{cf}}(b, z))] d\mu_1(b, z) \quad (194)$$

The first group contains households beginning at the borrowing limit, and households with positive assets form three groups of equal population ordered by initial assets. Figure 8 reports each group's percentage consumption change and its share $\Delta U_1(\mathcal{S})/\Delta U_1(\mathbb{R}_+ \times \mathbb{Z})$ of the impact utility gain.